

Title page

DELAY-RISK IDENTIFICATION AND PRIORITISATION FOR BRICK MASONRY WORK IN INDIAN BUILDING PROJECTS: AN EVIDENCE-TRACEABLE DECISION-SUPPORT TOOL

(Registered project title: “Framework for Brickwork Labor-Productivity and Delay-Risk Prediction Using Managerial Factors in Selected Indian Building Projects”. Section 1.6 explains how the work reported here relates to the registered title.)

A Final Year Project Report submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Civil Engineering

by

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Certificate

This is to certify that the project report entitled “**Delay-Risk Identification and Prioritisation for Brick Masonry Work in Indian Building Projects: An Evidence-Traceable Decision-Support Tool**” (registered title: “**Framework for Brickwork Labor-Productivity and Delay-Risk Prediction Using Managerial Factors in Selected Indian Building Projects**”) is a record of the work carried out by

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Declaration

We declare that this report is our own account of the project. Where we have used the work of others, we have named it and listed it in the References. Every numerical value in the report is either (a) copied from a cited published source, (b) computed from such values by a script or calculation that is named next to the result, (c) entered by a user of the tool as a labelled example, or (d) stated as an assumption of the tool and labelled as one. We have not collected, fabricated or altered any site data or survey responses. No site data and no expert survey were collected for this report, and the example case in the report is illustrative only.

We understand that any misrepresentation of the above is a breach of academic integrity.

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Declaration on the use of AI assistants

This project was carried out with substantial help from AI coding and writing assistants (Claude models from Anthropic, run as coding agents and chat sessions). We state what they did, what they did not do, and what the team is responsible for.

What AI agents did

Table 0.1. Tasks performed with AI assistance, and the check the team is asked to record for each.

Area	What the AI agents did	Check recorded by the team
Software	Wrote and edited most of the Python code of the tool (rules engine, risk matrix, register, exports, local web interface, PySide6 desktop wrapper, optional decision layer in app/analysis.py) and the automated tests.	[Team member, date: code read and tests run]
Literature search and notes	Ran keyword searches against open metadata sources, screened the resulting records by title and abstract, and wrote the research notes in Research_Notes/. The screening judgements are model output and were not independently re-read in full by a human (see Chapter 3 and docs/Literature_Database_Report.md).	[Team member, date: sample of records checked]
Data transcription	Transcribed relative importance index (RII) values from published papers into data/literature_seed.json. A later audit by AI agents compared all 182 rows with the printed tables and found the values matched; it also found and corrected metadata errors.	[Team member, date: rows spot-checked against the PDFs]
Audits and reviews	Ran a research-methods audit, an engine audit and four simulated committee reviews (engineering/UX, domain expert, methods, external examiner). These produced the list of withdrawn claims and corrected wording in docs/Audit_Corrections_2026-10-08.md and docs/Project_Handoff_Summary.md. They are not a substitute for review by the supervisor or an external examiner.	[Guide, date: reviewed]
Statistics	Wrote the scripts that compute the cross-study agreement statistics and sensitivity analyses (scripts/, analysis/).	[Team member, date: results re-computed]
Drafting	Drafted the text of this report from the repository files, with each number traced to a file or script output.	[Team member, date: report read in full]

What AI did not do

The AI assistants did **not** supply any probability, delay duration, cost, productivity rate, survey response or site observation. All probabilities, delays and costs that appear in the tool are typed in by a user and carry a Source label. The numbers taken from the literature are RII values copied from the published papers. The AI-suggestion feature of the tool can propose risk names, mechanisms and identifiers of evidence records only, and the tool rejects numeric fields in its output (guard rules described in Chapter 5). The example case in this report was written by the team for illustration and is labelled ILLUSTRATIVE wherever it appears.

What the team must verify

The team remains responsible for the content. Before submission the team should confirm, and record in Table 0.1, that: (1) the transcribed RII values and the factor-to-risk mapping have been checked by a human against the source papers; (2) the references exist and say what the report attributes to them, in particular the items listed as unverified in the reference assembly notes; (3) the statistics have been re-run by a team member; (4) the wording in the report about what was and was not done matches the real project history; and (5) the declaration above is accurate for each team member.

Acknowledgements

[To be written by the team. Suggested content: the project guide and any co-guide; the Head of Department; the faculty and practitioners who will be asked to rate risks in the expert-survey phase, if that phase goes ahead; family and friends. Do not name anyone who has not agreed to be named.]

Abstract

Delays in brick and block masonry recur in Indian building projects, and published delay-cause studies rank factors by relative importance index (RII) without carrying the ranking into a site-level risk register. This report describes an offline decision-support tool for delay risk in one activity, brick or blockwork masonry. The registered title promised labour-productivity and delay prediction from managerial factors in selected Indian projects. Because no site access was available, the scope was reduced to risk identification, a source-labelled risk register and a five-by-five probability-impact matrix, and the report states this change openly.

The tool holds 46 masonry delay risks, 22 rules that flag risks from site facts, and a register in which every number carries a Source label (Literature, Historical Data, User Input, Expert Judgment, Derived Calculation or Assumption). For 30 of the 46 risks an ordinal starting position is derived from 182 RII values transcribed from 11 published studies; five studies form the seed and six are held out. An RII measures importance, not probability or days, so this starting position is an assumption of the tool and is labelled as one. An optional Monte Carlo and expected-monetary-value layer uses only numbers the user enters.

Agreement between the published rankings is weak. Cross-prediction between seed studies gave Spearman correlations of 0.09 to 0.64, none significant, and agreement with three held-out surveys (0.28, 0.63 and -0.18) did not survive correction for multiple testing. No site data were collected and no expert survey was run, so the tool is verified as software but not validated on real projects, and the example case is illustrative. Site validation and expert elicitation are defined as future work.

Keywords

brick masonry; construction delay; risk register; probability-impact matrix; relative importance index; decision support; India

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List of abbreviations

Table 0.2. Abbreviations used in this report.

Abbreviation	Meaning
AAC	Autoclaved aerated concrete (block)
AI	Artificial intelligence
APA	American Psychological Association (citation style, 7th edition)
BIS	Bureau of Indian Standards
CPWD	Central Public Works Department (India)
CSV	Comma-separated values (file format)
DSS	Decision-support system
EMV	Expected monetary value
FYP	Final year project
GUI	Graphical user interface
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
LLM	Large language model
LOSO	Leave-one-study-out
MC	Monte Carlo (simulation)
NICMAR	The institution at which the project was carried out (see title page)
P-I	Probability-impact (as in P-I matrix)
PERT	Programme evaluation and review technique (three-point estimate)
RII	Relative importance index
SE	Standard error
WBS	Work breakdown structure
Seed	The default ordinal starting position of a library risk derived from RII values of the seed studies
Held-out study	A published survey whose RII values are not used to build the seed and are used only for comparison
ILLUSTRATIVE	Label for example numbers that are not evidence

Chapter 1. Introduction

Background

Brick masonry in Indian building construction

Brick and block masonry is labour-intensive hand work. In a framed residential or commercial building the walls are laid by gangs of masons and helpers after the structural frame is cast, and plastering, services and finishing trades cannot proceed in a room until its walls are up. A delay in masonry therefore delays the activities that follow it. The activity depends on the daily availability of skilled labour, on bricks or blocks and mortar materials arriving and being staged near the working face, on access and scaffolding, on weather, and on payment and supervision arrangements that determine whether a gang turns up and stays.

Three strands of published evidence, read for this project, describe how this work behaves on Indian sites. They are small studies and none of them is a representative sample of India, so they are used here only to describe the kind of problem, not to size it.

- Reddy and Chambrelin (2021) video-recorded brickwork at seven residential sites in Andhra Pradesh. Each site was observed once for roughly one to two hours, and the reported work efficiency ranged from 62.57% to 90.62% (Source: Literature; `Research_Notes/A_masonry_india.md`, card A01). The authors attribute low efficiency to lack of work targets, poor planning and monitoring,

material shortage and equipment breakdown. The sample is seven short observations and the area-based productivity column of the paper is not usable, as discussed in the research notes.

- Ponmalar et al. (2018) recorded the brick masonry activity on two three-storey residential buildings in Chennai. Both activities were planned as 20 working days. The authors report 6.14% and 8.17% work-hour overrun and that 7 and 8 of the 20 days respectively were ineffective (Source: Literature; card A02). The definitions are the authors' own and the sample is two projects.
- Loganathan and Kalidindi (2015) report, from the abstract of a pilot study on one project in Tamil Nadu, a 20% to 40% production variation between masonry crews (Source: Literature; card A03; the abstract does not define the variation measure, and the full text was not accessible).

The same research notes record a second set of studies that name the drivers of workforce unavailability in India. Loganathan and Kalidindi (2016) interviewed project managers and labour sub-contractors on 15 projects in six states and report illness, injury, lack of basic facilities and payment delays as significant contributors to absenteeism and turnover among migrant construction workers. Guha and Biswas (2008) used Monte Carlo simulation on a housing project in Kolkata and report that the monsoon increased project duration by about 5% and cost by about 12% (Source: Literature; abstract only).

Delay research in India and elsewhere

Most of the published work on construction delay asks practitioners to rate a list of candidate causes and then ranks the causes by an index, most often the relative importance index (RII). Chapter 2 reviews these studies in detail. Two features of this literature matter for the problem stated below. First, most surveys cover building or infrastructure construction in general, not one activity. Second, the output of a survey is a ranking of importance, whereas a site engineer who has to decide what to do on Monday morning needs to know which of the listed risks is present on the site, how likely it is to cause loss of working days, and what the loss would be. The ranking does not state these.

Risk registers and matrices in practice

Risk management standards describe a cycle of identifying, analysing, evaluating and treating risk (International Organization for Standardization [ISO], n.d.). In project practice this cycle is commonly recorded in a risk register and summarised on a probability-impact (P-I) matrix, in which each risk is placed by a probability class and an impact class and given a level. A matrix is easy to read and to produce, but it is a coarse device with known weaknesses (Cox, 2008), which Chapter 2 discusses because they shape several design decisions in this project.

Problem statement

A site engineer or planning engineer who wants to keep a risk register for the masonry activity of a building faces three practical problems.

1. **Identification is unaided.** There is no short, referenced list of the delay risks that are specific enough to masonry to be checked against a site, and general lists of delay causes mix project-level, contractual and activity-level items.
2. **Numbers have no source.** A register needs a probability and an impact for each risk. Practitioners can enter them, but a number typed into a spreadsheet usually carries no record of whether it came from past records, a published study, a colleague's opinion or a guess. A reviewer cannot tell which numbers are evidence and which are assumptions.
3. **Published rankings are not usable as probabilities.** Importance indices from delay surveys are the most accessible quantitative literature, but an RII is a rating of how important respondents consider a factor. It is neither a probability that the factor occurs nor a number of days lost. Using it as either, which is tempting because it is a number between 0 and 1, would present an opinion rating as a measurement.

The project therefore asks what a tool can honestly do for the masonry activity when the user has, at best, partial project data and the published literature offers rankings but not probabilities, and how far such a tool can be checked without site data.

Aim

To develop and verify an offline tool that helps site engineers identify, register and prioritise delay risks in a brick/block masonry activity, with every value traceable to its source and published survey evidence used only as a labelled ordinal starting point.

Objectives

1. Compile and verify a masonry delay-risk library from published delay/productivity surveys, with a documented and inter-coder-checked factor-to-risk mapping.
2. Implement site-condition rules, a source-labelled risk register and a 5x5 probability-impact matrix with all assumptions stated.
3. Analyse how consistent published RII rankings are (cross-study agreement, held-out comparison, sensitivity of matrix placement) to establish how far literature can stand in for project data.
4. Verify the software and evaluate it with practitioners (expert likelihood/impact ratings, face validity, usability), defining site validation as future work.

Two of these objectives are only partly met in the work reported here, and the report says so in the chapters concerned. For Objective 1, the mapping was done by a single coder, so the inter-coder check has not been carried out. For Objective 4, the software was verified by automated tests and by a hand-worked calculation, but no practitioner evaluation (expert rating or usability session) has been run. Table 1.1 records the status of each objective.

Table 1.1. Status of the objectives at the time of writing.

Objective	Status	Where reported
1. Library and mapping	Library of 46 risks built; 182 RII rows transcribed from 11 studies and compared with the printed tables; mapping done by one coder, second-coder check not done	Chapters 3 and 4
2. Rules, register, matrix	Implemented, with 22 site-fact rules; assumptions listed in an Assumption register	Chapter 5
3. Consistency of rankings	Cross-study agreement, leave-one-study-out cross-prediction, held-out comparison and sensitivity analyses done; results weak and not significant after correction	Chapter 4
4. Verification and evaluation	Software verification done; practitioner evaluation not done; site validation defined as future work	Chapters 6 and 8

Source: counts from `README.md` and `docs/Project_Handoff_Summary.md` section 9 (Literature / Derived Calculation from repository files); status entries are the authors' statement of work done.

Scope

In scope.

- One construction activity: brick or block masonry in building projects in India.
- Identification of delay risks from a literature-derived library and from rules applied to site facts that the user enters.
- A register in which each risk can carry a probability, a delay range, a Source label for each number, an owner, an early-warning trigger, a response type (avoid, reduce, transfer, accept), an action and a review date.
- A five-by-five probability-impact matrix with stated edges and level thresholds, and exports (HTML report, image, CSV, Markdown, case file).
- An ordinal literature starting position, labelled as an Assumption, for the 30 of 46 library risks that have a survey value in the seed studies.
- An optional quantitative layer (Monte Carlo schedule simulation, delay-cost and expected-monetary-value comparison of responses) that uses only numbers entered by the user, each with a Source.

- A secondary-data analysis of the 11 published surveys held in the repository.

Out of scope.

- Any prediction of labour productivity, and any use of CPWD or BIS labour output norms in the tool. Legacy code for these exists in the repository, is not connected to the interface and is not used in any result in this report.
- Collection of site data, and any claim about the accuracy of the tool on real sites.
- An expert survey (a form is described in the project records; it has not been administered) and a usability study.
- Other activities (plastering, concreting), project-level scheduling, and cloud or mobile versions.
- Any probability, delay or cost supplied by an AI model.

Limitations that apply to the whole report. No site data were collected and no expert survey was run. The example case used to demonstrate the tool is illustrative and its numbers are not evidence. The RII values come from general building-construction surveys in India and Sri Lanka, not masonry-specific surveys, and the transcription covers only a subset of the factors in several of the papers. These limits are repeated where they bear on a result.

Scope revision relative to the registered title

The registered title of the project is “Framework for Brickwork Labor-Productivity and Delay-Risk Prediction Using Managerial Factors in Selected Indian Building Projects”. The first project plan described a desktop application that would take daily site conditions, look up a delay probability and a mitigation cost in a rules matrix, run a Monte Carlo simulation of the schedule using a work breakdown structure held in a database, and output a financial command based on expected monetary value. On 28 September 2026 the scope was cut to identification, register and matrix, The project records give the date of the decision but no written rationale. The reasons on which this report relies are that no site data were available, and that the probabilities, delays and costs the larger design needs could not be taken from the literature, since the survey rankings the project holds are importance ratings and cannot supply them. The code of the larger version was kept on disk. On 8 October 2026, at the team’s request, a quantitative layer was reconnected that uses only user-entered numbers (Handoff section 8.14). The title has not been changed in the project records. Table 1.2 maps each term of the registered title and of the first plan to what was done, and why.

Table 1.2. Scope revision: registered title terms and first-plan features against what was done and what was deferred.

Term or feature	Status in this report	What was done	Why, and what would be needed
“Framework”	Done (as a process)	A risk-management process of identify, analyse, evaluate and record, following the cycle described in ISO 31000, implemented as library, rules, register, matrix and exports	No change of meaning; it is a procedure and a tool, not a new theory
“Brickwork”	Done	Library, rules and the register are written for a brick/block masonry activity. The survey seed is not masonry-specific (see Chapter 4)	The surveys are general building-construction surveys; masonry-specific ratings would need an expert survey
“Labor-Productivity”	Deferred	Not modelled. Legacy productivity models and an India labour-norms file exist in the repository and are not used in any result	Needs measured output data from sites; a known defect in the legacy norms file was found in the audit; CPWD norms are planning norms, not measured productivity
“Delay-Risk”	Done (identification and prioritisation)	Risks are identified, registered and placed on a 5x5 matrix; an optional layer can simulate delay from entered numbers	Probabilities and delays for real sites need project records or expert elicitation
“Prediction”	Not done; wording withdrawn	The tool orders risks; it does not predict delays	An RII is importance, not likelihood, and no site outcome data exist to test a predictive claim

Term or feature	Status in this report	What was done	Why, and what would be needed
“Managerial Factors”	Partly done	The library contains management-type risks (for example poor planning, supervision, site management, communication, labour discipline) and some site facts relate to them. They are not analysed as a separate group of factors	A managerial-factor analysis needs project-level data on supervision hours, crew ratios and material staging
“Selected Indian Building Projects”	Not done	No projects were selected or studied. The seed uses published surveys from India and Sri Lanka	Needs site access and permission; a site data-collection form is described in the project records but was not used
Desktop application (PySide6)	Done, with a qualification	A local web interface runs offline in a browser and inside a PySide6 desktop window	The PySide6 window hosts the same local interface; it is not a separate native interface
Rules matrix of 27 states giving delay probability and mitigation cost	Replaced	22 rules flag risks from site facts and give no numbers; the 5x5 matrix uses user-entered or labelled values	A lookup of probability and cost needs a source for each cell
Monte Carlo (10,000 runs), P90 delay, work breakdown database	Optional layer only	A Monte Carlo schedule simulation exists in the optional layer and runs on entered numbers. The database-based work breakdown is not used	No default distributions are supplied; results are only as good as the entered ranges
Expected monetary value and the “authorize mitigation / accept risk” command	Optional layer only	The layer compares entered response costs and effects with delay cost and returns a command worded as “preferred under the stated criterion”	No cost or effect size is supplied by the tool; real values need quotations and site records
CPWD labour norms	Not in the product	Context only, in Chapter 2	Legacy norms file has a documented defect; out of scope
Daily dynamic risk from site conditions	Partly done	Site facts entered by the user trigger rules that flag risks	Rules are logical conditions labelled Assumption; practitioner endorsement not yet collected
Publication in a Scopus-indexed journal	Not supported by the present work	Not claimed	See Chapter 8 for routes that the work could support

Sources: registered title and first-plan description from the project statement; scope history from docs/Project_Handoff_Summary.md sections 1, 8.13, 8.14 and 9; status of each item from README.md and the repository.

The decision on whether to change the registered title has not been recorded in the project files; a title that matches the work is proposed on the cover of this report, and the choice rests with the guide.

Research gap

The review in Chapter 2 supports the following statement of the gap, stated as modestly as the evidence allows. Indian delay-cause studies rank factors by RII but stop at a ranking that does not carry into site risk registers. General risk tools and templates require the user to enter probabilities and impacts and do not tie them to evidence. Within the search reported in Chapter 2 we did not find an activity-level tool for masonry in which site conditions feed a register and every value carries a source label. Absence from a keyword search of titles and abstracts is weak evidence, and the corpus was not searched in full text. The gap is therefore a practical one that this project addresses, and not a claim that no such tool exists.

Contribution

The contributions of the work are practical and modest.

1. A referenced library of 46 masonry delay risks, with the survey factors that support each one, and a documented rule for mapping survey factors to risks.
2. A verified secondary dataset of 182 RII rows from 11 studies, with a recorded source location and a list of the errors and omissions found in the transcription.

3. An offline tool in which every number in a register carries one of six Source labels, and in which assumptions of the tool (class edges, level thresholds, the equal-count bands, the one-class step for a rule flag) are named, listed and printed on every export.
4. An analysis, from published data only, of how far survey rankings agree with each other, which shows that the agreement is weak and so limits how much weight the literature starting position can bear.
5. A set of withdrawn claims, recorded with the reason, as a guide for later work that uses RII data in a risk register.

The project does not propose a new risk-analysis method. The probability-impact matrix, RII, Monte Carlo simulation and expected-value comparison are all established, and Chapter 2 cites published work that already combines several of them for construction schedules. The report makes no claim that the tool predicts delays, that it is accurate on real sites, or that it is the first of its kind.

Structure of the report

Chapter 2 reviews the literature on delay causes, the RII method and its limits, risk-management standards, the critique of risk matrices, quantitative schedule risk analysis as context, labour productivity norms (briefly, as out of scope) and decision-support tools, and ends with a summary of the gap. Chapter 3 describes the methodology: design approach, literature search and screening, data extraction and verification, the factor-to-risk mapping, the construction of the seed and the statistical plan. Chapter 4 presents the secondary-data analysis of the 11 surveys, including what it does not show. Chapter 5 describes the system design: requirements, architecture, case schema, library, rules, matrix logic, the Assumption register and the guard on AI output. Chapter 6 reports software verification, a worked example labelled illustrative, sensitivity of the matrix and the known defects. Chapter 7 discusses what can and cannot be claimed and the threats to validity. Chapter 8 gives conclusions, limitations and future work, including the data that the productivity, Monte Carlo and expected-value parts of the registered title would need. References and appendices follow, including the 182-row seed table, the risk library and the rule table.

A note on numbers. Every number in the report is labelled with its Source (Literature, Historical Data, User Input, Expert Judgment, Derived Calculation or Assumption) in the sentence, table note or caption where it appears. No value labelled Historical Data appears in this report, because no site records were obtained.

Chapter 2. Literature Review

Purpose and approach

This chapter reviews the published work that bears on one question: what can a risk register for brick masonry draw from the literature, and what must it leave to the user? The review covers delay-cause studies in India and elsewhere, the relative importance index (RII) that most of them use, the risk-management standards and the probability-impact matrix on which registers rest, quantitative schedule risk analysis as context, labour productivity norms (briefly, because they are out of scope), and decision-support tools. It ends with a table of the gap that the project addresses.

The review rests on the evidence corpus held in the project repository: the 11 RII surveys transcribed into `data/literature_seed.json`, the research notes in `Research_Notes/` (topic groups A to F and the P1 to P3 notes), a screened metadata database of open bibliographic records, and a set of references on risk matrices and statistics named in the project's methods review. Each paper cited below was read at one of three depths, and the depth limits what the review says about it: full text; abstract only; or, for a few items, title and metadata only. Where only an abstract was read, the text says so. Papers that the research notes list as unread leads are not cited for their findings.

The screened corpus

The project built a database of open bibliographic records (OpenAlex metadata) by keyword search and screening on titles and abstracts. The database report records 17,788 harvested records and 3,975 retained after screening (rules version 2026-10-07.1). Two other documents in the repository give different sizes (11,599 input and 3,550 included in the screening report; 21,902 harvested and 5,214 kept in the handoff summary), and

the three have not been reconciled. Chapter 3 reports one reconciled count and explains the difference. The figures used here are those of the database report, and they describe the database only. The keyword rules are the authors' own, the screening used titles and abstracts, and the relevance judgements on 4,981 records were made by model reviewers and not checked by a human, so the corpus is a map of what the search found and not a census of the literature.

Table 2.1. Composition of the screened corpus (N = 3,975 works).

Theme (a work may have several)	Works	Share of 3,975
Delay causes and factors	2,035	51.2%
Risk identification and assessment (RII, probability-impact, matrix)	1,557	39.2%
Labour productivity and masonry work	1,121	28.2%
Rule-based, expert-system and AI risk tools	512	12.9%

Source: counts from `docs/Corpus_Statistical_Analysis.md` and `docs/Literature_Database_Report.md` (Derived Calculation by regular-expression matching of title and abstract; shares are the counts divided by 3,975 and are not additive). Regular-expression matching measures how many abstracts mention a theme term, not how many studies used a method.

The corpus shows where the literature sits. About half of the works address delay causes, and fewer than 30% mention labour productivity or masonry. Themes overlap little: only 150 works carry both a delay-cause and a productivity-or-masonry tag, and the pairing of delay causes with rule-based or AI tools is also rare (204 works), partly because a title tags one theme (Source: `docs/Corpus_Statistical_Analysis.md`, section 3, with the caveat given there that the negative association is partly built into the tagging). Figure 2.1 shows the share of abstracts that mention each of several methods. Questionnaire surveys appear in 46.9% of abstracts and RII in 10.6% (423 works), whereas large language models appear in 0.4% (14 works) and machine learning in 5.7%; the machine-learning share rises from about 2% of abstracts in 1995 to 2009 to 10.4% in 2023 to 2026. India appears in 6.7% of abstracts (268 works), Sri Lanka in 1.2% and the Middle East in 6.8%.

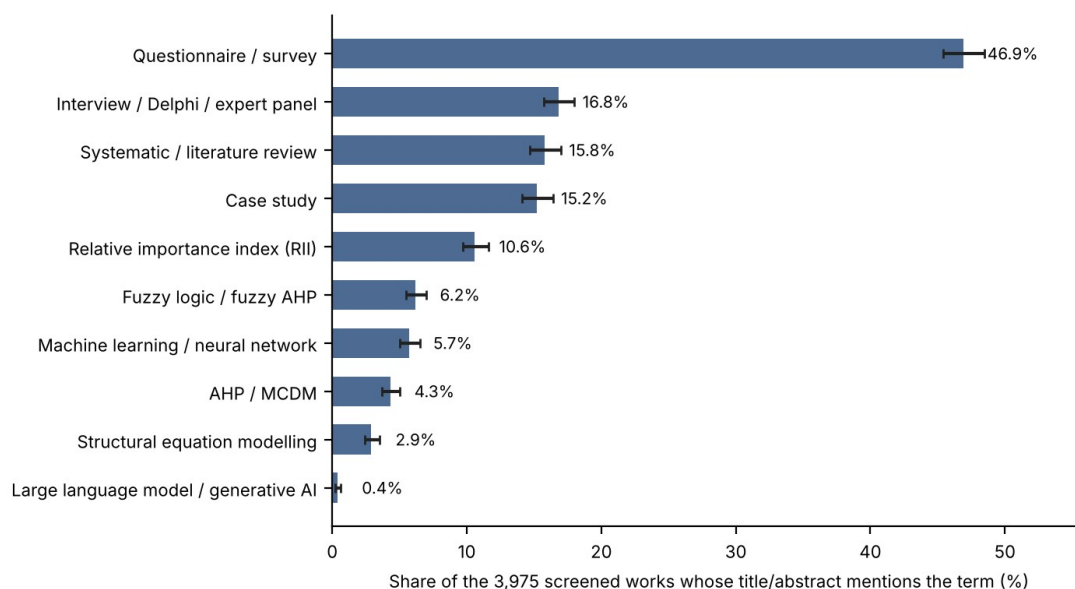


Figure 2.1. Share of screened works whose title or abstract mentions each method, with 95% intervals. Source: `docs/Corpus_Statistical_Analysis.md` section 6a (Derived Calculation, keyword matching; mentions are not evidence that the method was used).

Delay causes in building construction

What a delay-cause survey does

A typical delay-cause study compiles a list of candidate causes from earlier papers and interviews, asks a sample of contractors, consultants, clients or workers to rate each cause on a five-point or four-point scale, and ranks the causes by an index computed from the ratings. The RII is the most common index in the corpus and in the studies read here. The ranking identifies which causes respondents consider most important in general. It does not identify which cause occurred on a given project, how often, or how many days it cost.

Studies in India

Studies carried out in India and read for this project are of three kinds: RII surveys that enter the seed (Karthik & Rao, 2019a; Ponmalar et al., 2018; Dixit et al., 2019), RII surveys held out of the seed (Pinky Devi & Sindhu, 2025), and a larger group read at lower depth, collected in Table 2.2.

Karthik and Rao (2019a) surveyed 44 respondents from building firms in Telangana on factors affecting labour productivity. The ranked factors with RII of 0.700 or more include unsafe working conditions (0.809), poor planning and unrealistic scheduling (0.791 each), material shortage (0.777), payment delay (0.773) and work at height (0.764) (Source: Literature; `data/literature_seed.json`, M01). Only the 28 of 38 factors at or above 0.700 are listed in the paper's table, so lower-rated factors are absent, and the field case study in the paper concerns walls built from autoclaved aerated concrete (AAC) blocks, not bricks. Ponmalar et al. (2018) surveyed supervisors, project engineers and labourers on two residential buildings in Chennai; respondent numbers are not reported. Lack of experience (0.787), lack of construction material (0.780), lack of tools (0.760), absenteeism (0.751) and payment delays (0.7758) all rate highly, and weather is 0.700 (Source: Literature; M01 and A02 rows in the seed file; the paper's index uses weights of 1 to 4). Dixit et al. (2019) used 201 respondents across consultants, contractors, developers and academia in the Indian construction industry, with RII computed as total score divided by 5 x 201; the paper prints one factor twice with different values (Source: Literature, M06 note).

Pinky Devi and Sindhu (2025) surveyed 72 construction management professionals about delays in infrastructure (roads and bridges), not buildings, and ranked material-related issues first as a group (group mean 0.562), ahead of construction-site factors (0.555) and contractor-related factors (0.506) (Source: Literature; seed-file note for A10). This study is held out of the seed because it is not about buildings, and a repository audit recorded that about ten library risks overlap with its table, although those rows were not entered.

Table 2.2. Other Indian studies read for this project and not entered in the RII seed.

Study	Design and sample	Finding relevant to masonry	Depth read
Reddy & Chambrelin (2021)	Video time-and-motion study, seven residential sites in Andhra Pradesh; one observation of about 1 to 2 hours per site; crews of 2 to 4	Work efficiency 62.57% to 90.62%; causes named: lack of targets, poor planning and monitoring, material shortage, equipment breakdown	Full text
Loganathan & Kalidindi (2015)	Pilot on one project in Tamil Nadu	20% to 40% production variation between masonry crews (measure not defined in the abstract)	Abstract only
Loganathan & Kalidindi (2016)	Interviews on 15 projects in six states	Illness, injury, lack of facilities and payment delays are significant contributors to absenteeism and turnover of migrant workers	Abstract only
Guha & Biswas (2008)	Monte Carlo study of a housing project in Kolkata, contractor time and cost data	Monsoon increased project duration by about 5% and cost by about 12%	Abstract only
Soundarya et al. (2025)	Six residential sites in Chennai; frequency, severity and importance indices	Overall lag in schedule has frequency index 0.861 and severity index 0.819; rework (importance index 0.55) and inadequate skilled labour (0.53) rank next	Full text (summarised by fetch tool)
Mehta & Gaikwad (2017)	Questionnaire to 100 stakeholders; importance index, principal component analysis	Finance-related and labour-related problems dominate (values not in the abstract)	Abstract only

Study	Design and sample	Finding relevant to masonry	Depth read
Tankkar & Wanjari (2015)	85 respondents (33 clients, 10 consultants, 42 contractors)	Lack of human resources, poor site management and insufficient coordination are the top three	Abstract only
Noushad et al. (2023)	Delphi and RII, high-rise buildings in Kerala, 76 delay causes	No numbers in the preview read	Abstract only

Source: `Research_Notes/A_masonry_india.md`, cards A01, A03, A08, A09, A11, A12, A13 and A19 (Literature). The number of respondents in Soundarya et al. (2025) was not recorded in the notes.

These studies point the same way as the surveys: labour availability and skill, material supply, planning and management, payment and site conditions recur, and weather is cited but not always among the top factors. The masonry records are small. Reddy and Chambrelin's efficiencies come from seven short observations, Ponmalar et al.'s overrun figures from two projects, and Loganathan and Kalidindi's crew variation from one project, so none supports a probability or a delay distribution for a new site.

Studies outside India

Six further RII studies in the corpus are from outside India, and two of the five seed studies are from Sri Lanka. Abeysinghe and Jayathilaka (2022) received 163 responses to a questionnaire of 39 delay factors in Sri Lankan construction; shortage of labourers (0.8245), delay in delivering materials (0.8098) and fluctuation of material prices (0.7890) are among the top factors and bad weather conditions is 0.6528 (rank 30 of 39). About 26% of respondents worked on road projects, so the sample is not only buildings (Source: Literature; seed-file note for A14). Manoharan et al. (2022) surveyed 90 upper-grade contractor firms on labour-related factors in Sri Lankan building projects; of 32 factors, 21 exceed an RII of 0.7, with skills shortage first (0.82).

Adebowale and Agumba (2023) pooled RII values of 35 productivity factors from ten surveys in a meta-analysis labelled by the authors as Middle East, although the paper itself includes Malaysia and Vietnam among them. Almeida et al. (2021) surveyed 47 directors, managers and engineers of private building companies in Brasilia on delay causes. Loekito et al. (2026) surveyed 83 construction workers in East Java, Indonesia, on productivity factors. Ouansrimeang and Wisaeang (2024) analysed 30 delayed public projects in Thailand with RII from 380 respondents and ranked contractor financial issues (0.777) and labour issues (0.733 in the results; the abstract prints 0.773) near the top. Hassoon et al. (2025) is the only study in the corpus whose subject is brickwork specifically: its abstract, the only part read, reports RII of 0.874 for planning and 0.856 for team size for brickwork and housing skeleton tasks in Iraq (Source: Literature;

`Research_Notes/D_mitigation.md`, abstract only).

The 11 studies in the seed file

Table 2.3 lists the 11 studies from which the project transcribed RII values. Five serve as seed studies, and six are held out and used only to compare the ranking they imply with the seed order. Figure 2.2 shows the number of rows entered per study.

Table 2.3. The 11 RII studies in the seed file: location, sample, scale, method and role.

ID	Study	Location	Sample	Scale	Index / method	Masonry-specific	Role	Rows entered
M01	Karthik & Rao (2019a)	India (Telangana)	44 respondents	1-5	RII; field case on AAC walls	No	Seed	28
A02	Ponmalar et al. (2018)	India (Chennai)	Not reported	1-4	RII; separate masonry site record	No	Seed	14
M06	Dixit et al. (2019)	India	201 respondents	1-5 (inferred)	RII	No	Seed	40
A14	Abeysinghe & Jayathilaka (2022)	Sri Lanka	163 respondents	1-5	RII	No	Seed	5
A15	Manoharan et al. (2022)	Sri Lanka	90 firms	Not recorded	RII	No	Seed	5
M10	Adebowale & Agumba (2023)	Middle East (10 studies)	10 studies	Pooled RII	Meta-analysis, random	No	Held out	33

ID	Study	Location	Sample	Scale	Index / method	Masonry-specific	Role	Rows entered
					model			
M15	Almeida et al. (2021)	Brazil (Brasilia)	47 respondents	1-4	RII	No	Held out	24
M14	Loekito et al. (2026)	Indonesia (East Java)	83 workers	1-4 stated	RII printed as $W/(5N)$	No	Held out	23
A10	Pinky Devi & Sindhu (2025)	India (infrastructure)	72 respondents	1-5	RII	No	Held out	6
A21	Ouansrimeang & Wisaeang (2024)	Thailand (public projects)	30 projects; 380 respondents	1-5	RII with machine learning	No	Held out	2
HAS25	Hassoon et al. (2025)	Iraq	Not recorded	Not recorded	RII (abstract only)	Yes (brickwork)	Held out	2

Source: `data/literature_seed.json`, fields `studies` and `rows` (Literature for the study attributes; row counts are a Derived Calculation by counting rows; total 182, of which 92 are in seed studies and 90 in held-out studies). Dixit et al. (2019) state no scale in the text; 1-5 is inferred from the RII denominator. For Loekito et al. (2026) the printed index equals total score divided by 5N although the text gives $A = 4$.

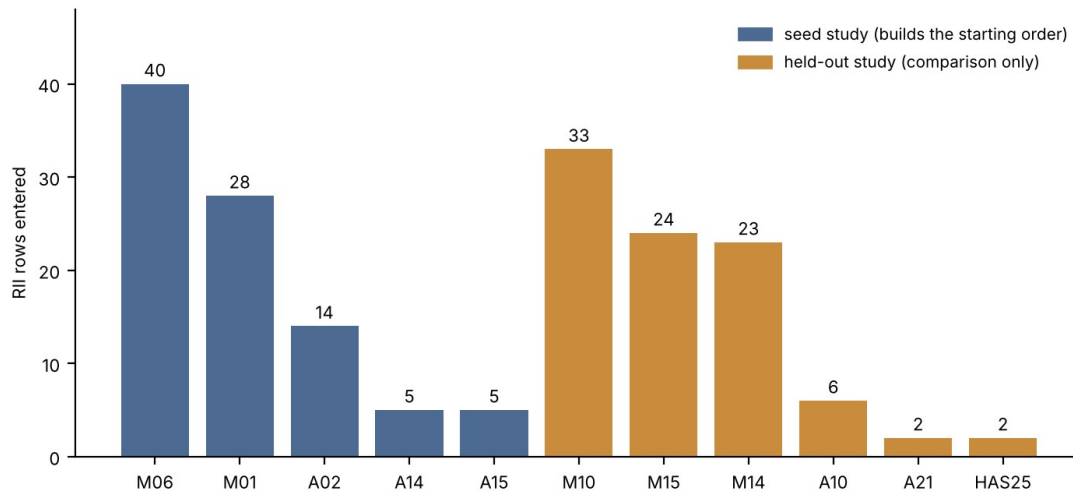


Figure 2.2. RII rows entered per study (182 rows from 11 studies). Source: `data/literature_seed.json` (Derived Calculation).

Three features of Table 2.3 limit how the studies can be used. First, none of the five seed studies is a masonry survey: M01 and A02 are general building labour-productivity surveys whose masonry content is a separate field record, and the two Sri Lankan studies include non-building respondents or only the top-ranked factors. Second, the transcription is a subset. A14 contributes 5 of its 39 factors, A15 5 of 32, A02 14 of 35, and M01 lists only the 28 factors at or above 0.700. A repository audit recorded that the omitted factors include some that rank higher than those entered (for example, financial difficulties of contractors, 0.8233, rank 3 in A14). Third, the studies use different scales and sample types, so absolute RII values are not comparable. M14, for example, prints an index that equals $W/(5N)$ on a scale the text describes as 1 to 4.

Masonry-specific productivity and delay literature

Direct evidence on masonry is thinner than on delay in general. The research notes record the following, mostly at abstract depth.

- Nepal: Lawaju et al. (2021) observed 35 sets of brick masonry output at seven sites in the Kathmandu Valley, chose 13 inputs from an RII survey of 44 factors and fitted an artificial neural network (a test set of five cases); Rana et al. (2023) used a Likert questionnaire with RII and neural-network sensitivity for bricklaying in Surkhhet and report that material-related factors form the highest-ranked group (Cronbach's alpha 0.976). Numbers other than these were not seen (abstract only).

- Turkey: Gerek et al. (2015) modelled the productivity of 147 masonry crews with two neural-network techniques using crew size, experience, hours, wall type and mortar type as inputs, with test mean absolute percentage error of about 14.3% to 15.0% (full text). The paper does not report descriptive statistics of crew productivity, so it cannot supply a distribution for an Indian crew.
- Indonesia: Djohim et al. (2024) simulated the schedule of a modest house type and report that wall work was the most influential activity for delay (abstract only; Source: Literature).
- Iraq: Hassoon et al. (2025), as above.

These studies identify which inputs and factors matter for masonry output. They do not provide India-specific, measured daily outputs or delay frequencies that a risk register could use as probabilities. One research note records that the area-based productivity column of Reddy and Chambrelin (2021) is physically implausible and must not be used; only its efficiency percentages are used here, as context.

Consolidating the factors

Across the seed studies, the same families of factors recur: labour availability, absenteeism and skill; material shortage and delivery; planning and scheduling; payment; site and equipment conditions; weather. The values differ by study. Weather, for instance, takes the values 0.700 (A02), 0.6528 (A14, rank 30 of 39), 0.764 (M01) and 0.6567 (M06) among the four seed studies that rate it, and labour shortage takes 0.751 (A02 absenteeism), 0.8245 (A14) and 0.723 (M01) (Source: Literature; docs/Project_Handoff_Summary.md section 8.2). An earlier statement of the project that weather is rated lower than other factors was a post-hoc finding on a ten-risk set, and the audit withdrew it (docs/Project_Handoff_Summary.md, section 8.6); it is not made in this report. The review draws only the weaker conclusion that the same families of factor appear in studies from different places, without agreement on their order.

From factor families to the risk library

The project's risk library groups the factor families into 11 categories. Table 2.4 shows how many of the 46 library risks fall in each category, how many have a survey value in the seed studies, and how many are tagged as project-level and not activity-level. A risk with no seed value stays off the matrix until a user enters numbers.

Table 2.4. The 46 library risks by category: total, with a survey (seed) value, and tagged project-level.

Category	Risks	With seed value	Without seed value	Tagged project-level
Labour	13	9	4	0
Material	6	4	2	0
Management	6	4	2	2
Design	4	3	1	1
Equipment	3	1	2	0
Financial	3	2	1	2
External	3	2	1	3
Safety	2	2	0	0
Weather	2	1	1	0
Quality	2	1	1	0
Site	2	1	1	0
Total	46	30	16	8

Source: data/risk_library_masonry.json (Derived Calculation by counting records by category, seed_status and scope). The 38 original risks were built from the factors transcribed from the 11 studies; the 8 added on 2026-10-08 cite evidence records and carry a masonry-specific form labelled Expert Judgment, to be confirmed by an expert survey.

The table shows the effect of the literature on the library: the seeded risks are the ones that surveys happen to ask about, so labour, material and management risks are well covered by survey values, while masonry-specific conditions such as vertical transport, scaffolding, mortar supply and work-front availability have no survey value and depend entirely on what the user enters.

Reviews of delay causes

Review papers in the corpus summarise this literature. Romzi and Doh (2022) review causes of project delay and list slow decision making, poor site management and supervision, labour shortage, scope change and late design approval, with number of studies and method not reported in the abstract. Abu Hasan et al. (2025) review 90 publications from 2007 to 2024, identify 93 common conflict factors, add 18 interviews and propose a conflict-delay model. Abd Aziz et al. (2022) review delay mitigation strategies in a systematic review. Ghaeb and Mahjoob (2023), reviewing 38 studies published from 1997 to 2020, observe that most work addresses risk identification and assessment and call for more work on risk response (the journal is modest and the review's method is thin). The reviews support a general point: identification and ranking of delay causes is well covered, and what is done with the ranking at the level of a single activity on a site is less so.

The relative importance index and its limits

Definition and a worked example

For a factor rated by N respondents on a scale from 1 to A, the relative importance index is

$$RII = (\text{sum of the ratings given by all respondents}) / (A \times N),$$

so an RII of 1 would mean every respondent gave the top rating. The studies in the seed state this formula in various ways; Dixit et al. (2019), for example, define RII as total score divided by 5×201 .

Worked example. In Karthik and Rao (2019a) the factor “bad leadership skill” received 1, 5, 5, 27 and 6 respondents at ratings 1 to 5 ($N = 44$). The total score is $1 \times 1 + 2 \times 5 + 3 \times 5 + 4 \times 27 + 5 \times 6 = 164$, so $RII = 164 / (5 \times 44) = 0.7455$ (Derived Calculation from the counts in `data/literature_seed.json`, note on M01). The paper prints 165 and 0.750 for this row; the repository keeps the printed value and records the inconsistency. The sample standard deviation of the 44 ratings is 0.92, so the standard error of the mean rating is $0.92 / \sqrt{44} = 0.139$ and the standard error of the RII is $0.139 / 5 = 0.028$ (Derived Calculation, treating the ratings as independent). An approximate 95% interval is therefore about 0.745 ± 0.055 . A single respondent changing one rating by one point moves the index by $1 / (5 \times 44) = 0.0045$.

This example shows how much of the RII scale the sampling error of one survey item occupies. In the seed table of the project, the first six risks lie within 0.019 of each other and 24 of 30 risks lie within 0.096, while the standard error of one M01 item is about 0.03 (Source: `docs/Project_Handoff_Summary.md` section 8.2, audit estimate). Differences of less than about 0.05 between two RII values from one survey of this size should not be read as differences in importance.

Importance is not probability or days

An RII answers the question “how important do respondents think this factor is?” It was designed to rank factors within a study. Three features separate it from a probability or a duration.

1. **It is a rating, not a count.** A factor can be rated important because its effect would be severe, because it occurs often, or because respondents have recently experienced it. The index does not separate frequency from severity. Soundarya et al. (2025) are unusual in reporting frequency and severity indices separately and combining them into an importance index: for the overall lag in schedule their frequency index is 0.861 and severity index 0.819, and the product of these two, 0.705, matches the reported importance index of 0.71 (Derived Calculation from values in `Research_Notes/A_masonry_india.md`). An RII on a single question cannot be decomposed this way.
2. **It is ordinal in use.** Moving from RII to a probability needs a mapping, and no validated mapping from importance to likelihood exists in the sources read. Research on elicitation records precedents for mapping qualitative levels to probability bands in project risk (Curto et al., 2022; Acebes et al., 2024) but not a validation of any mapping from survey importance.
3. **It depends on the respondent group and the scale.** Loekito et al. (2026) surveyed workers, whereas most of the other studies surveyed managers, engineers or consultants. The agreement check in Chapter 4 found a weak negative correlation between the worker survey and the seed order ($\rho = -0.18$, 13 risks, $p = 0.55$, not significant); this is consistent with workers and managers rating factors

differently but cannot be distinguished from noise with these data. Scales of 1 to 4 and 1 to 5 give different absolute RII values for the same opinions.

Other limits recorded in the project's audit

Beyond the points above, the project's audit recorded the following limits of the secondary data. Truncation: M01 reports only factors with RII of at least 0.700 (28 of 38). Selection: the seed transcribes only a subset of factors from several papers. Single-study support: 14 of the 30 seeded risks rest on one study. Instability: when the ranking built from the other seed studies is compared with a left-out seed study's own order, the Spearman correlations are 0.09 (M01, $n = 13$), 0.37 (A02, $n = 10$), 0.64 (M06, $n = 9$) and 0.60 (A14, $n = 4$), none significant, and A15 cannot be tested with $n = 2$ (Source: docs/Project_Handoff_Summary.md section 8.4; Derived Calculation). Agreement with held-out studies is weak and not significant after correction for multiple testing (section 8.5 of the same document). Chapter 4 gives the method and numbers; they are summarised here because they determine how the RII may be used in a tool. It may order risks coarsely when nothing else is known; it cannot stand in for site data.

Risk management standards

ISO 31000

ISO 31000 is the international standard that gives principles and guidelines for risk management and describes a process in which risk is identified, analysed, evaluated and treated, with communication, monitoring and review throughout (ISO, n.d.). The standard is used here as the organising frame of the tool and not as a source of numbers. The wording of the process stages follows the project's own methodology description (docs/Project_Handoff_Summary.md, section 2) and should be checked against the current edition of the standard before submission (the edition year is not recorded in the repository).

IEC 31010

IEC 31010 is the companion standard on techniques for risk assessment (International Electrotechnical Commission [IEC], n.d.). It is cited because the techniques of this project, a consequence-and-likelihood matrix, a register, and (optionally) Monte Carlo simulation, are among the kinds of technique that such standards catalogue. The content of the standard has not been reviewed in detail for this report, and the claim that it lists these techniques should be checked against the standard text.

Mapping the tool onto the process

Table 2.5 shows which part of the tool serves each stage of the ISO 31000 process, and what the tool does not do.

Table 2.5. Stages of the risk-management process and the corresponding parts of the tool.

Process stage	Part of the tool	What the tool does not do
Establish context	Case record: project, activity, planned duration, site facts	Does not set the organisation's risk appetite
Identify	Library of 46 risks; 22 site-fact rules; evidence-cited AI suggestions of names and mechanisms	Does not discover risks outside the library unless the user adds them
Analyse	User-entered probability and delay range with Source label; ordinal literature tier for 30 risks without entered numbers	Does not supply probabilities or delays
Evaluate	5x5 probability-impact matrix with stated edges and thresholds (Assumptions); prioritisation by score	Does not decide which level is acceptable
Treat	Register fields for owner, trigger, response type, action, review date; optional comparison of responses using entered costs	Does not recommend a response or supply effect sizes
Monitor and review	Review date field; exports for the review meeting	Does not collect actual outcomes

Source: README.md, docs/Project_Handoff_Summary.md sections 2, 8.14 and 9 (descriptive).

Risk matrices and their critique

Use of the matrix

A probability-impact matrix assigns each risk to a probability class and an impact class and reads a level from the cell. In the tool, the classes run from 1 to 5, the score is the product of the two class numbers, and the level is Low for a score up to 5, Moderate for 6 to 10, High for 11 to 15 and Extreme for 16 or more. These thresholds are Assumptions of the tool, not values from a source. Table 2.6 shows the resulting scores and levels.

Table 2.6. Score (probability class x impact class) and level in the tool's 5x5 matrix.

Probability class Impact class	1	2	3	4	5
5	5 Low	10 Moderate	15 High	20 Extreme	25 Extreme
4	4 Low	8 Moderate	12 High	16 Extreme	20 Extreme
3	3 Low	6 Moderate	9 Moderate	12 High	15 High
2	2 Low	4 Low	6 Moderate	8 Moderate	10 Moderate
1	1 Low	2 Low	3 Low	4 Low	5 Low

Source: Derived Calculation from the thresholds in `docs/Project_Handoff_Summary.md` section 4 (Assumption). Counting cells gives 10 Low, 7 Moderate, 4 High and 4 Extreme of 25.

Cox (2008) and Duijm (2015)

Cox (2008) examined risk matrices and concluded that they have four problems: poor resolution, errors (a matrix can rate a quantitatively smaller risk higher than a larger one), suboptimal resource allocation and ambiguous inputs and outputs, and that they should be used with caution and with careful explanation of the embedded judgements (this review read the abstract only). Duijm (2015) gives recommendations on the use and design of risk matrices (this review read the reference details only; the paper's detailed recommendations should be reviewed before the chapter is finalised).

Table 2.6 illustrates the range-compression point that follows from Cox's critique. Because the level is a function of the product, the whole first row and first column are Low whatever the other class is: a risk in probability class 1 and impact class 5 scores 5 and is Low, and so is a risk in probability class 5 and impact class 1. Only 8 of 25 cells are High or Extreme. The thresholds are a design choice: a different cut at 4 or 6 would move cells between levels. Chapter 6 reports a sensitivity analysis on how many risks of the example case change level when the class edges and thresholds change. Two consequences are drawn for the design.

1. The matrix is presented as an aid to ordering, and the thresholds are printed on every export as Assumptions.
2. Risks with impact class 5 and an entered probability are also listed separately, as a filter and not as a score, so that a rare, severe risk is not hidden in the Low band.

Beyond the matrix

Several papers in the corpus compare or combine matrices with simulation. Acebes et al. (2024) propose a quantitative methodology for risk prioritisation as an alternative to probability-impact matrices; the research notes record that rankings from a matrix and from Monte Carlo analysis differ in their example. Koulinas et al. (2021) use a risk matrix to set probability and impact for each activity and then run Monte Carlo simulation for an artificial-lake project in Greece, with a qualitative mitigation loop. Starczyk-Kolbyk and Jedras (2025) compare a matrix and Monte Carlo on cost for twelve risks of a high-rise office and report a high level of consistency between the methods (abstract only). Canesi et al. (2025) integrate a probabilistic risk matrix with Bayesian Monte Carlo simulation for cost overruns in infrastructure (abstract only). The papers show that matrix and simulation are used together, and that the relation between their rankings depends on the case.

Quantitative schedule risk analysis: context

Monte Carlo and three-point estimates

Monte Carlo schedule risk analysis samples activity durations and risk events many times and reports the distribution of the project finish. Activity durations are often given as three-point (optimistic, most likely,

pessimistic) estimates under a beta-PERT assumption. The literature read for this project is critical of the assumptions behind that choice: the PERT variance is fixed by the range alone (Hahn, 2008), the mean and variance formulas do not correspond to a beta whose mode is the elicited value (Herrerias-Velasco et al., 2011), and Trietsch et al. (2012) argue for a lognormal core and report a Parkinson effect in recorded activity times. Flyvbjerg (2006) documents optimism bias in forecasts for large infrastructure projects, which implies that three-point estimates elicited from project teams may be too narrow. Song and Vanhoucke (2025) treat the interaction between risks in schedule risk analysis.

Risk-event simulation in construction schedules

Monte Carlo simulation with risk events and mitigation has been applied to construction projects, including in India. Gheewala et al. (2025) simulated an elevated metro rail project with a baseline schedule of 711 days, 17 risks prioritised by a probability-impact approach and 1,000 simulations; all risks together gave a mean duration of 842 days, and a mitigation plan costing Rs 3.32 million reduced the mean and maximum durations by 26 and 76 days (abstract only). Ichsan et al. (2025) simulated the schedule of the Jakarta Central Railway Station project at the level of nine critical activities and compared pre- and post-mitigation (residual) risk, with response effects from a focus group of five panel members. Guha and Biswas (2008) applied Monte Carlo to monsoon effects on a Kolkata housing project. Koulinas et al. (2020) built a simulation-based expert system for delay risk in a hotel renovation. Hadad and Keren (2025) combine Monte Carlo simulation with budget-constrained selection of risk responses on an office building. Prins et al. (2022) model risk occurrence as a binary variable with a probability and apply Beta-PERT to durations in a mitigation-planning method for a tunnel project; this is a preprint. Table 2.7 sets these studies beside the present project.

Table 2.7. Closest published work combining risk events, simulation and mitigation, and how this project differs.

Study	Level	What it does	Difference from this project
Prins et al. (2022)	Project (tunnel)	Binary risk occurrence, Beta-PERT, mitigation optimisation	No single-activity masonry focus; inputs from a project risk register
Ichsan et al. (2025)	Nine critical activities (railway station)	Pre- and post-mitigation schedule simulation	Response effects from a five-person focus group; not masonry
Gheewala et al. (2025)	Project (metro, India)	Matrix-prioritised risks, simulation, mitigation plan cost versus days saved	Project level; mitigation modelled as one plan, not as alternatives
Hadad & Keren (2025)	Project (office building)	Simulation with response budgeting	Dollar-based, project level, no schedule network
Koulinas et al. (2021)	Activity (lake project)	Risk matrix, simulation, qualitative mitigation loop	Not masonry; mitigation is a qualitative loop without residual quantification
Djohim et al. (2024)	Activity-level schedule of a house	Simulation identifies wall work as most influential	No mitigation

Source: `Research_Notes/F_novelty_validation.md`, items F01, F03, F04, F05, F11 and F12 (Literature; abstracts or full text as marked in the notes).

The same research notes conclude that the individual mechanics of risk-event simulation with mitigation are published and not new, and the project therefore makes no claim to novelty in them. Two conditions apply to the simulation studies above. Their inputs (probabilities, ranges, response effects) come from project registers, contractor records, expert panels or focus groups, and the research notes found no construction paper that reports a calibration or back-test of a simulated range against actual outcomes (the closest, Ichsan et al., 2025, claims an accuracy figure in its abstract without a back-test in the text read). Khodabakhshian et al. (2023), in a systematic review of 48 papers on deterministic and probabilistic risk management approaches, note that experience, prior knowledge and historical data are required and often not documented or easily accessible. This is the data-scarcity problem that confronts a user of the optional layer in this project: the simulation can run, but the numbers it needs must be entered by the user and carry their own Source labels.

Labour productivity norms (context only; out of scope)

The registered title mentions labour productivity. This section records what the research notes found, briefly, to show why productivity was not modelled.

India's planning norms for masonry labour are published as labour output constants. IS 7272 (Part 1), adopted in 1974 and reaffirmed in 2010, gives labour-days per unit of work for building works in north India, derived by the Central Building Research Institute from work-measurement observations on sites in Delhi and Roorkee (Bureau of Indian Standards, 1974). For brickwork in walls thicker than one brick the constant is 0.94 mason-days per cubic metre (with 1.80 labourer-days and 0.20 water-carrier-days). The CPWD Analysis of Rates for Delhi 2007 gives the same mason labour quantity for brickwork in superstructure, 0.47 plus 0.47 mason-days per cubic metre (Central Public Works Department, 2008). The research notes record that the CPWD brick masonry labour constants are identical across the 2007, 2012 and 2014 editions read and equal the 1974 constants, so they are long-standing planning norms and not fresh measurements. A state schedule read (Works Department, Government of Odisha, 2006) gives a mason total about 1.5 to 1.9 times the CPWD figure, with larger bricks, so published norms differ between sources by a large factor. The reading of the unit of the one-brick-wall row of IS 7272 is an inference from a low-resolution scan (Source: Literature; `Research_Notes/P3_india_norms.md`, records P3-01 to P3-04).

A norm is a planning constant with allowances built in; it is not a measurement of what a crew achieves on a given day, and it gives no distribution. Measured productivity studies read for this project are small (Reddy & Chambrelin, 2021; Loganathan & Kalidindi, 2015) or come from other countries (Gerek et al., 2015). A productivity model would need measured daily output by crew on Indian sites, which the project does not have. A defect in the legacy norms file in the repository was also recorded in the audit. Productivity is therefore listed in Chapter 8 as future work with its data prerequisites, and no CPWD number is used in any result in this report.

Elicitation, verification and validation

The tool leaves probabilities and delays to the user. Where they come from experts, the elicitation literature applies, and where the tool itself is to be judged, so does the literature on verification and validation. This section records the sources the project draws on for its future-work plan (Chapter 8); none of them has been applied in the work reported here.

Elicitation. Bedford et al. (2006) review expert elicitation for reliability design and state that accurate subjective probabilities are not obtained by simply asking for a number, naming four standard biases (motivational, cognitive, anchoring and availability) and recommending structured elicitation with feedback. Williams et al. (2021) compare aggregation methods on a clinical example with three experts and ten seed questions and find that all three aggregation methods outperformed the individual experts, with equal weights a defensible default when experts and calibration questions are few. Dharmarathne et al. (2022) point out that tests of expert calibration on a modest number of questions have poor power. Ameyaw et al. (2016) reviewed 88 Delphi papers in construction engineering and management: panel sizes ranged from 3 to 93 with 8 to 20 the most common, rounds from 2 to 6, and a Likert scale was used in 41 papers; Delphi output is ordinal, and the review says nothing about converting Likert scores to probabilities. Curto et al. (2022) and Acebes et al. (2024) attach a probability range to each qualitative level (for example, "High" as a uniform range of 20% to 35% in one project) and simulate within the range; the ranges are set by each project's expert committee and are not transferable. These sources imply that, in this project, any mapping from a rating to a probability would be an explicit Assumption with a sensitivity check, and that an expert survey should report its sample size and agreement, not only a mean.

Verification and validation. Sargent (2013) describes approaches to deciding model validity, conceptual model validity, model verification, operational validity and data validity, and a procedure for documenting validation (the research notes record a discrepancy in the venue metadata of this item, which should be checked before submission). Verification, meaning that the software computes what its specification says, is possible without site data and is what Chapter 6 reports. Operational validity, meaning that outputs are accurate enough for their purpose on real cases, needs site outcomes, and is not claimed. Fowler et al. (2024) offer a template for face validity of a decision-support tool by surveying subject-matter experts (in a cybersecurity setting, not construction); a similar trial with site engineers is the nearest evaluation step available to this project and is proposed in Chapter 8.

Decision-support tools for delay risk

Tools in the literature

The tools reported in the corpus fall into three groups. The first is schedule risk software used in case studies, such as Primavera P6 with Risk Analyzer in Gheewala et al. (2025) and @RISK in Ichsan et al. (2025), which run simulation on a schedule and rely on the user to supply the distributions and probabilities. The second is expert systems and data-driven models: Koulinas et al. (2020) describe a simulation-based expert system for delay risk, and Zorrilla et al. (2026) report a decision-support system that applies clustering, anomaly detection and explainable machine learning to a dataset of construction documents, with an F1 score of 0.73 in bottleneck detection (abstract; no simulation of mitigation). The third is the use of large language models for risk identification and classification.

Large language models

Vilibic et al. (2026) applied several large language models to risk identification in one project and conclude that human expertise remains crucial for prioritisation, contextual interpretation and mitigation. Erfani and Khanjar (2025) compare configurations for risk classification and report that few-shot GPT-4 reaches an F1 of 0.81 against 0.86 for the best classical model (BERT with a support vector machine). Martin et al. (2025) examine ChatGPT limitations in risk identification, mitigation strategies and user experience (the notes record the title only). A systematic review of large language models in smart construction (Gao et al., 2026) lists hallucination control and domain adaptation among persisting challenges. In the corpus, large language models appear in 14 abstracts, 0.4% of 3,975 works (Figure 2.1). The use in this project is limited to proposing risk names, mechanisms and evidence identifiers, and the design rejects numeric output (Chapter 5); the literature supports caution about letting such models supply numbers.

What the tools do not provide

Within the corpus read, the tools take probabilities and durations as user input and do not record, for each number, where it came from. They are project-level more often than activity-level. The papers above also say little about how the tool user, a site engineer, would see which risk conditions are present on a particular day. A register kept as a plain spreadsheet has only the columns its author gives it; whether it carries a source for each number depends on the author. These observations are about the studies read and are not a survey of commercial software, which this project did not carry out. Chapter 7 compares features of the tool with the tools that could be checked.

Summary and the research gap

Table 2.8. Summary of what the literature provides, what it lacks, and how this project responds.

Need of a site engineer	What the literature provides	What it lacks	How this project responds
Know which delay risks to check for masonry	Many delay-cause surveys; ranked lists; reviews	Few masonry-specific surveys; most are general building or infrastructure; mixed project-level and activity-level items	A library of 46 risks, each with supporting survey factors, tagged by scope; 22 site-fact rules
Tell how important a risk is when nothing is entered	RII values from India and Sri Lanka	RII is importance, not probability or days; studies disagree weakly; one study per risk for 14 of 30	An ordinal starting position labelled as an Assumption and replaced by entered values; the weak agreement reported
Record a probability and impact with their source	Matrix and register practice; Cox (2008) on matrix weaknesses	No recorded source for each number in the tools reviewed	Six Source labels on every number; assumptions printed on every export
Quantify schedule effect	Published simulation methods and Indian and international case studies	Inputs from teams or panels; no back-test on masonry activities	Optional layer using only entered numbers; no defaults
Use AI assistance safely	Studies of language models for identification and classification	Risk of hallucination; numbers from models unverified	AI may propose names and evidence identifiers only; numeric output rejected
Judge whether the tool works	Verification and validation guidance	No site data and no practitioner evaluation in this project	Software verification now; expert rating, usability and site validation defined as future work

Source: synthesis of sections 2.2 to 2.9 of this chapter and the cited sources (descriptive).

The gap, in the form given in Chapter 1, is practical: published rankings do not carry into site risk registers, general tools ask for probabilities and impacts without tying them to evidence, and within this search we did not find an activity-level masonry tool in which each value carries its source. The search was by keyword on titles and abstracts, and the finding is not a statement that no such tool exists. The review also shows what the project cannot rely on: the survey rankings are weak and unstable as a base for probabilities, masonry-specific measured data are scarce, and simulation and matrix methods are established and not new. The project therefore limits itself to identification, a source-labelled register and prioritisation, with simulation as an optional layer that depends entirely on what the user enters.

Chapter 3. Methodology

This chapter states how the evidence was gathered and checked, how the literature seed and the risk matrix are defined, which statistics are used and why, how the software was verified, and what a future expert survey and second-coder check would involve. It describes what was done, not what was hoped for. No site data have been collected and no expert survey has been run; every analysis in Chapter 4 therefore uses published survey values (Literature) or an illustrative example case whose inputs are placeholders (Assumption). Chapter 4 reports the results; this chapter fixes the rules before they are applied.

3.1 Research design and scope

The work is a tool-development project with a secondary-data analysis attached to it. The tool is an offline register for delay risks in one activity, brick and block masonry. It supports risk identification from a library, a risk register, and an ordinal probability-impact matrix. The secondary-data analysis asks how much weight the published survey values can carry when they are used as the starting position of a risk on that matrix. The design follows the stages of the ISO 31000 risk-management process in a reduced form: identification (risk library and site-fact rules), analysis (ordinal classes and a score), evaluation (matrix levels) and recording (register and exports). The risk-matrix technique itself is one of the assessment techniques catalogued in IEC 31010. Communication, treatment and monitoring are outside what this project can evaluate without site users.

Five questions guide the analysis. They are questions about the order of risks and about the behaviour of the software's classification rule, not about delay in days or probability of occurrence.

1. How is a risk's seed class obtained from the survey values, and with which assumptions?
2. How much does a risk's class change when the banding rule, the data variant or the set of seed studies changes?
3. Which risks keep a high or low class under study resampling, and which cannot be assessed because they rest on one study?
4. Do the surveys agree with each other on the order of the risks they share, and by how much could a check of this kind detect agreement?
5. How does the matrix level of a risk with entered numbers react to its inputs and to the assumed edges and thresholds?

The design deliberately excludes four things. Monte Carlo schedule simulation, expected monetary value and the productivity models exist in the repository as optional or legacy code (the decision layer was re-connected on 2026-10-08 and uses only numbers the user enters). They are not analysed here because no input for them has a source: no delay distribution, cost or mitigation effect has been measured. A survey importance index is never converted into a probability or a number of days. The original project title promised labour-productivity and delay-risk prediction; Chapter 1 explains the scope revision, and nothing in this chapter or the next claims prediction.

3.2 The evidence corpus and the corpus count

Three documents in the repository give three different corpus sizes, and the repository also contains a fourth number, the count of curated evidence records that the application loads. The audit logged the conflict as finding F08 and left it as a decision for the supervisor (Audit Corrections, 2026-10-08, F08 and Section 5). This report does not reconcile the figures and does not choose one. Table 3.1 lists all four with their sources.

Table 3.1. Corpus and record counts found in the repository (Literature-search bookkeeping; n as stated per row; source labels: counts copied from the files named, not recomputed except the last row)

Version	Harvested or input	Kept	Where it is stated
1	21,902 harvested	5,214 construction-relevant	docs/ Project_Handoff_Summary.md Section 5 and 8.9 (data/openalex_corpus.jsonl, not in the repository)
2	17,788 harvested	3,975 in the database	docs/ Literature_Database_Report.md, docs/Corpus_Statistical_Analysis.md (data/literature.db, not in the repository; screening rules 2026-10-07.1)
3	11,599 input	3,550 included	data/screening_report.json (screening rules 2026-10-07.1)
4	not applicable	472 curated evidence records	Derived Calculation: build_full_store() in app/ai_layer/corpus.py, run on 2026-10-08 on this checkout; it reports “TOTAL: 472 evidence records”

Source: the files named in the last column; row 4 re-run for this report. The same figure (472) was seen in the application’s health check during the committee review.

Three points matter for how the corpus is used in this report.

First, the 472 records are not a screened search result. They are the records the application can retrieve: 48 from `Literature_Evidence_Package.xlsx`, records parsed from the research notes in `Research_Notes/`, benchmark and model registries, and 54 full-text review files. The OpenAlex bulk file (`data/openalex_corpus.jsonl`) and the screening database are excluded from version control by `.gitignore`, so a fresh clone loads 472 records and not thousands. The relationship between the 472 records and versions 1 to 3 has not been documented, and no relationship is assumed here.

Second, none of the three screening counts can be turned into a single PRISMA flow with confidence. Versions 2 and 3 carry the same rule version label (2026-10-07.1) but differ by 6,189 records in input and 425 in records kept, and no script output in the repository explains the difference. A PRISMA-style flow diagram would require re-running the harvest and screening scripts against one frozen input, which has not been done. The counts are therefore reported as a conflict and the corpus is treated as a source of background reading, not as the evidence base of any statistic in this report.

Third, the corpus statistics in `docs/Corpus_Statistical_Analysis.md` describe a database of $N = 3,975$ records (version 2) and use regular-expression matching on titles and abstracts. They show how many abstracts mention a term, not how many studies used a method, and the “sufficiency” bootstrap in that document is a subsample illustration of a census, not evidence that the corpus is large enough (audit correction, F19). Where Chapter 2 quotes those figures it must carry the same limit. The statistics that matter for this report are computed on a much smaller object: the 182 rows of `data/literature_seed.json`, described next.

3.3 Study selection for the RII data

The quantitative evidence is the relative importance index (RII) values that published surveys report for delay or productivity-loss factors. RII, as defined in these papers, is the sum of the respondents’ ratings divided by the highest possible rating times the number of respondents:

where w is the rating weight, n the number of respondents giving it, $w_{\max}$ the highest weight and N the number of respondents. It lies between 0 and 1 and measures how important respondents rated a factor. It is not a probability of occurrence and not a number of days.

Eleven studies are in the dataset. The seed file records no formal inclusion criteria, so none is claimed. What the entries show is that each study reports RII (or an RII-like importance index on a stated scale) for factors that

map to at least one library risk, and that the team had its full text or abstract; the PDF folders named in the file (5_Masonry_Productivity_Delay, 6_Ranking_Studies_OpenAccess) are collections the team assembled, not the output of a systematic screen. Two further sources were recorded but not used: Soundarya et al. (2025), because it reports a frequency-times-severity index, not RII, and Karthik and Rao (2022 or 2019, the year differs between sources), for which only a factor-group average of 0.784 is available from the abstract and which may report the same survey as M01 (other_indices in the seed file).

Each study carries a role in the dataset: five are **seed** studies (M01, A02, M06, A14, A15) and six are **held-out** studies (M10, M15, M14, A10, A21, HAS25). The seed file stores the held-out role under the older label **validation**; this report uses “held-out” because the check is an agreement check, not a validation (Audit Corrections, F05). Seed studies are Indian or Sri Lankan surveys; held-out studies come from the Middle East (pooled), Brazil, Indonesia, India (infrastructure), Thailand and Iraq. Only seed studies feed the placement of a risk. Held-out studies are used only to compare orders.

The date at which these roles were assigned relative to the first agreement calculation is not recorded. The earliest committed version of the seed file (commit **ed97aab**, 2026-10-07) already contains the roles, but the handoff summary records agreement results on a ten-risk set dated 2026-10-06 and 2026-10-07, and the library was widened from 10 to 38 risks after that first look. The role assignment therefore cannot be shown to precede all analysis, and the agreement results in Chapter 4 are labelled exploratory for that reason. Table 3.2 lists the studies.

Table 3.2. The 11 studies in `data/literature_seed.json` (Literature; n = 11 studies, 182 rows)

ID	Study (as recorded in the seed file)	Country	N respondents	Scale	Masonry-specific?	Role	Rows	Entered subset
M01	Karthik and Rao (2019a)	India (Telangana)	44	1-5	No (field case: AAC blocks)	seed	28	only factors with RII $\geq$ 0.700 are printed (28 of 38); truncated list
A02	Ponmalar et al. (2018)	India (Chennai)	not reported	1-4	No	seed	14	14 of 35 factors
M06	Dixit et al. (2019)	India	201 (206 in text)	1-5	No	seed	40	40 table rows
A14	Abeyasinghe and Jayathilaka (2022)	Sri Lanka	163	1-5	No (26.4% of respondents on road projects)	seed	5	5 of 39 factors
A15	Manoharan et al. (2022)	Sri Lanka	90 firms	not recorded	No	seed	5	5 of 32 factors; two-decimal summary values
M10	Adebawale and Agumba (2023)	Middle East, pooled	10 studies	pooled RII	No	held-out	33	Table 7, 33 rows
M15	Almeida et al. (2021)	Brazil	47	1-4	No	held-out	24	not stated
M14	Loekito et al. (2026)	Indonesia	83 (workers)	1-4 stated; printed RII = $W/(5N)$	No	held-out	23	not stated
A10	Pinky Devi and Sindhu (2025)	India (infrastructure)	72	1-5	No (roads, bridges)	held-out	6	subset; about 10 tool risks overlap
A21	Ouansrimeang and Wisaeang (2024)	Thailand	380 (30 projects)	1-5	No	held-out	2	subset; more items overlap
HAS25	Hassoon et al. (2025)	Iraq	not recorded	not recorded	Yes (abstract only)	held-out	2	abstract

Source: `data/literature_seed.json` (study fields, corrected 2026-10-08) and `docs/Audit_Corrections_2026-10-08.md`, Sections 3. Row counts: Derived Calculation from the file (n = 182). N for M06: 201 in the abstract and Table 3, 206 in Section 3.1 of the paper.

Three features of this table limit what the seed can mean. Only one study (HAS25) is masonry-specific, and it contributes two rows from an abstract. M01 and A02 are general building-construction surveys; masonry appears only in their field observations. Scales differ (1-4, 1-5, pooled, not recorded), and M14 prints RII as although its text states , so its absolute RII level is not comparable with other studies. Ranks within M14 are unaffected.

3.4 Extraction and verification protocol

Each RII value was copied by the team from the published table into one row of the seed file with the fields study, factor, RII, printed rank, mapped risk and mapping strength (**direct**, **related** or none). The source location is recorded per study, not per row (for example “Table 3, p.61” for M01), so the claim that every value carries its table and page is withdrawn; the accurate statement is 182 rows (178 factor values mapped to a risk, four unmapped or group-level rows), each with a study-level source location (Audit Corrections, F25).

The audit then compared the rows with the papers. Its summary states that all 182 **rii** and **rank** values match the printed papers, that no seed row is absent from its paper, and that the 182 rows were identical before and after the audit’s edits (only metadata and notes changed). A test, `test_g3_seed_rows_are_frozen`, pins a SHA-256 hash of the study, factor, RII and rank fields so that a later edit cannot change a value silently.

One discrepancy must be stated. The audit’s description of its source list names eight studies for the row-level check against PDFs: M01, M06, A14, M10, M15, M14, A10 and A21. Those eight studies hold 161 of the 182 rows. The seed file records A02 (14 rows) and A15 (5 rows) as taken from research notes, with the PDF not in the project folder, and HAS25 (2 rows) as taken from an abstract. This report therefore treats 161 rows as checked against PDFs and 21 rows as checked only against notes or an abstract, until the audit’s author confirms otherwise. A02 and A15 matter: A02 contributes 13 direct values to the seed and A15 four. Table 3.3 gives the position study by study.

Table 3.3. Provenance of the 182 seed rows (Derived Calculation from the seed file’s **obtained** field; n = 182 rows)

Provenance recorded in the seed file	Studies	Rows
Read from the PDF	M01 (28), M06 (40), A14 (5), M10 (33), M15 (24), M14 (23), A10 (6), A21 (2)	161
From research notes (PDF not in project folder)	A02 (14), A15 (5)	19
From the abstract only	HAS25 (2)	2

Known inconsistencies inside the papers were kept as printed and noted in the seed file. Examples: M01 “Bad leadership skill” prints a sum of 165 and RII 0.750, while the paper’s own counts over give and ; M06 gives as 201 and 206 in different places; M15 prints two DOIs; M10’s Table 7 has 33 rows and two further items appear only in text. None of these changes a seed value by more than 0.005 where it can be quantified. The values of A11 and of the group average P1-23 were not verified and are not used.

The seed file is a deliberate subset of its papers. A14 enters 5 of 39 factors, and the omitted ones include financial difficulties of contractors (RII 0.8233, rank 3) and Covid-19 (0.7902, rank 5); the audit estimated that adding the omitted A14 rows would move 14 of 35 classes. The statement that the library was “built from every factor in the 11 studies” is withdrawn; the accurate statement is “from the factors transcribed so far”.

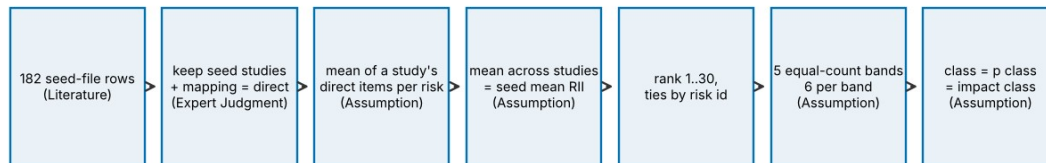
3.5 Mapping protocol

Each factor was assigned by the team to a library risk with a strength: **direct** (the factor is the risk or a clear synonym), **related** (an overlapping concept), or none (group-level or unmappable). The library holds 46 risks, of which 30 have at least one direct value in a seed study; 16 have none and stay off the matrix until numbers are entered (eight of the original 38 and the eight added on 2026-10-08). Only direct mappings from seed studies enter the seed: 66 values (M01 25, A02 13, M06 20, A14 4, A15 4), which combine into 30 per-risk means.

The mapping is one coder’s judgment (Expert Judgment, single coder). The audit found inconsistent decisions, for example for “poor site management” in M10, “training” in M06, “age” in A02 and “changing jobs” in M14 (finding F12), and recommended a second-coder check before any mapping value changes. That check has not been run. Until it is, every statistic that depends on the mapping carries the label “single coder, agreement not measured”. The planned protocol is in Section 3.10.

3.6 Seed construction

The procedure is the one implemented in `app/literature_seed.py`. Figure 3.1 summarises it and the following steps state it exactly.



Derived Calculation | procedure of `app/literature_seed.py` (n = 30 seeded of 46 library risks) | RII is importance, not probability or days

Figure 3.1. Steps from the 182 seed-file rows to a seed class (Derived Calculation; n = 30 seeded of 46 library risks; labels show the source label of each step).

1. **Select rows.** Keep rows whose study has role seed and whose mapping is **direct** and which carry a risk id. This gives 66 rows.
2. **Average within a study.** For each risk and each seed study, take the arithmetic mean of that study’s direct items. A study then counts once for the risk, however many of its items map to it (Assumption, AS-9 below).
3. **Average across studies.** The seed mean RII of a risk is the arithmetic mean of its study means, with equal weight for each study (Assumption, AS-8).
4. **Rank.** Order the 30 seeded risks by seed mean RII, highest first. Ties are broken by risk id in alphabetical order (Assumption, AS-12). Exactly one tie exists, R-MTH and R-STO at 0.7000.
5. **Band.** For rank among risks the class is

so ranks 1 to 6 are class 5, ranks 7 to 12 class 4, and so on down to class 1 (ranks 25 to 30); five equal-count bands of six (Assumption, AS-5). The number of bands, five, is chosen to match the five classes of the matrix axes.

6. **Place.** The class is used as both the probability class and the impact class of the matrix, because a survey importance index measures neither probability nor days (Assumption, AS-6). A rule flag marked “elevated” raises the probability class by one, capped at 5 (Assumption, AS-7). A seeded placement is an ordinal starting position. A user who enters both a probability and a delay for the risk replaces it.

Two worked examples follow. The first shows the RII definition applied to counts. The second applies steps 2 to 5 to one risk.

Worked example 3.1: RII from counts (Literature counts, Derived Calculation). M01 “Bad leadership skill”: 44 respondents on a 1-5 scale with counts 1, 5, 5, 27 and 6 for ratings 1 to 5. The weighted sum is , the maximum possible sum is , so RII . The paper prints 165 and 0.750; the seed uses the printed 0.750.

Worked example 3.2: from RII to class for R-LAB (labour absenteeism or shortage). Direct items: A02 “Absenteeism” 0.751; A14 “Shortage of labourers (skilled, semi-skilled, unskilled)” 0.8245; M01 “High workforce absenteeism” 0.723. Each study has one direct item, so the study means are 0.751, 0.8245 and 0.723.

The seed mean RII is (code: 0.7662). In the order of 30 means, 6 risks have a higher mean, so the rank is 7. Then class . The code returns rank 7 and class 4. The same calculation for a risk with several items in one study, R-SUP in M01 (0.759, 0.759, 0.714), gives a study mean of , which equals the value in the seed table. The check was run by `report/scripts/ch34_build.py`, which asserts both results against `seed_table()`.

3.7 Matrix definitions and thresholds

The matrix classifies each risk on two axes with five classes each and combines them into a level. The definitions, all from `app/engine/matrix.py`, are:

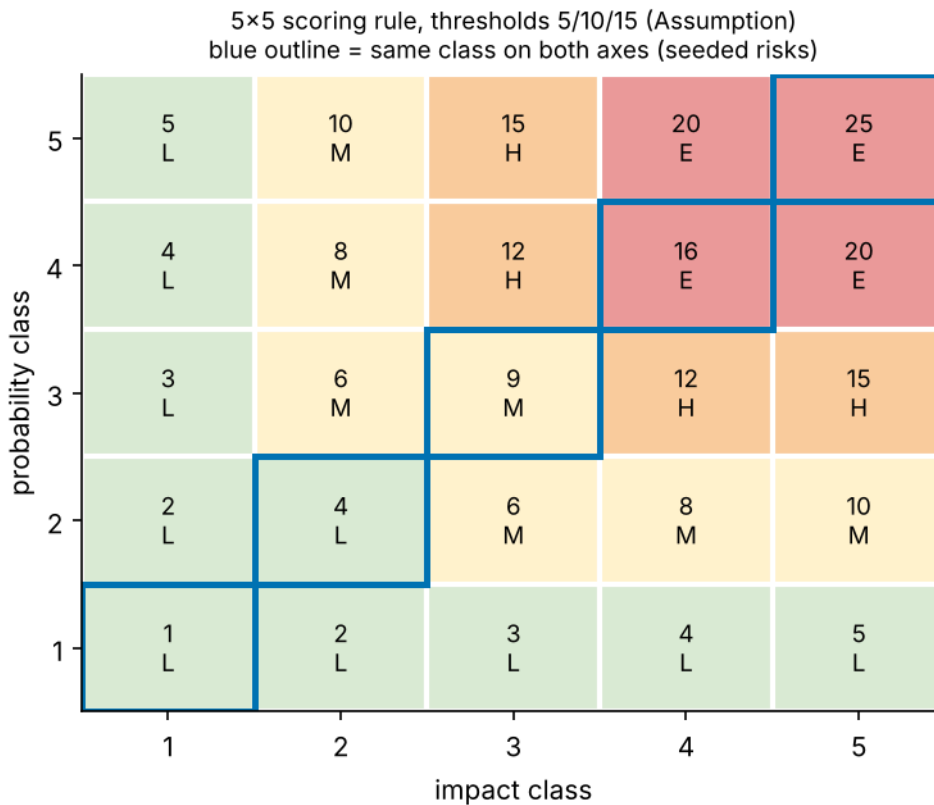
- **Probability class** from a probability with edges 0.2, 0.4, 0.6, 0.8: class = 1 + the number of edges that has reached. A value on an edge goes to the higher class (lower-edge-inclusive), with a relative tolerance of so that binary rounding does not move a value that is meant to lie on an edge. The edges are equal-width bins (Assumption, AS-1).
- **Impact class** from the conditional expected delay divided by the planned activity duration, with four ascending edges entered by the user. The tool has no default, and the example case uses 0.02, 0.05, 0.10 and 0.20 (Assumption, ILLUSTRATIVE, AS-3). The same lower-edge-inclusive convention and tolerance apply (AS-4).
- **Score** is the product of the two classes.
- **Level** is Low for scores up to 5, Moderate for 6 to 10, High for 11 to 15 and Extreme for 16 and above (thresholds 5, 10, 15, Assumption, AS-2).

Table 3.4 and Figure 3.2 show the level of all 25 cells. The structure follows from the rule and is a property of the chosen scoring rule, not a finding about risk. Row 1 and column 1 are always Low, because the largest score on them is . High needs both classes of at least 3. The score is symmetric, so the rule treats probability and impact as interchangeable. Cox (2008) discusses these properties of multiplicative matrices (poor resolution and range compression), and Duijm (2015) gives recommendations for matrix design; the choices here follow neither as a standard, and the sensitivity analyses of Chapter 4 test how far the thresholds matter.

Table 3.4. Score and level of each cell of the 5x5 rule (Derived Calculation from `matrix_level`; n = 25 cells; L = Low, M = Moderate, H = High, E = Extreme)

	i class 1	i class 2	i class 3	i class 4	i class 5
p class 5	5 L	10 M	15 H	20 E	25 E
p class 4	4 L	8 M	12 H	16 E	20 E
p class 3	3 L	6 M	9 M	12 H	15 H
p class 2	2 L	4 L	6 M	8 M	10 M
p class 1	1 L	2 L	3 L	4 L	5 L

Source: `app.engine.matrix.matrix_level`, thresholds 5/10/15 (Assumption).



Derived Calculation | n = 25 cells | L=Low M=Moderate H=High E=Extreme | source: app.engine.matrix.matrix_level

Figure 3.2. Level map of the 5x5 scoring rule; the blue outline marks the cells a seeded risk can occupy (Derived Calculation; n = 25 cells; thresholds are an Assumption).

With the same class on both axes, a seeded risk can only land on the diagonal: scores 1, 4, 9, 16 and 25, hence levels Low, Low, Moderate, Extreme and Extreme. Six risks in each class give 12 Low, 6 Moderate, 0 High and 12 Extreme (counted in Chapter 4). The effect is that the rank is effectively squared and given an absolute-sounding label, which the audit raised as finding F02. The application now labels seeded placements as “literature tier (Assumption)”, draws them as grey dashed chips and excludes them from the level totals; whether to hide the level colour entirely remains a supervisor decision.

The Assumption register

The labels AS-1 to AS-16 are local to this chapter and Chapter 4; the full register in Appendix H uses its own numbering (AS-01 to AS-18), and the two are not the same list. Table 3.5 lists every assumption that changes a number in this report, where it is used, why it was made, and which analysis tests it. Items AS-1 to AS-7 define the tool; AS-8 to AS-13 define how literature values are combined; AS-14 to AS-16 concern the analysis itself.

Table 3.5. Assumption register (Source label of every row: Assumption)

ID	Assumption	Where used	Reason	Tested by
AS-1	Probability edges 0.2/0.4/0.6/0.8 (equal width)	probability class of risks with entered p	simplest neutral bins; no source	S2 (example case)
AS-2	Level thresholds 5/10/15 on the product of classes	matrix level	no source	S2 (example), A1 extra (seeded, 969 triples)
AS-3	Impact = expected delay / planned duration with four user-entered edges (example 0.02/0.05/0.10/0.20, ILLUSTRATIVE)	impact class of risks with entered delay	puts delay on a duration-relative scale	S2
AS-4	Lower-edge-inclusive classes, relative tolerance	both class functions	removes float artefacts at an edge	S0 probes; S1 break-even (R-WX)
AS-5	Five equal-count bands, six risks per band	seed class	five classes on the axes	S3 band variants
AS-6	Same class on both axes	seeded placements	RII is neither probability	structural analysis

ID	Assumption	Where used	Reason	Tested by
			nor days	(Chapter 4); not testable with these data
AS-7	A rule flag adds one probability class, cap 5	seeded risks with an elevated rule	lets site facts move a seeded risk; no source for the step size	structural analysis (Chapter 4); not testable with these data
AS-8	Each seed study has equal weight	seed mean RII	avoids weighting by N across unequal designs	S3 (study bootstrap, drop-one-study)
AS-9	A study's direct items are averaged first	seed mean RII	one study counts once for a risk	not varied
AS-10	Only direct mappings enter the seed	seed rows	related rows are weaker matches	S3 (direct + related)
AS-11	Raw RII from 1-4 and 1-5 scales is averaged	seed mean RII	no common scale is available	S3 (rescaled; percentile rank)
AS-12	Ties are ordered by risk id	rank	needs a deterministic rule	one tie (R-MTH, R-STO), reported
AS-13	Seed and held-out roles as listed in the seed file	agreement analysis	see Section 3.3	S4 (non-circular comparison)
AS-14	Sensitivity ranges (x0.5 to x2, edge shifts, cut-off triples), the 80% tier rule, B = 2000, seed 20261008	all of Chapter 4	analysis design; fixed before reporting	all variants reported
AS-15	Planning SD of one Likert point for survey precision	Section 3.10	M01's counts give about 0.96	not applicable (planning only)
AS-16	Every probability, delay and the planned duration of the example case	S1, S2	placeholders for a demonstration	S1, S2

3.8 Analysis plan

The committee's methods review fixed one family of analyses, A1 to A7, before the results of the final scripts were produced. The scripts are in `analysis/` (S0 to S5) and one further script, `report/scripts/ch34_build.py`, adds the structural counts of A1, the cut-off analysis for the seeded placements, the tier rule, the critical correlations and the worked-example checks. Table 3.6 maps the plan to the scripts.

Table 3.6. Analysis family A1 to A7 and where each is implemented (Source: methods plan of 2026-10-08 and `analysis/README.md`)

ID	Question	Method	Implemented in	Input label
A1	What does the 5x5 rule imply by itself?	enumerate 25 cells; count levels; apply to the seeded diagonal	ch34_build.py; S0 probes	Assumption
A2	How does the level of a risk with entered numbers respond to its inputs?	one-at-a-time and joint scaling (x0.5, 0.8, 1.2, 1.5), 5x5 joint grid, 5000 random joint draws, break-even multipliers	S1	Assumption (ILLUSTRATIVE)
A3	How do edges and thresholds matter?	shift each probability edge, scale impact edges, alternative sets, 5000 random edge draws, cut-off triples	S2 (example); ch34_build.py (seeded: 969 triples)	Assumption
A4	How stable are seed classes under banding and data variants?	14 variants, drop-one-study, boundary gaps	S3	Literature
A5	Which risks keep a coarse tier under study resampling?	study-level bootstrap (2000), exact jackknife, tier rule	S3; tier rule in ch34_build.py; spread in S5	Literature
A6	Do studies agree on order, without circularity?	pairwise Spearman with n, permutation p, Holm; study vs seed from the other studies; Kendall W blocks	S4	Literature
A7	Are the RII values comparable across studies?	scale and selection checks reported as limits	S3 (rescaled, percentile, drop-one), S5; Table 3.2	Literature

Statistics used.

- *Spearman's rho* between two orders over the risks that both orders contain. Every correlation is reported with its n . A correlation with fewer than five shared risks is not tested.
- *Permutation p-values*. For n up to 8 the two-sided p is exact, by enumeration of all $n!$ permutations. For larger n , 20,000 random permutations are drawn and $p = (\text{count} + 1)/(20,000 + 1)$, where count is the number of permutations with $\cdot$. Monte Carlo error is about $\cdot$. Permutation tests are preferred to the t -approximation because n is small and the ranks contain ties.
- *Kendall's W* (Kendall & Babington Smith, 1939) for m studies ranking n risks that all of them contain, with a tie correction. The test statistic is W , where W is the sum of squared deviations of the risk rank sums and the tie term. The p -value is a permutation p (20,000 permutations, one-sided, with the +1 correction), because the chi-square approximation is unreliable at n of 5 to 10. W measures consensus about order, not reliability.
- *Holm's step-down correction* (Holm, 1979) within each family of tests: the sorted p -values are multiplied by n and made monotone. Three separate families are used: the 13 pairwise correlations with n of at least 5; the leave-one-study-out comparisons; and the 14 Kendall W blocks. The W blocks share studies and risks, so their adjusted p -values are descriptive.
- *Study-level bootstrap*. The five seed studies are resampled with replacement 2000 times, a study drawn twice has weight 2, and the seed is rebuilt each time. A risk is absent from a resample if all its studies were left out; class shares are therefore conditional on presence, and the share of resamples in which a risk is present is reported. Bands are cut over the risks present in the resample. With five clusters a percentile bootstrap is crude, so the exact leave-one-study-out (jackknife) is reported beside it, and the bootstrap is descriptive.
- *Tier rule* (Assumption, AS-14). A risk is in Tier 1 if it is in class 5 in at least 80% of the resamples in which it is present and it rests on at least two seed studies; "Tier lower" is the same with classes 1 and 2; a risk that meets a threshold but rests on one study is flagged "not assessable"; all others are "unresolved".
- *Critical correlations*. For each n the smallest $\cdot$ whose two-sided permutation p is at most 0.05, so that the reader knows what a check of that size could detect.

All random draws use one generator seed, `numpy.random.default_rng(20261008)`. All variants named in the plan are reported, whether or not they look favourable. Software: Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, Matplotlib 3.11.2 (versions read from the interpreter used for this report). The command `PYTHONPATH=/usr/local/lib/python3.13/dist-packages python analysis/run_all.py` clears `analysis/out/`, reruns S0 to S5 in about 32 seconds, asserts that every output carries the label "Derived Calculation" and an n , and writes a manifest of SHA-256 hashes. The command was run twice for this report and the two manifests were identical. Because `analysis/out/` is excluded from version control, the result could not be compared with an earlier run of the same files; the check that remains is the identical second run and agreement with the figures of the committee plan where those are quoted.

What was and was not pre-specified. The family was written down in the committee plan on 2026-10-08, after the ten-risk analyses and after the library was widened to 38 risks. It is therefore a specification made after the data were known, not a pre-registration. Its protection against forking paths is that the variants are enumerated, none is dropped and every one is reported. The 38-risk run is a second look at data first examined on 10 risks and is exploratory.

Excluded on purpose. The earlier scripts computed a number of statistics that this report does not repeat: item-level Kruskal-Wallis and Mann-Whitney tests (pseudo-replication and post hoc), any pooled order that contains the seed studies compared with the seed (circular), "mean pairwise rho" as an agreement statistic, a two-way model whose headline p depends on single-study risks, and any conversion of RII into a probability. The audit's findings F01, F05, F09 and F10 list them.

3.9 Software verification method

Verification asks whether the software does what its specification says. Validation would ask whether its output helps a site manager or matches site reality, and no validation has been done. The verification used four methods.

1. **Automated tests.** The pytest suite covers the engine, matrix, rules, service, exports, AI guard and the optional analysis layer. On 2026-10-08, with the command `python -m pytest -q`, it reported 513 passed and 3 skipped in about 29 seconds; the three skips need the screening database, which is not in the repository. This count is reported as read at that time, not as a fixed property: other counts (46 library risks, 30 seeded, 182 seed rows, 11 studies, 5 seed studies, 22 site-fact rules) are pinned through `tests/test_relevance_guard.py`, which reads each count from the data files and fails when a document or data file changes without a conscious update.
2. **Guard tests.** Ten guards G-1 to G-10 check that counts agree across data and documents, that the seed rows are frozen (a hash), that withdrawn wording is absent from current documents, that every exported number has a source, that the example stays labelled ILLUSTRATIVE, and that unseeded risks carry a status and no seed rows and the AI path carries no numbers.
3. **Hand calculation against the tool.** Chapter 6 compares hand calculations with tool output for a worked case. Within this report, Worked examples 3.2 and 4.1 recompute a seed class and a matrix placement by hand and compare them with the code's output.
4. **Cross-implementation checks inside the analysis package.** S3 re-implements the RII-to-class rule and asserts equality with `seed_table()` for all 30 seeded risks. S1 and S2 use a faster path through the matrix functions and assert equality with `run_case` on all 84 one-at-a-time runs, on 100 random draws, on the baseline and on every impact-edge variant. No mismatch occurred (`fast_path_vs_run_case_mismatches` is 0).

S0 probes small behaviours of the repository code. Results relevant here: `run_case` does not expose the probability edges (classification uses the default edges, although a docstring suggests otherwise), so S2 varies them through the matrix functions and asserts agreement at the defaults; the example case's R-WX, with $p = 0.20$, lies exactly on the first probability edge; and switching the seed basis to "all" (held-out studies and related mappings pooled) changes the class of 24 of the 30 common risks. The S0 output also stores a finding text about a missing probability tolerance, which is stale: the tolerance was added after the committee review (Audit Corrections, Section 6), and the probe's recorded class for $0.1 + 0.7$ is 5, as the tolerant rule expects.

Engine defects found by the audit (H1, M1 to M10, L1 to L14) were partly fixed on 2026-10-08. Those still open are listed in the audit and in Chapter 6; NaN and boolean inputs are now rejected by the case validator. None of them affects the seed or the analyses reported here.

3.10 Future expert survey and second-coder design

Neither of the following has been done. They are described so that the analysis can be pre-registered before any data arrive.

3.10.1 Expert survey

The design is a short questionnaire of 15 neutral-worded causes of masonry delay (for example bricks or blocks not on site, masons absent, hoist unavailable, rain stopping external masonry), each rated for **frequency** (1 never or almost never to 5 on almost every masonry activity) and for **severity** (1 no noticeable slip to 5 holds up following trades), with "not applicable" and "don't know" options. The anchor wording is the team's own (Assumption) and is never converted to days or probability. Item order is randomised, the literature ranks and the tool's matrix are not shown, one attention-check item is included, team members do not respond, and a three-cause ranking question gives a check independent of the rating scales. The respondent profile records role, years of masonry supervision, building types, unit types, region and employer type, without names. A pilot with two or three site engineers tests wording.

Pre-registration. A dated file in the repository, written before data arrive, will fix the items, the item-to-library mapping, the exclusion rules (failed attention check, under one year of masonry supervision), the handling of N/A (pairwise, never imputed) and the analysis list below.

Analysis list.

- Per item: n answering, median, IQR, frequency index (FI) and severity index (SI) by the RII formula, importance FI x SI, with a respondent-bootstrap 95% interval (B = 5000), and ranks with bootstrap rank intervals.
- Kendall's W across respondents on complete cases, separately for frequency and severity, and by group (contractor-side, client or consultant side) when a group has at least 8 respondents.
- Reliability of the item order, not only Cronbach's alpha (Cronbach, 1951): random split-half of respondents, Spearman of the two halves' FI orders corrected by the Spearman-Brown formula, averaged over 1000 splits; alpha is reported for convention only, because the 15 causes are a checklist and not one reflective construct.
- Group differences per item by Mann-Whitney with Holm over 15 items and a rank-biserial effect size, only if each group has at least 10 respondents.
- Comparison with the seed by Spearman rho on the items that have a pre-registered one-to-one mapping (about 10 expected), exact p; labelled "agreement check, exploratory".

Precision and sample size. The sample will be purposive or snowball, so no population margin of error is claimed. With a planning SD of one Likert point (Assumption; M01's counts give about 0.96), the standard error of one item's index is about : Table 3.7 gives the values and the half-width of a 95% interval. The gaps between neighbouring seed risks are 0.001 to 0.02, so an expert sample of 10 to 15 is a pilot and cannot resolve order inside a band. A target of at least 30, with response rate and respondent profile reported, is stated in advance. With n = 10 shared items, the smallest that would reach p of 0.05 is 0.648 (Table 3.8), so the detectable agreement is stated before the survey, not after.

Table 3.7. Planning precision of one importance index (Derived Calculation from the Assumption SD = 1 Likert point; formula)

n respondents	SE of one index	Half-width of 95% interval (1.96 x SE)
15	0.052	0.101
30	0.037	0.072
44 (M01's N)	0.030	0.059

Table 3.8. Smallest absolute Spearman correlation reaching two-sided permutation p <= 0.05 (Derived Calculation; exact for n up to 9, 200,000 random permutations with seed 20261008 above that; n = number of shared items or risks)

n shared risks	Smallest absolute rho with two-sided permutation p <= 0.05
5	1.000
6	0.886
7	0.786
8	0.738
9	0.700
10	0.648
11	0.618
13	0.566
15	0.521
19	0.460

Source: `report/scripts/ch34_build.py`. Values are discrete for small n; n = 5 admits only a perfect rank agreement.

Frequency answers and the optional "out of the last 10 activities" counts will not be converted into probabilities in the analysis. They can be entered later, risk by risk, as Expert Judgment with n stated.

3.10.2 Second-coder protocol for the mapping

1. Freeze a coding manual first: the risk definitions (name, scope, inclusion and exclusion examples), the decision rule for direct, related and none, and a rule for “unmappable”.
2. The second coder is a person who did not build the mapping (a team member who did not code, or a faculty member). The coder is blind to the first coder’s assignments and to the RII and rank columns; only the factor text and study context are exported.
3. Calibrate on 15 to 20 rows from a study that is not in the agreement sample (for example A10 or A21 items not yet entered), discuss, amend the manual once and freeze it.
4. Code all 178 mapped factor rows and any newly entered rows.
5. Statistics: Cohen’s kappa (Cohen, 1960) for risk assignment across the library categories plus none, with percent agreement and a bootstrap 95% interval over rows; linearly weighted kappa for strength (direct, related, none); and the same for the subset that feeds the seed (seed studies, direct). Report prevalence. Landis and Koch (1977) descriptors are used only as labels.
6. Pre-committed rule: kappa below 0.60 means revise the manual and recode a fresh sample; otherwise resolve disagreements by discussion and log every change (row, old, new, reason).
7. Re-run A4 to A6 once on the reconciled mapping and report original and reconciled results side by side.

3.11 Ethics

No human participants have been involved in the work reported here. All data are published survey values, and the example case is made up. The planned expert survey will involve adults who volunteer information about their own professional experience. Before it starts, the team will: obtain whatever institutional approval or exemption NICMAR requires for a questionnaire study (the requirement has not been confirmed with the supervisor and no approval number exists); show an information sheet stating purpose, voluntary participation, the right to stop, the anonymity of responses and the uses of the data; obtain informed consent as the first item of the form; collect no names, employer names or contact details in the response file (invitations are sent separately); store the response file on institutional or team-controlled storage and share only aggregated results; and report n, response rate and respondent profile truthfully. A site data collection phase would need a separate approval and the consent of the site owner. The AI evidence layer of the tool is not used to produce numbers.

3.12 Threats to validity

Table 3.9 collects the threats by type, with what has been done and what remains open. Chapter 7 returns to them.

Table 3.9. Threats to validity (Assumption and Literature labels as in the text)

Type	Threat	Handling in this report	Status
Construct	RII measures importance, not likelihood or days; the same band is used on both matrix axes; multiplicative matrices compress range (Cox, 2008)	seed labelled ordinal and Assumption; A1 and S2 show what the rule implies; no RII-to-probability conversion	open by design
Internal	One coder for the mapping	kappa protocol designed	open
Internal	Analysis choices after seeing the ten-risk data; many variants	one enumerated family, all reported, 38-risk run called exploratory	mitigated, not removed
Internal	M01 prints only RII ≥ 0.70 ; A14 and A15 give top factors only; selection on outcome pulls these risks up	dropping A14 and A15 together is one variant in S3	reported
Internal	Mixed scales, M14 printed basis, A15 scale not recorded	rescaling and percentile-rank variants	reported
Statistical conclusion	5 seed studies; 14 of 30 seeded risks rest on one study; overlaps of 1 to 19 risks give low power; multiple testing	exact or permutation p, Holm, n for every statistic, critical rho	reported
Statistical conclusion	Intervals ignore mapping and	stated with each interval	open

Type	Threat	Handling in this report	Status
	scale uncertainty, so they understate it		
External	General building surveys, not masonry-specific; two Sri Lankan studies; road projects in A14; manager-dominated respondents; dated surveys	study table and limits stated	open
Reliability of the tool	Engine defects (NaN, booleans, edge rounding)	partly fixed, tests, S0 probes	partly open (NaN, booleans, edge rounding fixed)
Ecological	No site data; example is ILLUSTRATIVE; no user evaluation	no claim of accuracy on real sites	open
Corpus	Three corpus counts and a fourth record count unreconciled	reported, not resolved	open (supervisor decision)

3.13 Integrity rules and source labels

Every number in this report has a source label: **Literature** (as printed in a published study), **Expert Judgment** (the team's single-coder mapping, and, in future, survey ratings with n), **Derived Calculation** (computed from inputs of the other kinds, which are named; a value derived from Assumption inputs inherits the label ILLUSTRATIVE) and **Assumption** (a choice made by the team and listed in Table 3.5). Historical Data and User Input do not occur in this report, because no site data or user entries exist. Tables and figures of Chapter 4 carry the label and the n in their captions. RII is described as importance and never as probability or days; the seed as an ordinal starting position; the held-out comparison as an agreement check that is weak, low-powered and inconclusive. The wording withdrawn in the audit (Handoff Section 8.11) is not used.

Chapter 4. Secondary-Data Analysis

This chapter reports every analysis defined in Section 3.8, with the numbers as the scripts produced them. All tables and figures are Derived Calculations: their inputs are Literature values (the 182 seed rows) or the Assumption inputs of the illustrative example case, as each caption states, and every caption gives n. The scripts are in `analysis/` and `report/scripts/`; the outputs are in `analysis/out/`; the figures are copies of the files there, with chapter-based names. Two rules govern how the results are worded. RII is an importance rating, so every analysis concerns the order of risks and never a probability or a number of days. The surveys are few and overlap little, so a result that is not significant is described as low-powered and inconclusive, not as evidence that agreement is absent.

4.1 What was run

Table 4.1 gives the inventory. The rows follow the order of the sections below, which is the order of the question set in Section 3.1.

Table 4.1. Analyses reported in this chapter (Derived Calculation; counts of risks, runs, draws or resamples as shown)

Section	Analysis	Inputs	n
4.2	Seed table and two worked examples	66 direct seed values	30 seeded risks
4.3	Behaviour of the bands and of the 5x5 rule	seed classes; thresholds	30 risks; 969 cut-off triples
4.4	Example-case sensitivity (S1)	ILLUSTRATIVE example, Assumption inputs	7 risks; 84 one-at-a-time runs; 25 grid cells; 5000 draws
4.5	Edge and threshold sensitivity (S2)	same example	7 risks; 55 scenarios; 5000 draws
4.6	Seed-band sensitivity (S3)	Literature	30 risks; 14 variants
4.7	Bootstrap and jackknife tiers (S3)	Literature	30 risks; 5 seed studies; 2000 resamples
4.8	Non-circular inter-study agreement (S4)	Literature	11 studies; 55 pairs; 13 tested
4.9	RII uncertainty (S5)	Literature	30 seeded risks; 11 studies

4.2 The 30-risk literature seed

Table 4.2 is the seed as `app.literature_seed.seed_table()` returns it, with the library name of each risk. The study means are the per-study averages of Step 2 of Section 3.6. The “Level” column applies the 5x5 rule to a risk whose probability class and impact class are both equal to its seed class (Assumption, AS-6); it is shown to document what the tool does and is not a finding about the risk.

Table 4.2. The literature seed, 30 seeded risks (Derived Calculation from Literature values; n = 30 risks, 66 direct values from 5 seed studies; classes and levels are Assumptions of the tool)

Rank	Risk	Library name	Seed studies (n)	Seed mean RII	Class	Level if p-class = i-class	Study means (Literature, averaged per study)
1	R-MAT	Brick / block / mortar material shortage or late delivery	3	0.7889	5	Extreme	A02 0.7800, A14 0.8098, M01 0.7770
2	R-SAFE	Unsafe conditions or work at height	1	0.7865	5	Extreme	M01 0.7865
3	R-SKILL	Unskilled or unqualified labour	3	0.7831	5	Extreme	A02 0.7870, A15 0.8033, M01 0.7590
4	R-PLAN	Poor planning or unrealistic scheduling	2	0.7771	5	Extreme	M01 0.7910, M06 0.7632
5	R-PAY	Payment delay	3	0.7733	5	Extreme	A02 0.7758, M01 0.7730, M06 0.7711
6	R-DIS	Poor labour discipline or weak site leadership	2	0.7700	5	Extreme	A15 0.7900, M01 0.7500
7	R-LAB	Labour absenteeism or shortage	3	0.7662	4	Extreme	A02 0.7510, A14 0.8245, M01 0.7230
8	R-SCP	Unclear scope or inadequate project formulation	1	0.7642	4	Extreme	M06 0.7642
9	R-SITE	Poor site access, clearance or conditions	2	0.7611	4	Extreme	A02 0.7510, M06 0.7711
10	R-DES	Design changes or incomplete documentation	2	0.7585	4	Extreme	M01 0.7320, M06 0.7851
11	R-RWK	Rework due to workmanship or defects	2	0.7561	4	Extreme	M01 0.7360, M06 0.7761
12	R-TOOL	Tools and equipment delay	2	0.7528	4	Extreme	A02 0.7600, M01 0.7455
13	R-INC	Inadequate wages or incentives	1	0.7520	3	Moderate	M01 0.7520
14	R-STR	Labour strikes or disputes	1	0.7493	3	Moderate	M06 0.7493
15	R-CTR	Contractual disputes and claims	1	0.7468	3	Moderate	M06 0.7468
16	R-SUP	Inadequate supervision or monitoring of workers	1	0.7440	3	Moderate	M01 0.7440
17	R-MOT	Low motivation, morale or	1	0.7360	3	Moderate	M01 0.7360

Rank	Risk	Library name	Seed studies (n)	Seed mean RII	Class	Level if p-class = i-class	Study means (Literature, averaged per study)
		labour turnover					
18	R-MGT	Poor site management and coordination	2	0.7331	3	Moderate	M01 0.7090, M06 0.7572
19	R-CST	Cost control and budget updating problems	1	0.7184	2	Low	M06 0.7184
20	R-OVT	Overtime and long working hours	2	0.7063	2	Low	M01 0.7250, M06 0.6876
21	R-ACC	Accidents and injuries on site	2	0.7045	2	Low	A02 0.6500, M01 0.7590
22	R-MTH	Construction method or design complexity	1	0.7000	2	Low	M01 0.7000
23	R-STO	Poor material storage or handling	1	0.7000	2	Low	A02 0.7000
24	R-WX	Rain / monsoon stoppage	4	0.6934	2	Low	A02 0.7000, A14 0.6528, M01 0.7640, M06 0.6567
25	R-COM	Poor communication or misunderstanding on site	2	0.6793	1	Low	A02 0.6800, M06 0.6786
26	R-PRC	Material price escalation	2	0.6695	1	Low	A02 0.5500, A14 0.7890
27	R-ECO	Economic conditions, inflation or interest rates	1	0.6612	1	Low	M06 0.6612
28	R-WAT	Water shortage	1	0.6500	1	Low	A02 0.6500
29	R-FAT	Worker fatigue, age or personal problems	1	0.6100	1	Low	A02 0.6100
30	R-SOC	Social or community environment	1	0.5940	1	Low	M06 0.5940

Source: `app.literature_seed.seed_table()` (basis “seed”);

`data/literature_seed.json`; risk names from `data/risk_library_masonry.json`. R-MTH and R-STO tie at 0.7000 and are ordered by id.

Six features of the table matter for everything that follows.

The spread is small. The seed means run from 0.594 (R-SOC) to 0.789 (R-MAT), a range of 0.195. Ranks 1 to 6 lie within 0.019 of each other (0.7889 to 0.7700), and ranks 1 to 24 within 0.096. The median gap between adjacent ranks is 0.0034 (S5). The class of a risk therefore depends on small differences in a mean of at most four numbers.

Fourteen of the 30 risks rest on a single study: R-CST, R-CTR, R-ECO, R-FAT, R-INC, R-MOT, R-MTH, R-SAFE, R-SCP, R-SOC, R-STO, R-STR, R-SUP and R-WAT (S0). For those the “average across studies” is one study’s number.

Sixteen library risks have no seed. R-CLI, R-DEF, R-FIN, R-GOV, R-PRD, R-QLT, R-SUB and R-TRN have no direct value in the seed studies, and eight further risks added on 2026-10-08 (R-FRONT, R-VT, R-MORT, R-GPAY, R-OPEN, R-SCAF, R-FEST, R-HEAT) were written without seed rows. They stay off the matrix

until numbers are entered. Several of the omitted risks (work-front release, scaffolding, the hoist) are the ones a site manager might most expect, and no seed study has a direct value for them.

Six per class is a rule, not a result. Equal-count banding puts exactly six risks in each class whatever the data (S0 confirms 6, 6, 6, 6, 6).

Seed studies differ in kind. M01 lists only items rated 0.70 or higher; A14 and A15 contribute four values each; A02 is on a 1-4 scale. The class of a risk reflects which of these studies happened to report it.

One tie exists (R-MTH and R-STO at 0.7000), broken alphabetically.

Worked example 4.1: one risk placed on the matrix by hand. R-LAB has seed class 4 (Worked example 3.2). Both axes take the class: probability class 4, impact class 4. The score is . The level thresholds are 5, 10 and 15; a score of 16 exceeds 15, so the level is Extreme (the code takes `bisect_right([5,10,15], 16 - 1)`, which is 3, the fourth name). If a site-fact rule marks R-LAB as elevated, the probability class becomes , the score , and the level stays Extreme. The returned placement agrees: `seed_for("R-LAB")` gives p class 4, impact class 4, and `seed_for("R-LAB", elevated=True)` gives p class 5, impact class 4, with `raised_by_rule` true. For contrast, a risk with entered numbers is placed from its inputs. In the illustrative example R-MAT has $p = 0.5$ and a PERT delay with minimum 1, most likely 3 and maximum 8 days, so the expected delay if it occurs is days. The planned duration is 16 days, so the ratio is . With impact edges 0.02, 0.05, 0.10 and 0.20, the ratio has reached all four, so the impact class is 5. With probability edges 0.2, 0.4, 0.6 and 0.8, $p = 0.5$ has reached two, so the probability class is 3. The score is 15, which is High. The tool returns p class 3, impact class 5, score 15, High (S1, base case). All inputs of the second calculation are Assumptions (ILLUSTRATIVE); the first uses Literature values and the assumed rule.

4.3 Behaviour of the bands and of the 5x5 rule

Section 3.7 showed that the rule has 10 Low, 7 Moderate, 4 High and 4 Extreme cells (Derived Calculation, $n = 25$ cells; `ch34_build.py` recounts them from `matrix_level`). Applied to the seeded risks, with the same class on both axes, the effect is stark (Table 4.3).

Table 4.3. Levels of the 30 seeded risks under the repo rule (Derived Calculation; $n = 30$ risks; classes from Table 4.2; same class on both axes is an Assumption)

Case	Low	Moderate	High	Extreme
Placed as the tool does, no rule flag	12	6	0	12
Hypothetical: every seeded risk carries an “elevated” flag (probability class + 1, cap 5)	6	6	6	12

Source: `ch34_build.py` (facts in `report/scripts/out/facts.json`). The second row is a structural illustration of Assumption AS-7, not a scenario that occurs in the example case.

With no flag, no seeded risk can be High, and risks in classes 4 and 5 are all Extreme, so the twelve most important risks by seed rank are all “Extreme” although their seed means differ by at most 0.0361 (R-MAT 0.7889 to R-TOOL 0.7528). The boundary between ranks 12 and 13 shows the label’s arbitrariness: R-TOOL (0.7528, class 4, Extreme) and R-INC (0.7520, class 3, Moderate) differ by 0.00075 in mean RII, which is smaller than the change in an M01 RII caused by one respondent moving one Likert point (0.0045 at $N = 44$; Handoff Section 8.2). One flag lifts a class 3 risk from Moderate to High and a class 2 risk from Low to Moderate, so in the flagged row six risks are High.

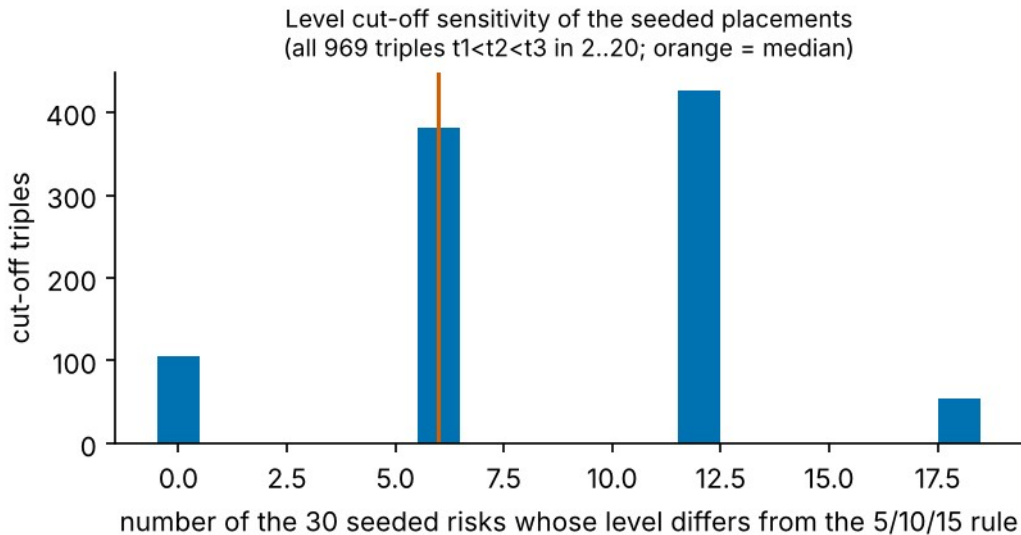
The levels also depend on the cut-offs. Table 4.4 gives the effect of six named alternatives on the seeded placements, and Figure 4.1 shows the distribution over all 969 admissible triples in 2 to 20.

Table 4.4. Levels of the 30 seeded risks under alternative level cut-offs (Derived Calculation; $n = 30$ risks per row; cut-offs are the upper score of Low, Moderate and High; same class on both axes is an Assumption)

Cut-offs (Low/Mod/High upper score)	Risks changing level (of 30)	Low	Moderate	High	Extreme
5/10/15 (repo)	0	12	6	0	12

Cut-offs (Low/Mod/High upper score)	Risks changing level (of 30)	Low	Moderate	High	Extreme
4/9/15	0	12	6	0	12
4/9/16	6	12	6	6	6
4/8/12	6	12	0	6	12
6/12/16	6	12	6	6	6
3/8/14	12	6	6	6	12
5/10/20	6	12	6	6	6

Source: ch34_build.py.



Derived Calculation | $n = 30$ risks $\times$ 969 triples | deterministic | same class on both axes is an Assumption

Figure 4.1. Number of the 30 seeded risks whose level changes when the cut-offs change, over all 969 triples (Derived Calculation; $n = 30$ risks $\times$ 969 triples).

Over the 969 triples the median number of changed levels is 6 of 30 (interquartile range 6 to 12, range 0 to 18); in 10.8% of triples no seeded level changes. Because six risks share each diagonal score, a cut-off that moves past one diagonal score changes six labels at once. The levels of seeded placements are therefore a function of the assumed thresholds as much as of the evidence, which is consistent with the application's decision to draw them as an ordinal tier and exclude them from level totals.

4.4 Example-case sensitivity (S1)

S1 uses the illustrative example case, which has 7 risks, a planned duration of 16 days, probabilities from 0.10 to 0.50 and PERT delays of 1 to 8 days, all placeholders (Assumption, ILLUSTRATIVE; Table 4.5). It tests how the classification rule responds to its inputs. It says nothing about how a real site behaves. With 7 risks, the counts below are descriptions of one small case, not a sample.

Table 4.5. The example case as classified (Derived Calculation from Assumption inputs, ILLUSTRATIVE; $n = 7$ risks; planned duration 16 days; impact edges 0.02/0.05/0.10/0.20 of planned duration)

Risk	p (Assumption)	Distance of p to nearest p edge	Expected delay / planned duration	Distance of ratio to nearest impact edge	p class	Impact class	Level
R-MAT	0.5	0.1	0.2188	0.0938	3	5	High
R-LAB	0.35	0.05	0.1562	0.2188	2	4	Moderate
R-RWK	0.25	0.05	0.1458	0.2708	2	4	Moderate
R-WX	0.2	0.0	0.1667	0.1667	2	4	Moderate
R-PLAN	0.1	0.1	0.0833	0.1667	1	3	Low
R-SAFE	0.3	0.1	0.1562	0.2188	2	4	Moderate
R-SKILL	0.22	0.02	0.1458	0.2708	2	4	Moderate

Source: analysis/out/s2_edge_margins.csv; inputs from examples/example_case.json. Expected delay is the mean of the PERT delay $(a + 4m + b)/6$.

Five risks (R-LAB, R-RWK, R-WX, R-SAFE, R-SKILL) are probability class 2 and impact class 4 and are Moderate (score 8). R-MAT is High (score 15) and R-PLAN Low (score 3). R-WX has $p = 0.20$, exactly on the first probability edge, so under the lower-edge-inclusive rule it is class 2 and any shift of that edge or of p flips it. This single placeholder drives many of the counts below.

One at a time. Each risk's p , its delay days (a , m , b together) or both were multiplied by 0.5, 0.8, 1.2 and 1.5 (p capped at 1), with the other six risks unchanged. 84 runs were made. No run changed the level of any other risk (`oat_other_risks_ever_changed` is 0); the matrix is per-risk and independent.

Table 4.6. Number of the 7 risks whose level changes when one input is scaled, one risk at a time summed over risks (Derived Calculation from Assumption inputs, ILLUSTRATIVE; $n = 7$ risks $\times$ 4 factors per input; 84 runs in all)

Input changed	x0.5	x0.8	x1.2	x1.5
p	6	2	2	3
delay	0	0	0	0
p +delay	6	2	2	3

Source: `analysis/out/s1_summary.json`. Each cell counts risks, 7 at most, whose own level changed when that risk alone was scaled by the factor shown.

The level responds to probability and not to delay. Scaling every risk's delay by 0.5 to 1.5 changes no level, because the probability class caps the score: a risk of probability class 2 scores at most, still Moderate. Delay changes the impact class (for example R-MAT drops from impact class 5 to 4 at 0.91 times its delay) but not the level. Table 4.7 gives the break-even multiplier, the smallest scaling of one input within 0.30 to 2.00 (step 0.01) at which a class or level first changes.

Table 4.7. Break-even multipliers, one input of one risk at a time (Derived Calculation from Assumption inputs, ILLUSTRATIVE; $n = 7$ risks; grid 0.30 to 2.00 in steps of 0.01)

Risk	Input	Up: first multiplier changing class	Down: first multiplier changing class	Up: first multiplier changing level	Down: first multiplier changing level
R-MAT	p	1.2	0.79	1.2	0.79
R-MAT	delay	none in 0.30-2.00	0.91	none in 0.30-2.00	0.45
R-LAB	p	1.15	0.57	1.15	0.57
R-LAB	delay	1.28	0.63	none in 0.30-2.00	0.31
R-RWK	p	1.6	0.79	1.6	0.79
R-RWK	delay	1.38	0.68	none in 0.30-2.00	0.34
R-WX	p	2.0	0.99	2.0	0.99
R-WX	delay	1.2	0.59	none in 0.30-2.00	none in 0.30-2.00
R-PLAN	p	2.0	none in 0.30-2.00	2.0	none in 0.30-2.00
R-PLAN	delay	1.2	0.59	none in 0.30-2.00	none in 0.30-2.00
R-SAFE	p	1.34	0.66	1.34	0.66
R-SAFE	delay	1.28	0.63	none in 0.30-2.00	0.31
R-SKILL	p	1.82	0.9	1.82	0.9
R-SKILL	delay	1.38	0.68	none in 0.30-2.00	0.34

Source: `analysis/out/s1_breakeven.csv`. "none in 0.30-2.00" means no change anywhere on the grid.

Probability changes a level at multipliers between 0.57 and 2.0, risk by risk: R-MAT at +20% or -21%, R-LAB at +15%, R-SAFE at +34%. R-WX flips down at 0.99, because it sits on the edge. Delay changes a level only for decreases of 55% to 69% (multipliers 0.31 to 0.45), and not at all within the grid for R-WX and R-PLAN. Delay increases up to double change no level.

Joint changes. Scaling all probabilities and all delays together on a 5x5 grid (25 cells) and drawing each input's factor independently from $U(0.5, 1.5)$ 5000 times (seed 20261008) give the same picture as the single-input runs (Figure 4.3; grid and per-risk shares in `analysis/out/s1_joint_grid_summary.csv` and `s1_joint_random_by_risk.csv`). On average 2.48 of the 7 levels change per draw, and in 4% of draws none does.

S1 One-at-a-time sensitivity, ILLUSTRATIVE example case
[Derived Calculation] n = 7 risks (example case); 84 one-at-a-time runs; 5000 random joint draws

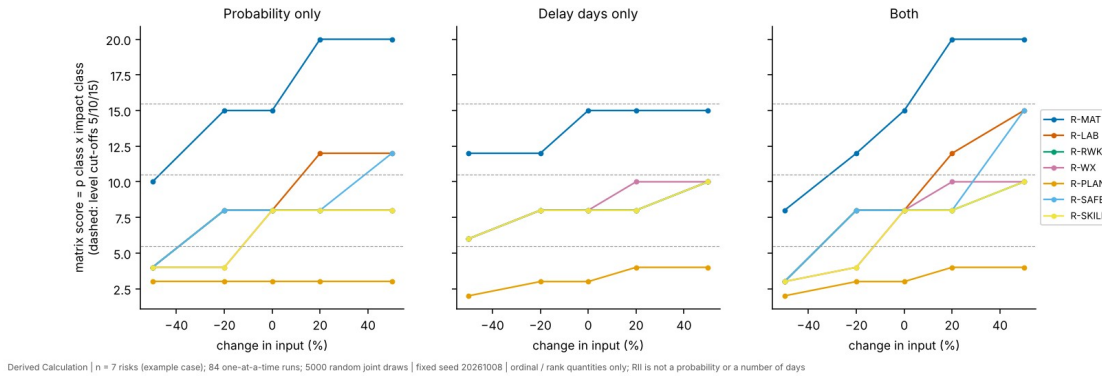


Figure 4.2. One-at-a-time sensitivity of the matrix score (Derived Calculation from Assumption inputs, ILLUSTRATIVE; n = 7 risks; 84 runs). Dashed lines are the level cut-offs.

S1 Joint sensitivity, ILLUSTRATIVE example case
[Derived Calculation] n = 7 risks (example case); 84 one-at-a-time runs; 5000 random joint draws

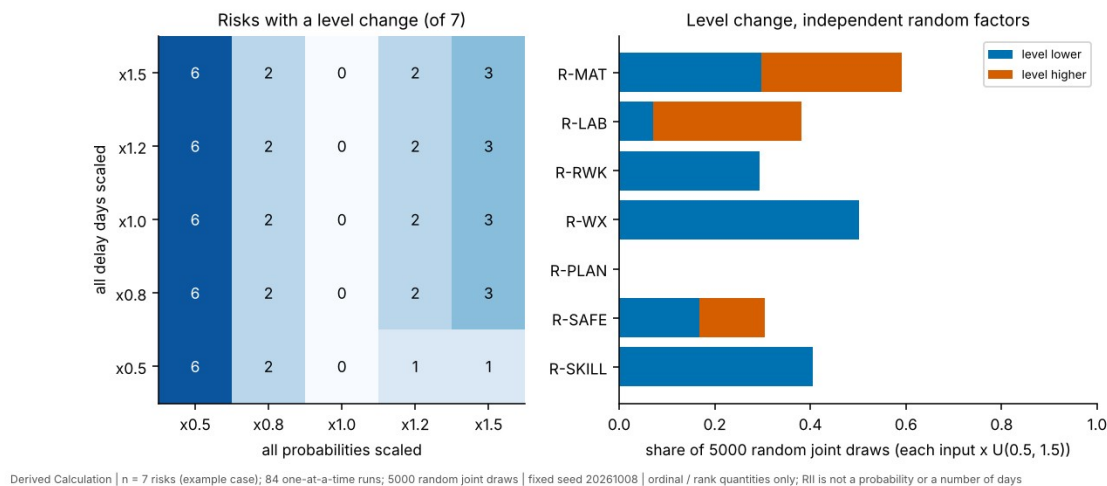


Figure 4.3. Joint sensitivity: risks with a level change on the 5x5 grid (left) and share of random draws with a level change by risk (right) (Derived Calculation from Assumption inputs, ILLUSTRATIVE; n = 7 risks; 25 cells; 5000 draws).

The grid and the draws agree with the single-input results. Scaling all probabilities by 0.5 changes 6 of 7 levels whatever the delay factor; by 0.8, 2; by 1.2, 2 (1 at a delay factor of 0.5); by 1.5, 3 (1 at a delay factor of 0.5). Delay only matters at the margins of the grid: a probability factor of 1.2 or 1.5 together with a delay factor of 0.5 gives 1 changed level instead of 2 or 3, because the shorter delay offsets the higher probability for one risk. In the random draws, R-PLAN never changes level ($p = 0.10$ and a short delay keep it Low), R-MAT changes in 59% of draws, R-WX in 50% (its edge position), and R-LAB and R-SAFE can move up or down.

What this shows is how the rule responds to inputs: the ordinal level is governed by the probability class, and a change of the assumed probabilities by 20 to 50% changes the level of between 2 and 6 of 7 risks. It does not show that any real risk is sensitive to probability, and the example's numbers are placeholders.

4.5 Edge and threshold sensitivity (S2)

S2 varies the classification edges and thresholds on the same example, in 55 named scenarios and 5000 random edge draws. For each scenario it counts how many of the 7 risks change class and how many change level. Both counts are reported, because counting levels alone hides class changes (Table 4.8).

Table 4.8. Level changes under edge and threshold scenarios (Derived Calculation from Assumption inputs, ILLUSTRATIVE; n = 7 risks per scenario; 55 scenarios in 7 groups)

Scenario group (ILLUSTRATIVE example, 7 risks)	Scenarios	Levels changed: min- median-max (of 7)	Max p classes changed	Max impact classes changed
One p edge shifted by - 0.10, -0.05, +0.05 or +0.10	16	0-0-3	3	0
All four p edges shifted together by the same amounts	4	1-2.5-4	4	0
Alternative p-edge sets (4)	4	2-4.5-7	7	0
One impact edge scaled (x0.5 to x2)	16	0-0-0	0	3
All four impact edges scaled (x0.5 to x2)	5	0-0-0	0	7
Alternative impact-edge sets (3)	3	0-0-6	0	7
Alternative level cut-offs (7 named triples)	7	0-0-1	0	0

Source: `analysis/out/s2_summary.json` and `s2_threshold_scenarios.csv`. Median of the single-edge group is 0.

Probability edges. Shifting all four edges by -0.10 changes four levels (R-MAT, R-LAB, R-PLAN, R-SAFE); by +0.10, three (R-RWK, R-WX, R-SKILL). A low-skewed alternative set (0.05, 0.15, 0.35, 0.65) changes all seven. Shifting a single edge matters only when a risk sits near it: the first edge (0.2) at -0.10 flips R-PLAN ($p = 0.10$) and at +0.05 flips R-WX and R-SKILL ($p = 0.20$ and 0.22). The fourth edge (0.8) never matters here because no example risk has p above 0.5.

Impact edges. Scaling the impact edges changes the impact class of up to all seven risks (six at x0.5, seven at x2), but changes no level. Again the probability class caps the score: with probability class 2 or lower, even impact class 5 gives at most 10, Moderate, and probability class 1 gives at most 5, Low. Only one alternative changes levels, equal-width impact edges (0.2, 0.4, 0.6, 0.8), which put six risks in a lower level because every impact class falls to 1 or 2 (the delay ratios, 0.08 to 0.22, mostly lie below the first edge of 0.2: six of the seven are 0.17 or lower and only R-MAT, at 0.22, is above it). The statement “the matrix is robust to the impact edges” would therefore be wrong in general: it holds here because the placeholder probabilities are low.

Level cut-offs. Of seven named triples, two change one level (R-MAT, at 4/8/12 and 3/8/14), and five change none. Over all 2,024 admissible triples with cut-offs between 1 and 24 (S2) the median is 5 of 7 risks changing level (interquartile range 1 to 6). The seeded placements in Section 4.3 are more exposed, with a median of 6 of 30.

Random edges. Jittering each probability edge independently by up to 0.05 changes at least one level in 48% of 5000 draws (mean 0.77 changed levels, maximum 2); jittering impact edges by up to 50% changes none; jittering both gives the same as probability jitter. By risk, the levels that move are R-WX (48% of draws) and R-SKILL (29%), the two risks whose p lies within 0.02 of an edge (Table 4.5, column 3).

S2 Threshold sensitivity by scenario, ILLUSTRATIVE example case
 [Derived Calculation] n = 7 risks (example case); 55 named scenarios; 5000 random edge draws



Derived Calculation | n = 7 risks (example case); 55 named scenarios; 5000 random edge draws | fixed seed 20261008 | ordinal / rank quantities only; RII is not a probability or a number of days

Figure 4.4. Level and class changes across the 55 named edge and threshold scenarios (Derived Calculation from Assumption inputs, ILLUSTRATIVE; n = 7 risks; 55 scenarios).

The result is a description of how fragile the level label is near an edge. The margin table (Table 4.5) shows which placeholders sit close to an edge: R-WX at 0.00 and R-SKILL at 0.02 from a probability edge, the rest at 0.05 to 0.10. A real user's numbers will have their own margins.

4.6 Seed-band sensitivity (S3)

S3 asks how many seeded risks change class when the banding rule or the data are changed. The 14 variants of Table 4.9 were fixed in advance and are all reported. Percentages are of the risks common to the variant and the base rule (30 unless the variant drops a risk). Spearman's rho is between the variant's per-risk means and the base means; Kendall's tau between the classes.

Table 4.9. Class changes under banding and data variants (Derived Calculation from Literature values; n = 30 seeded risks, 5 seed studies; one row per variant)

Variant	Risks	Changed class	% changed	Spearman of mean RII vs base	Kendall tau of classes	Class counts (class:risks)	Dropped	Added
repo rule: 5 equal-count bands (6 per band)	30	0	0.0	1.00	1.00	1:6 2:6 3:6 4:6 5:6	-	-
3 equal-count bands (tertiles), mapped to classes 1/3/5	30	12	40.0	1.00	0.86	1:10 3:10 5:10	-	-
4 equal-count bands, mapped to 1-5	30	9	30.0	1.00	0.90	1:7 2:8 4:7 5:8	-	-
5 equal-width bands on mean RII	30	22	73.3	1.00	0.89	1:2 2:3 3:6 4:6 5:13	-	-
fixed RII cut-offs 0.2/0.4/0.6/0.8	30	24	80.0	1.00	0.23	3:1 4:29	-	-
fixed RII cut-offs 0.60/0.65/0.70/0.75	30	23	76.7	1.00	0.86	1:1 2:1 3:5 4:10 5:13	-	-
direct + related mappings	32	8	26.7	0.96	0.88	1:6 2:6 3:7 4:6 5:7	-	R-CLI R-QLT
>= 2 seed studies per risk (re-banded)	16	6	37.5	1.00	0.91	1:3 2:3 3:3 4:3 5:4	R-CST R-CTR R-ECO R-FAT R-INC R-MOT R-MTH R-SAFE R-SCP R-SOC R-STO R-STR R-SUP R-WAT	-
scale-rescaled (RII-1/k)/(1-1/k)	30	2	6.7	0.99	0.97	1:6 2:6 3:6 4:6 5:6	-	-
within-study percentile rank (studies with >= 3 risks)	30	11	36.7	0.92	0.80	1:6 2:6 3:6 4:6 5:6	-	-
drop seed study A02	27	13	48.1	0.78	0.67	1:5 2:5 3:6 4:5 5:6	R-FAT R-STO R-WAT	-
drop seed study A14	30	2	6.7	0.97	0.97	1:6 2:6 3:6 4:6 5:6	-	-
drop seed study A15	30	3	10.0	0.99	0.92	1:6 2:6 3:6 4:6 5:6	-	-

Variant	Risks	Changed class	% changed	Spearman of mean RII vs base	Kendall tau of classes	Class counts (class:risks)	Dropped	Added
drop seed study M01	25	9	36.0	0.95	0.83	1:5 2:5 3:5 4:5 5:5	R-INC R-MOT R-MTH R-SAFE R-SUP	-
drop seed study M06	24	6	25.0	0.96	0.89	1:4 2:5 3:5 4:5 5:5	R-CST R-CTR R-ECO R-SCP R-SOC R-STR	-

Source: `analysis/out/s3_band_variants.csv`. Variants with fewer risks list the risks dropped or added.

Three groups of variants differ in meaning.

Banding rules change which risks share a class even though the order of the risks is unchanged (Spearman 1.00 in every row). With tertiles mapped to classes 1, 3 and 5, 12 of 30 classes change; with four equal-count bands, 9; with five equal-width bands on mean RII, 22, because 13 risks fall in the top band; with fixed RII cut-offs of 0.2, 0.4, 0.6 and 0.8, 24, because 29 of 30 risks fall in class 4 (all means lie between 0.59 and 0.79). A probability-like fixed scale applied to RII is not informative here. Fixed cut-offs of 0.60, 0.65, 0.70 and 0.75 change 23. These changes are by construction: they show that a class belongs to the rule, not to the risk.

Data variants. Adding **related** mappings (32 risks, adding R-CLI and R-QLT) changes 8 of 30 classes (Spearman 0.96, largest rank shift 10). Restricting to risks with at least two seed studies re-bands 16 risks and changes 6 of them. Rescaling each study's RII as for scale points k changes 2 (Spearman 0.99): mixing 1-4 and 1-5 scales has a small effect on the order. Replacing RII by within-study percentile rank changes 11 of 30 (Spearman 0.92, largest rank shift 8). Switching the repository's seed basis to "all" (held-out studies and related mappings pooled; S0) changes the class of 24 of the 30 common risks.

Dropping a seed study. Dropping A02 changes 13 of the 27 remaining risks (48%; Spearman 0.78, largest rank shift 23) and removes R-FAT, R-STO and R-WAT. Dropping M01 changes 9 of 25, M06 6 of 24, A15 3 of 30 and A14 2 of 30. The class of a seeded risk depends on which studies are present, and most on A02, a study whose N is not reported, whose scale is 1-4 and whose values come from the research notes.

Across all 14 variants between 2 and 24 of the common risks change class (6.7% to 80%); the data variants that keep five bands change between 2 and 13 risks. The order of the means is more stable than the classes: every banding variant keeps Spearman 1.00, and the data variants keep 0.78 to 1.00.

Table 4.10. Gaps in seed mean RII at the four band boundaries of the repo rule (Derived Calculation from Literature values; $n = 30$ risks)

Band boundary (ranks)	Upper risk	Lower risk	Gap in seed mean RII
6-7	R-DIS	R-LAB	0.00383
12-13	R-TOOL	R-INC	0.00075
18-19	R-MGT	R-CST	0.01470
24-25	R-WX	R-COM	0.01408

Source: `analysis/out/s3_band_boundary_gaps.csv`.

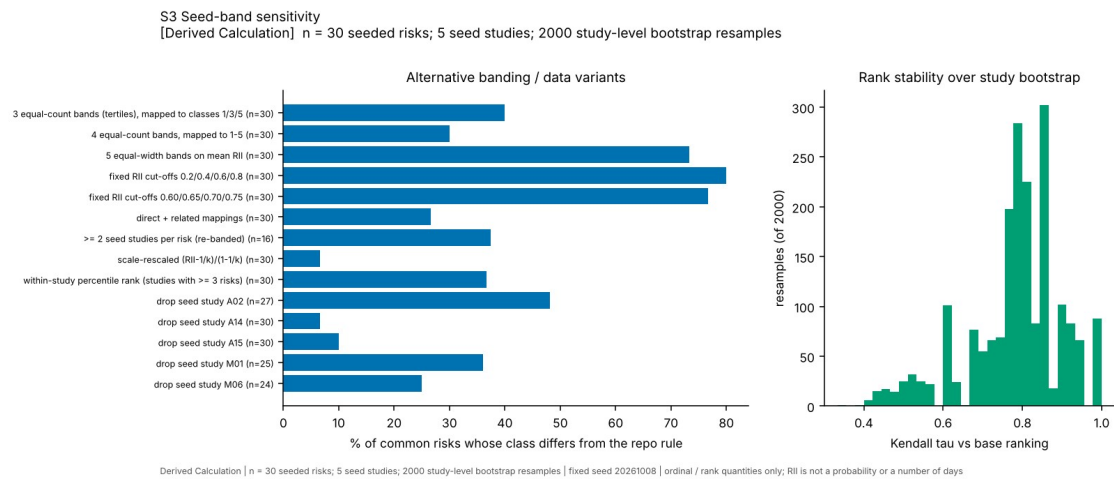


Figure 4.5. Percentage of risks changing class under each variant (left) and Kendall tau of the study-bootstrap ranking against the base ranking (right) (Derived Calculation from Literature values; n = 30 seeded risks; 5 seed studies; 2000 resamples).

The boundary between ranks 6 and 7 (R-DIS to R-LAB) is 0.0038 wide; between 12 and 13 it is 0.00075; the two outer boundaries are about 0.014 wide. Two risks on either side of the narrower boundaries differ by less than the effect of one respondent in the smallest-N seed survey with a printed N.

4.7 Bootstrap and jackknife tiers (S3)

The seed was rebuilt 2000 times from five studies drawn with replacement. The Kendall tau between the resampled mean ordering and the base ordering averages 0.784 (2.5th to 97.5th percentile 0.476 to 1.000) and Spearman's rho 0.884 (0.60 to 1.00). On average 24.5 of the 30 risks are present in a resample (range 14 to 30), and the share of present risks keeping their base class averages 0.664 (0.316 to 1.000). Five risks keep their base class in fewer than half of the resamples where they are present: R-LAB, R-DES, R-RWK, R-INC and R-DIS. Table 4.11 gives every risk, and Figure 4.6 shows the class distribution.

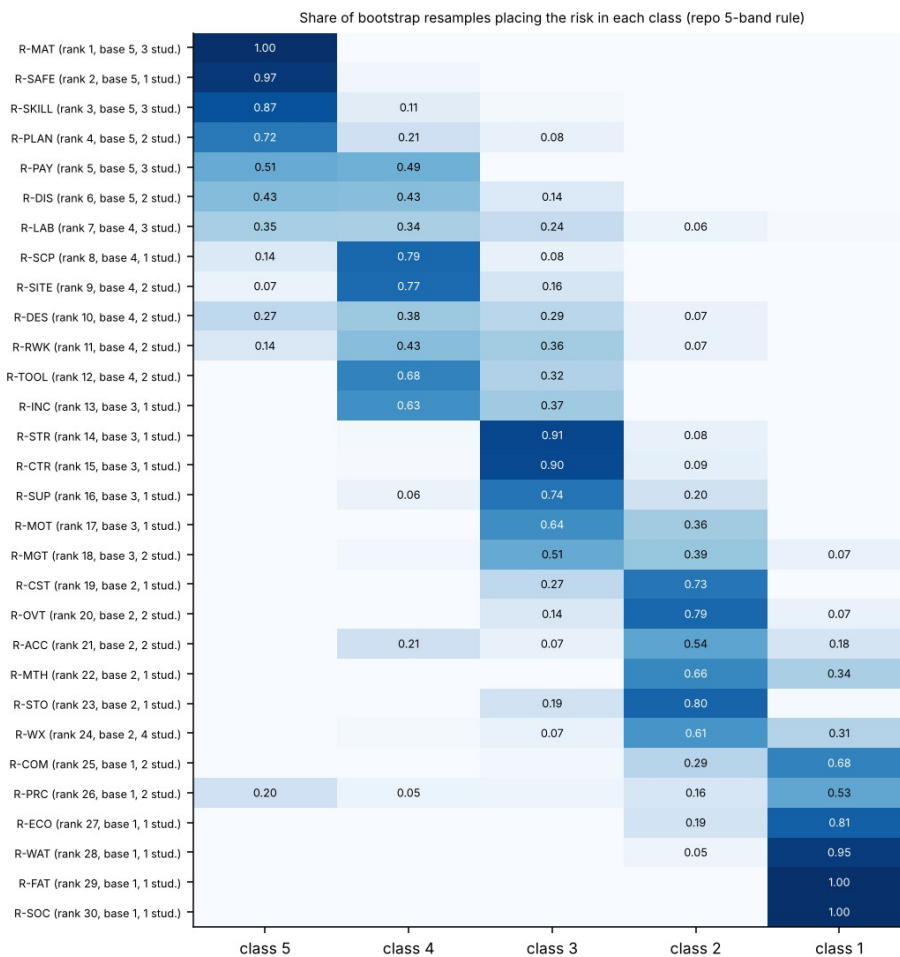
Table 4.11. Class stability of each seeded risk under study resampling (Derived Calculation from Literature values; n = 30 risks, 5 seed studies, 2000 resamples; shares are conditional on the risk being present in the resample)

Rank	Risk	Base class	Seed studies	Present	P(same class)	P(class 5)	P(top 10)	Rank in resamples (2.5%-median-97.5%)
1	R-MAT	5	3	0.99	1.00	1.00	1.00	1-2-4
2	R-SAFE	5	1	0.67	0.97	0.97	1.00	1-2-5
3	R-SKILL	5	3	0.99	0.87	0.87	0.94	1-3-10
4	R-PLAN	5	2	0.91	0.72	0.72	0.89	1-4-11
5	R-PAY	5	3	0.99	0.51	0.51	0.93	4-5-8
6	R-DIS	5	2	0.92	0.43	0.43	0.72	2-6-14
7	R-LAB	4	3	0.99	0.34	0.35	0.64	1-7-19
8	R-SCP	4	1	0.67	0.79	0.14	0.79	6-8-11
9	R-SITE	4	2	0.93	0.77	0.07	0.57	6-8-11
10	R-DES	4	2	0.91	0.38	0.27	0.52	3-10-15
11	R-RWK	4	2	0.91	0.43	0.14	0.44	4-11-14
12	R-TOOL	4	2	0.93	0.68	0.00	0.27	6-10-17
13	R-INC	3	1	0.67	0.37	0.00	0.30	7-10-14
14	R-STR	3	1	0.67	0.91	0.00	0.01	11-13-15
15	R-CTR	3	1	0.67	0.90	0.00	0.00	12-14-16
16	R-SUP	3	1	0.67	0.74	0.00	0.00	10-15-18
17	R-MOT	3	1	0.67	0.64	0.00	0.00	12-17-20
18	R-MGT	3	2	0.91	0.51	0.00	0.00	10-17-20
19	R-CST	2	1	0.67	0.73	0.00	0.00	13-19-21
20	R-OVT	2	2	0.91	0.79	0.00	0.00	15-17-22
21	R-ACC	2	2	0.93	0.54	0.00	0.09	8-19-24

Rank	Risk	Base class	Seed studies	Present	P(same class)	P(class 5)	P(top 10)	Rank in resamples (2.5%-median-97.5%)
22	R-MTH	2	1	0.67	0.66	0.00	0.00	18-22-24
23	R-STO	2	1	0.69	0.80	0.00	0.00	8-20-24
24	R-WX	2	4	1.00	0.61	0.00	0.00	9-19-25
25	R-COM	1	2	0.93	0.68	0.00	0.00	9-21-25
26	R-PRC	1	2	0.93	0.53	0.20	0.25	1-20-30
27	R-ECO	1	1	0.67	0.81	0.00	0.00	16-26-27
28	R-WAT	1	1	0.69	0.95	0.00	0.00	12-23-28
29	R-FAT	1	1	0.69	1.00	0.00	0.00	13-24-29
30	R-SOC	1	1	0.67	1.00	0.00	0.00	18-27-30

Source: `analysis/out/s3_bootstrap_tier_membership.csv`. “Present” is the share of resamples that contain at least one study reporting the risk. “Top 10” is membership of the top third.

S3 Class-membership frequency under study resampling
[Derived Calculation] n = 30 seeded risks; 5 seed studies; 2000 study-level bootstrap resamples



Derived Calculation | n = 30 seeded risks; 5 seed studies; 2000 study-level bootstrap resamples | fixed seed 20261008 | ordinal / rank quantities only; RII is not a probability or a number of days

Figure 4.6. Share of resamples placing each risk in each class (Derived Calculation from Literature values; n = 30 seeded risks; 5 seed studies; 2000 study-level bootstrap resamples).

Reading the table: R-MAT is in class 5 in 99.7% of resamples where present and in rank 1 to 4 in 95% of them, and rests on three studies. R-SAFE is class 5 in 96.8% of resamples but is present in only about 67% of them, because it rests on M01 alone. R-LAB, class 4 at rank 7, is class 5 in 34.5% of resamples, class 4 in 33.9% and ranges over ranks 1 to 19. R-PRC (two studies, 0.55 in A02 and 0.789 in A14) is class 5 in 20% of resamples and class 1 in 53%.

Because five clusters are too few for a reliable percentile interval, the exact leave-one-study-out (jackknife) results are given beside it.

Table 4.12. Exact leave-one-study-out rank stability of the seed (Derived Calculation from Literature values; n = 30 risks; five rebuilds)

Seed study dropped	Risks still seeded	Spearman vs full seed order	Kendall tau	Risks changing class (of 30)
A02	27	0.78	0.77	13
A14	30	0.97	0.91	2
A15	30	0.99	0.96	3
M01	25	0.95	0.84	9
M06	24	0.96	0.87	6

Source: `analysis/out/s3_leave_one_study_out_rank_stability.csv`.

Applying the tier rule of Section 3.8 (class 5 or class 1 to 2 in at least 80% of present resamples, and at least two seed studies) gives Table 4.13, which lists the risks that are not “unresolved”.

Table 4.13. Tier assignments under the 80% rule, risks that meet a threshold (Derived Calculation from Literature values; n = 30 risks, 2000 resamples; the rule is an Assumption)

Risk	Base rank	Seed studies (n)	Present in resamples	P(class 5), when present	P(class 1-2), when present	Tier by rule
R-MAT	1	3	0.993	0.997	0.000	Tier 1 (high)
R-SAFE	2	1	0.665	0.968	0.000	high-class but single study (not assessable)
R-SKILL	3	3	0.988	0.868	0.000	Tier 1 (high)
R-OVT	20	2	0.912	0.000	0.856	Tier lower
R-MTH	22	1	0.665	0.000	1.000	low-class but single study (not assessable)
R-STO	23	1	0.685	0.000	0.812	low-class but single study (not assessable)
R-WX	24	4	1.000	0.000	0.916	Tier lower
R-COM	25	2	0.932	0.000	0.976	Tier lower
R-ECO	27	1	0.668	0.000	1.000	low-class but single study (not assessable)
R-WAT	28	1	0.685	0.000	1.000	low-class but single study (not assessable)
R-FAT	29	1	0.685	0.000	1.000	low-class but single study (not assessable)
R-SOC	30	1	0.668	0.000	1.000	low-class but single study (not assessable)

Source: `ch34_build.py` applied to `s3_bootstrap_tier_membership.csv`.

Only two risks, R-MAT and R-SKILL, meet the rule for the top tier with at least two studies. R-SAFE meets the 80% threshold but rests on one study and so cannot be assessed. Three risks (R-OVT, R-WX, R-COM) meet the rule for the lower tier, and six single-study risks (R-MTH, R-STO, R-ECO, R-WAT, R-FAT, R-SOC) are low but not assessable. The remaining 18 are unresolved. Two cautions apply to the lower tier. R-WX is in class 1 or 2 in 92% of resamples, yet its four seed studies range from 0.653 to 0.764 (M01 ranks “weather conditions” 8th of 28 printed items, A14 30th, M06 37th). The tier states where the mean of the studies falls in the order of the 30 risks, not that weather is unimportant, and the earlier claim that weather is rated lower than

the rest was withdrawn (Audit Corrections, F09). Second, the rule ignores mapping uncertainty and scale differences, so it understates uncertainty.

The data therefore support, at most, coarse tiers: two risks in a high tier, a few in a low tier, and an unresolved middle in which the order is not determined by these studies.

4.8 Non-circular inter-study agreement (S4)

S4 compares studies without letting a study into the reference it is compared with. For a seed study X the reference is the seed built from the other four seed studies; for a held-out study it is the full five-study seed. A correlation needs at least five shared risks to be tested at all.

Coverage. Little overlap exists (`analysis/out/s4_study_coverage.csv`). Of 55 study pairs, 13 share five or more risks; the other 42 share four or fewer (13 pairs share none, 15 share one, 9 share two, 2 share three, 3 share four). The studies contribute these numbers of risks with a direct mapping (items in brackets): M01 18 (25), A02 13 (13), M06 15 (20), A14 4 (4), A15 2 (4), M10 23 (27), M15 13 (18), M14 15 (18), A10 2 (3), A21 2 (2) and HAS25 1 (1).

Pairwise agreement. Table 4.14 gives the 13 testable pairs with n , ρ , the two-sided permutation p and the Holm-adjusted p over the 13 tests.

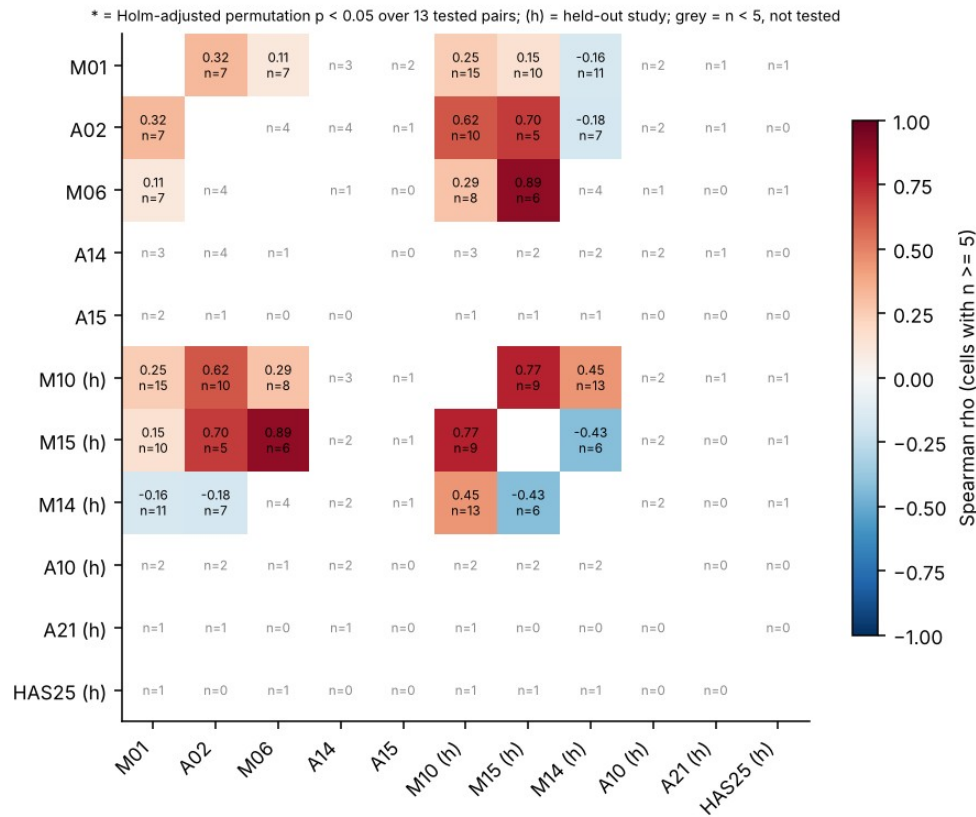
Table 4.14. Spearman agreement between studies on shared risks, pairs with $n \geq 5$ (Derived Calculation from Literature values; $n = 5$ to 15 shared risks per pair; 13 tests; exact p for $n \leq 8$, otherwise 20,000 permutations)

Pair	Roles	n shared risks	Spearman ρ	Two-sided p	p method	Holm p (13 tests)
M01 - A02	seed / seed	7	0.32	0.482	exact	1.000
M01 - M06	seed / seed	7	0.11	0.840	exact	1.000
M01 - M10	seed / held-out	15	0.25	0.383	20000 permutations	1.000
M01 - M15	seed / held-out	10	0.15	0.679	20000 permutations	1.000
M01 - M14	seed / held-out	11	-0.16	0.634	20000 permutations	1.000
A02 - M10	seed / held-out	10	0.62	0.063	20000 permutations	0.694
A02 - M15	seed / held-out	5	0.70	0.233	exact	1.000
A02 - M14	seed / held-out	7	-0.18	0.713	exact	1.000
M06 - M10	seed / held-out	8	0.29	0.501	exact	1.000
M06 - M15	seed / held-out	6	0.89	0.033	exact	0.400
M10 - M15	held-out / held-out	9	0.77	0.021	20000 permutations	0.270
M10 - M14	held-out / held-out	13	0.45	0.129	20000 permutations	1.000
M15 - M14	held-out / held-out	6	-0.43	0.419	exact	1.000

Source: `analysis/out/s4_pairwise_spearman.csv`. Values within ± 0.003 for permutation p .

The median ρ over the 13 tested pairs is 0.29 (range -0.43 to 0.89). Only two pairs have a raw p below 0.05, M10 with M15 (ρ 0.77, $n = 9$, $p = 0.021$) and M06 with M15 (ρ 0.89, $n = 6$, $p = 0.033$); after Holm the smallest adjusted p is 0.270 and none is significant. The two seed-seed pairs that can be tested are weak: M01 with A02 0.32 ($n = 7$) and M01 with M06 0.11 ($n = 7$). Of the untestable seed pairs, A02 with M06 and A02 with A14 show 0.74 and 0.60 on four shared risks each, which a test could not distinguish from chance, and M01 with A14 shows -0.50 on three. The M15-M14 pair is negative (-0.43, $n = 6$), consistent with the manager-versus-worker explanation that has been offered post hoc, which this data set cannot test.

S4 Pairwise study agreement on shared risks
[Derived Calculation] n = 11 studies, 55 pairs; n per cell = shared risks



Derived Calculation | n = 11 studies, 55 pairs; n per cell = shared risks | fixed seed 20261008 | ordinal / rank quantities only; RII is not a probability or a number of days

Figure 4.7. Pairwise Spearman agreement (cell text: rho and number of shared risks; grey = n below 5, not tested) (Derived Calculation from Literature values; n = 11 studies, 55 pairs; 13 tested).

Study against a seed built without it. Table 4.15 compares each study with a seed that does not contain it. The last two columns show what would have been reported had the study been left inside its own seed (circular): 0.54, 0.91 and 0.87 for M01, A02 and M06, against 0.09, 0.37 and 0.64 without them. The inflation is why the earlier “leave-one-study-out” figures of 0.78 to 0.99 were withdrawn.

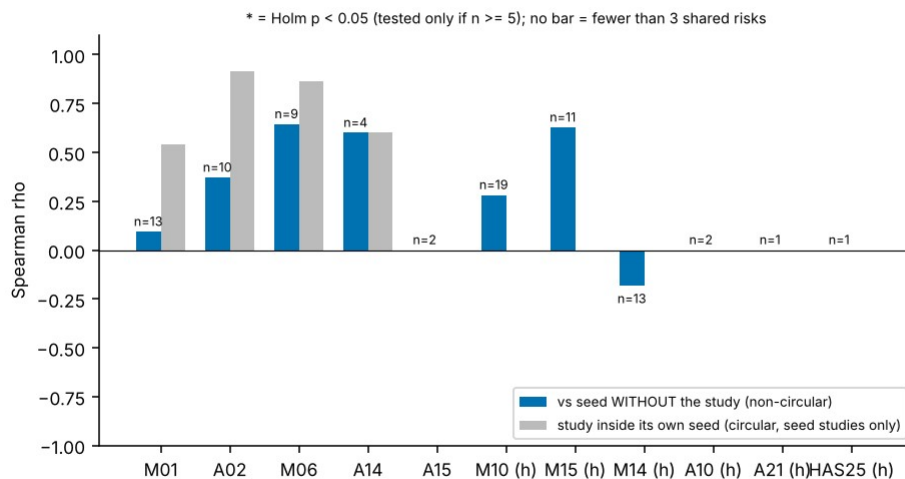
Table 4.15. Agreement of each study with a seed built without it (Derived Calculation from Literature values; n = 1 to 19 shared risks per study; 6 tests in the Holm family)

Study X	Role	n shared risks (X vs seed built from the other 4)	Spearman rho	Two-sided p	Holm p	n (circular)	rho if X is inside its own seed (circular, shown only as a warning)
M01	seed	13	0.09	0.765	1.000	18	0.54
A02	seed	10	0.37	0.290	0.959	13	0.91
M06	seed	9	0.64	0.066	0.329	15	0.87
A14	seed	4	0.60	not tested	-	4	0.60
A15	seed	2		not tested	-	2	-
M10	held-out	19	0.28	0.240	0.959	19	-
M15	held-out	11	0.63	0.045	0.267	11	-
M14	held-out	13	-0.18	0.549	1.000	13	-
A10	held-out	2		not tested	-	2	-
A21	held-out	1		not tested	-	1	-
HAS25	held-out	1		not tested	-	1	-

Source: `analysis/out/s4_leave_one_study_out.csv`. For held-out studies the reference is the five-study seed, which does not contain them, so there is no circular column.

None of the six testable comparisons is significant after Holm correction over the six (smallest adjusted p 0.267, M15). M15's raw permutation p is 0.045 (the t-approximation in earlier documents gave 0.039); M06 is 0.066. Restricting the family to the three held-out studies, as the handoff did, the adjusted values computed from the printed raw p-values are 0.134 (M15), 0.479 (M10) and 0.549 (M14), still none below 0.05. A15 (n = 2), A10 (2), A21 (1) and HAS25 (1) share too few risks to test. The agreement of the three held-out surveys with the seed is therefore rho 0.28, 0.63 and -0.18: weak, low-powered and inconclusive. M14 is negative, and its basis, W/(5N), is not comparable in level with the others.

S4 Leave-one-study-out rank agreement
[Derived Calculation] n = 11 studies; n per bar = risks shared with the comparison seed



Derived Calculation | n = 11 studies; n per bar = risks shared with the comparison seed | fixed seed 20261008 | ordinal / rank quantities only; RII is not a probability or a number of days

Figure 4.8. Agreement with the seed built without the study (blue) and the circular value (grey) (Derived Calculation from Literature values; n = 11 studies; the n above each bar is the number of shared risks).

Detectable agreement. With 5 to 19 shared risks only large correlations could have been detected. The smallest absolute rho that reaches a two-sided permutation p of 0.05 is 0.52 at n = 15, 0.57 at 13, 0.62 at 11, 0.65 at 10, 0.70 at 9, 0.74 at 8, 0.79 at 7, 0.89 at 6 and only a perfect 1.00 at 5, and 0.46 at 19 (Table 3.8). Most overlaps in Table 4.14 are at or below 10, so only a rho of about 0.65 or more could have been called significant for a typical pair. The check is low-powered, and a non-significant result is neither evidence for nor against agreement.

Kendall W. Table 4.16 gives W for every complete block of at least three studies and at least five risks, 14 blocks in all.

Table 4.16. Kendall W over complete study-by-risk blocks (Derived Calculation from Literature values; m = 3 or 4 studies, n = 5 to 10 risks per block; 14 blocks; one-sided permutation p, 20,000 permutations; Holm over the 14 blocks, descriptive because blocks overlap)

Studies (m)	Complete risks (n)	Kendall W	Permutation p	Holm p (14 blocks)
M01 M10 M14	10	0.488	0.126	1.000
M01 M10 M15	9	0.539	0.079	0.871
A02 M10 M14	7	0.429	0.278	1.000
M01 A02 M10	7	0.733	0.010	0.122
M01 M06 M10	6	0.467	0.231	1.000
M01 M15 M14	6	0.289	0.571	1.000
M10 M15 M14	6	0.441	0.275	1.000
A02 M10 M15	5	0.889	0.004	0.059
M01 A02 M14	5	0.200	0.717	1.000
M01 A02 M15	5	0.638	0.083	0.871
M01 M06 M15	5	0.467	0.252	1.000
M01 M10 M15 M14	6	0.400	0.153	1.000

Studies (m)	Complete risks (n)	Kendall W	Permutation p	Holm p (14 blocks)
M01 A02 M10 M14	5	0.263	0.411	1.000
M01 A02 M10 M15	5	0.699	0.009	0.112

Source: `analysis/out/s4_kendall_w.csv`.

The block that the earlier analysis reported, M01, M10, M15 and M14 on six risks, gives $W = 0.400$ with permutation $p = 0.153$, the same conclusion as the chi-square p of 0.156: not distinguishable from chance. Two blocks that include A02 and M10 reach raw permutation p of 0.004 (A02, M10, M15; five risks; $W = 0.889$) and 0.009 (M01, A02, M10, M15; five risks; $W = 0.699$), but their Holm-adjusted p -values are 0.059 and 0.112, and the highest W values come from the blocks with the fewest risks, where a large W is easiest to obtain by chance and the 14 blocks use the same studies repeatedly. No block is significant after correction. W describes consensus on order among the studies in the block; it is not reliability.

Summary of agreement. The non-circular evidence is: agreement between studies on shared risks is weak (median ρ 0.29), the raw p -values below 0.05 (two pairs, M10 with M15 and M06 with M15, and M15 against the seed) do not survive correction, and the held-out comparison is weak and inconclusive. The agreement that the earlier, circular calculations suggested (pooled rank against the seed 0.73; z order against the seed 0.40 and 0.53) does not appear when the seed studies are excluded from the reference: the held-out-only values are 0.08 and 0.09 (Handoff Section 8.3, quoted, not recomputed here).

4.9 RII uncertainty (S5)

S5 compares the spread of RII for the same risk across studies with the gaps between adjacent ranks. The spread is between-study, on unequal scales; it is not the sampling error of any survey.

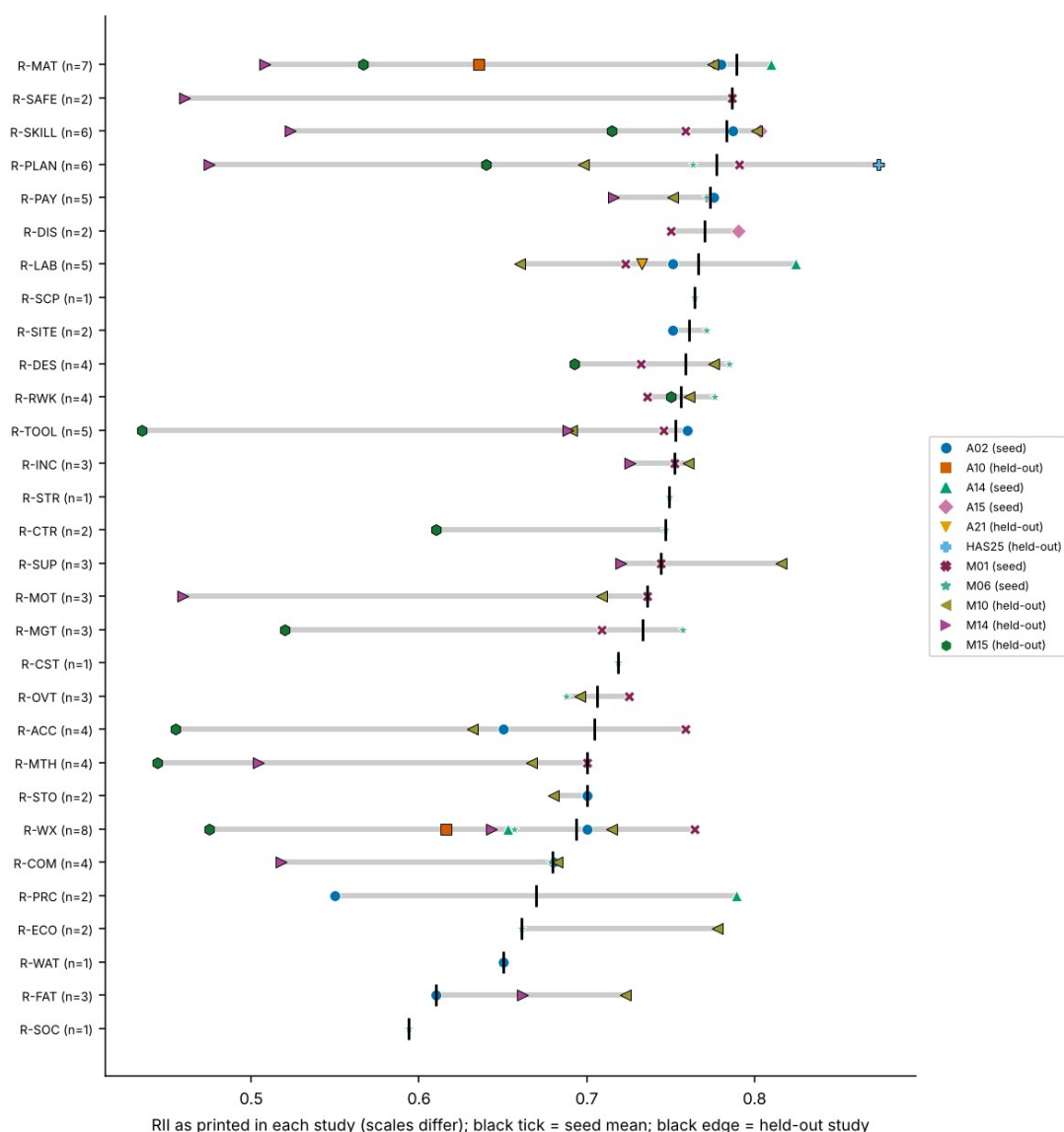
Table 4.17. Spread of RII across studies, seeded risks with at least two seed studies (Derived Calculation from Literature values; $n = 16$ risks of 30; RII as printed, scales differ)

Seed rank	Risk	Seed studies	Seed mean RII	Range of seed study means	All studies reporting	Min RII (all)	Max RII (all)	Range (all)
1	R-MAT	3	0.7889	0.0328	7	0.508	0.810	0.302
3	R-SKILL	3	0.7831	0.0443	6	0.523	0.803	0.280
4	R-PLAN	2	0.7771	0.0278	6	0.475	0.874	0.399
5	R-PAY	3	0.7733	0.0047	5	0.716	0.776	0.060
6	R-DIS	2	0.7700	0.0400	2	0.750	0.790	0.040
7	R-LAB	3	0.7662	0.1015	5	0.660	0.825	0.165
9	R-SITE	2	0.7611	0.0201	2	0.751	0.771	0.020
10	R-DES	2	0.7585	0.0531	4	0.693	0.785	0.093
11	R-RWK	2	0.7561	0.0401	4	0.736	0.776	0.040
12	R-TOOL	2	0.7528	0.0145	5	0.435	0.760	0.325
18	R-MGT	2	0.7331	0.0482	3	0.520	0.757	0.237
20	R-OVT	2	0.7063	0.0374	3	0.688	0.725	0.037
21	R-ACC	2	0.7045	0.1090	4	0.455	0.759	0.304
24	R-WX	4	0.6934	0.1112	8	0.475	0.764	0.289
25	R-COM	2	0.6793	0.0014	4	0.518	0.682	0.164
26	R-PRC	2	0.6695	0.2390	2	0.550	0.789	0.239

Source: `analysis/out/s5_rii_spread_by_risk.csv`.

The median gap between adjacent seed ranks is 0.0034. For the 16 risks with two or more seed studies the median range of the study means is 0.040, and 15 of the 16 (94%) have a range larger than the median adjacent gap. The widest seed ranges are R-PRC 0.239 (A02 0.55, A14 0.789), R-WX 0.111 (four studies), R-ACC 0.109, R-LAB 0.101 and R-DES 0.053. Using all 11 studies, the median between-study standard deviation over the 26 risks reported by at least two studies is 0.082. The order inside the middle of the list, where the gaps are thousandths, is therefore finer than the studies disagree with each other.

S5 Between-study spread of RII per risk (rows ordered by seed rank)
 [Derived Calculation] n = 30 seeded risks, 11 studies; n per row = studies reporting the risk



Derived Calculation | n = 30 seeded risks, 11 studies; n per row = studies reporting the risk | fixed seed 20261008 | ordinal / rank quantities only; RII is not a probability or a number of days

Figure 4.9. RII of each seeded risk in every study that reports it (marker per study, black tick = seed mean; scales differ)
 (Derived Calculation from Literature values; n = 30 seeded risks, 11 studies).

Sampling error within a study is a separate and smaller component. The audit estimated about 0.03 for one M01 RII from that study's own counts, and the planning arithmetic of Table 3.7 gives 0.030 at N = 44 for an SD of one Likert point (Assumption). Against a median adjacent gap of 0.0034, that is roughly nine times larger. No standard error was computed for A02 and A15 (no counts or SDs) or for the other studies in this report; a within-study error draw (analysis A5, part ii) was planned but is not implemented in the analysis package, so the tiers above carry between-study uncertainty only.

4.10 Summary of results

Table 4.18. What each analysis shows, in the words the evidence allows (Derived Calculation; n as in each section)

Section	Result	Wording to use
4.2	30 seeded risks, 14 on one study, 16 unseeded; top six within 0.019	the seed is an ordinal starting position
4.3	12 Low, 6 Moderate, 0 High, 12 Extreme with the same class on both axes; median 6 of 30 levels change over 969 cut-off triples	level labels of seeded risks are artefacts of the rule and thresholds
4.4	level driven by p, not by delay; p x0.5 changes 6 of 7 levels	shows the rule's response, not site behaviour
4.5	impact edges change 0 levels but up to 7 classes; p edges change up to 7 levels	level labels depend on assumed edges
4.6	2 to 24 of 30 classes change across variants; order of means more stable (Spearman 0.78 to 1.00)	the class depends on banding and on which studies are present
4.7	tau 0.78 under resampling; only R-MAT and R-SKILL are Tier 1	at most coarse tiers; middle unresolved
4.8	median pairwise rho 0.29; none significant after Holm; held-out 0.28, 0.63, -0.18; W 0.40, p 0.15	weak, low-powered, inconclusive
4.9	between-study range exceeds the adjacent gap for 15 of 16 risks	order inside the middle is finer than the data

4.11 What this analysis does not show

- It does not validate the seed or show that it predicts anything. RII is importance, not probability or days, and no site data were used.
- It does not show that the seed order is wrong. The agreement check is low-powered; a non-significant correlation on 5 to 19 shared risks is compatible with real agreement and with none.
- It does not give independent evidence that the seed is corroborated. The only independent comparisons are the held-out studies (rho 0.28, 0.63, -0.18) and none survives correction.
- It does not show what drives delay in masonry on Indian sites. Ten of the eleven studies are not masonry-specific, the seed studies are general building or construction surveys, and one held-out study (A10) is infrastructure.
- It does not show that the matrix level of any real risk is stable or unstable. S1 and S2 use one illustrative case of seven risks with placeholder numbers; their counts describe that case and are not a sample.
- It does not make the bootstrap tiers confidence statements. Five clusters, uneven coverage (14 risks on one study), mapping uncertainty ignored and scale differences ignored mean the tiers understate uncertainty.
- It does not measure within-study sampling error beyond the M01 estimate; the third source of error, the single-coder mapping, has not been measured.
- It does not support the withdrawn claims: that weather is rated lower than the rest, that studies agree modestly, or that the pooled order agrees with the seed.
- It is a second look at data first examined on ten risks, so it is exploratory; the 38-risk library was defined after the first look.
- The planned expert survey, second-coder check and site data are the next evidence. Until they exist, the seed is a transparent default ordering and nothing more.

Chapter 5. System Design and Implementation

This chapter describes the tool as it exists in the repository on 8 October 2026. Everything stated here can be checked against a file in the repository; where a statement rests on a test, the test is named. Section 5.1 sets out the requirements the design answers, Sections 5.2 to 5.4 the architecture, data model and risk library, Sections

5.5 to 5.7 the rule engine, the register and matrix logic and the literature-tier display, Sections 5.8 and 5.9 the AI evidence layer and the exports, Section 5.10 the interface, Section 5.11 the desktop wrapper and the build, and Section 5.12 the optional decision layer. Chapter 6 reports how the tool was verified.

Two points frame the chapter. First, the tool does not predict delay. It organises delay risks for one activity (brick or block masonry), places them on a probability-impact matrix as an ordinal aid, and keeps every number it shows tied to a Source label. Second, no site data and no expert survey were collected for this project, so every probability, delay, cost and planned duration shown in the screenshots below is an ILLUSTRATIVE placeholder labelled Assumption. The scope change from the original project title is explained in Chapter 1.

5.1 Requirements

5.1.1 How the requirements were obtained

The requirements below were reconstructed from the project’s own design documents (`README.md`, `docs/Project_Handoff_Summary.md`, `docs/case_schema.md`), from the integrity rules the team adopted (Handoff section 3), and from four review passes carried out on 8 October 2026 (engineering and usability, domain, research methods, examiner). They were not elicited from site managers. No requirements interview, site visit or usability session was held, and the requirements therefore express what the team decided the tool should do, not what practitioners asked for. This is a limitation of the work and is repeated in Chapter 6.

5.1.2 Functional requirements

Table 5.1. Functional requirements and where each is implemented. Source: repository files named in the last column.

ID	Requirement	Implemented in
FR1	Describe one masonry activity: project details, activity name, planned duration in working days with a Source	<code>gui/static/js/screens/case.js</code> ; <code>app/service.py</code> (<code>parse_case</code>)
FR2	Accept site facts (Yes / No / Unknown, or a number) and evaluate rules on them; a missing fact must never be guessed	<code>app/engine/rules.py</code> ; <code>gui/static/js/screens/rules.js</code>
FR3	Offer a curated library of masonry delay risks, each with the evidence record IDs that show the risk exists	<code>data/risk_library_masonry.json</code> ; register screen side panel
FR4	Keep a risk register: probability and a delay range per risk, each with a Source, plus status and text-only owner, early-warning sign, response type, action and review date	<code>app/service.py</code> ; <code>gui/static/js/screens/risks.js</code>
FR5	Place risks that have entered numbers on a 5 x 5 probability-impact matrix and state the level	<code>app/engine/matrix.py</code> ; <code>gui/static/js/screens/matrix.js</code>
FR6	Place risks without entered numbers at a labelled literature tier, never as a probability	<code>app/literature_seed.py</code> ; <code>app/service.py</code> (<code>run_case</code>)
FR7	Search a cited evidence corpus and show the record behind each citation	<code>app/ai_layer/evidence.py</code> , <code>corpus.py</code> ; evidence screen
FR8	Propose candidate risks from the corpus (offline retrieval, optionally an LLM) without any number coming from the model	<code>app/ai_layer/guard.py</code> ; <code>gui/api.py</code> (<code>/api/suggest-risks</code>)
FR9	Export the register and matrix as <code>register.md</code> , <code>register.csv</code> , <code>report.html</code> , <code>matrix.svg</code> , <code>matrix.png</code> , and save and reopen the case as JSON	<code>app/reporting/*</code> ; <code>gui/api.py</code>
FR10	Reject wrong input with a plain message and keep the last valid result on screen	<code>app/service.py</code> ; <code>gui/static/js/problems.js</code>
FR11	Optional: compare responses by simulated schedule and cost using only user-entered numbers	<code>app/analysis.py</code> , <code>app/engine/*</code>
FR12	Run offline on a site laptop, as a browser page or a desktop window, with a Windows build	<code>gui/_main_.py</code> , <code>gui/qt_app.py</code> , <code>.github/workflows/build-exe.yml</code>

5.1.3 Non-functional requirements and integrity constraints

Table 5.2. Non-functional requirements and constraints. Source:

docs/Project_Handoff_Summary.md sections 3 and 9; gui/README_GUI.md. RG-n refers to the relevance guard test `test_gn_*` in `tests/test_relevance_guard.py` (Section 6.3).

ID	Requirement	How it is met
NFR1	Every number carries a Source label (Literature, Historical Data, User Input, Expert Judgment, Derived Calculation, Assumption)	Source enumeration in <code>app/engine/models.py</code> ; a number without a Source is a problem; guard test RG-5 checks that every number in the exports has its Source
NFR2	A survey importance index (RII) is never presented as a probability or as days	literature tier labelled Assumption; excluded from level totals; no p or delay shown for it
NFR3	AI never supplies a probability, delay, cost or effect size	guard rules G2, G7, G9 (Section 5.8); guard test RG-10
NFR4	Deterministic and reproducible: same case, same output	determinism checked in the QA run (Section 6.7); Monte Carlo uses a fixed seed
NFR5	Offline for the core path; nothing leaves the machine unless the user adds an AI key	server binds 127.0.0.1 only; host-header check; key stored outside the project folder
NFR6	Calm failure: no stack trace to the user, no silent defaults	CaseError problems list; JSON error handlers in <code>gui/api.py</code>
NFR7	Readable at 1280 px and 800 px, light and dark theme, no horizontal page scroll	checked by screenshots on 8 October 2026 (Handoff section 9, item 7)
NFR8	Wording stays within what the work supports (no “validated”, “predicts”, “optimal”)	guard tests RG-4 and the analysis test on optimality wording

5.2 Architecture

5.2.1 Layers

The tool is a single Python package with a small local web interface. Figure 5.1 shows the layers. The interface (HTML, CSS and JavaScript in `gui/static`) is served by a Flask application (`gui/api.py`) bound to `127.0.0.1`. The interface holds the case in the browser, and on every edit it posts the case to the server, which returns the register and the matrix. The interface computes nothing of its own: the service function `app.service.run_case` produces every figure on screen (`gui/README_GUI.md`).

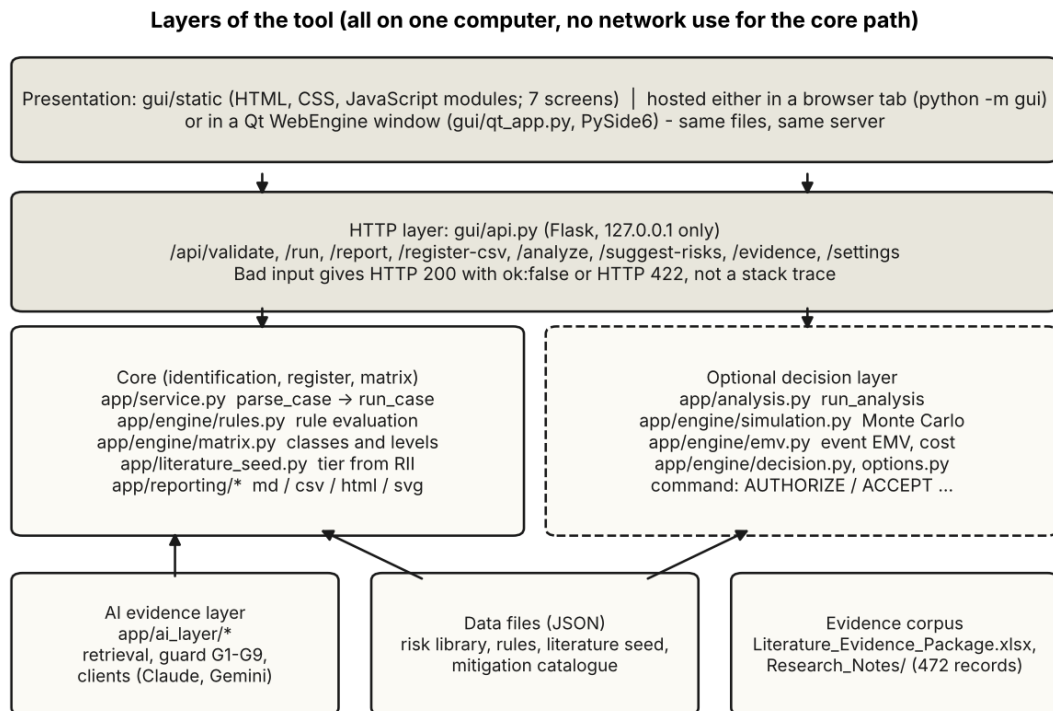


Figure 5.1. Layers of the tool. Dashed box: optional decision layer. Drawn by `report/scripts/make_ch5_diagrams.py` from the module structure of the repository.

This arrangement was chosen so that the same code is exercised by the command-line tool (`python -m app.cli`), the browser version, the desktop window and the tests. It keeps the engine testable without a browser and the interface free of numeric logic that could disagree with the engine.

Table 5.3. Size of the code base, counted with `wc -l` on 8 October 2026 over `.py`, `.js`, `.css` and `.html` files (pycache excluded). Source: command run for this report.

Part	Lines
app/ (engine, service, AI layer, reporting, decision layer, legacy productivity models)	5,060
of which app/productivity/ (legacy, not used by the interface)	420
gui/ Python (server, entry point, Qt wrapper)	761
gui/static/ (interface: HTML, CSS, JavaScript)	3,928
tests/	3,297

Line counts say nothing about quality; they are given so the reader can judge the size of what was verified.

5.2.2 Data flow of one edit

Figure 5.2 follows one edit through the service. `parse_case` checks types and the presence of a Source on every number and builds a list of problems. If there are problems, the run stops and the interface keeps the last valid result on screen, marked as stale. Otherwise the rule engine evaluates the site facts, the matrix classifies every risk that has entered numbers, and the literature seed places the remaining library risks that have a seed value. The result object carries the register rows, the matrix rows, the summary counts, the matrix settings with their Source, the rule outcomes, the warnings, and the rendered Markdown and SVG.

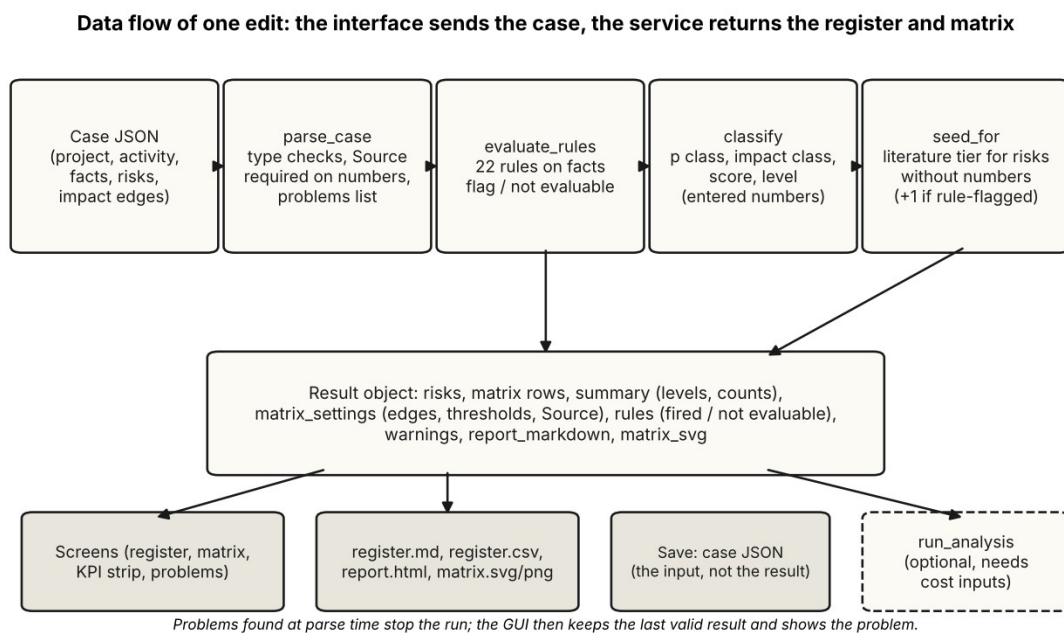


Figure 5.2. Data flow of `run_case (app/service.py)`. Dashed box: the optional decision layer takes the same case plus cost inputs.

5.2.3 Module map

Table 5.4. Main modules. Source: repository tree.

Module	Role
app/service.py	Parsing and validation of the case, <code>run_case</code> , example and template cases, evidence lookup, suggestion entry point
app/engine/models.py	Source, Param, DelayDist (PERT, triangular, uniform, fixed), Risk, Mitigation, CostModel
app/engine/rules.py	Rule objects, typed evaluation, rule-base consistency check, priority and conflict resolution
app/engine/matrix.py	Probability class, impact class, score and level
app/literature_seed.py	Literature tier from the seed file
app/ai_layer/	Evidence store, retrieval, guard, LLM clients, settings store
app/reporting/	register.md, register.csv, report.html, matrix.svg writers

Module	Role
app/analysis.py, app/engine/{simulation,emv,decision,options}.py	Optional decision layer
gui/api.py, gui/__main__.py, gui/qt_app.py	Local server, entry points, desktop wrapper
data/*.json	Risk library, rules, literature seed, mitigation catalogue
app/productivity/, data/productivity_models.json, data/india_labour_norms.json	Earlier labour-productivity models; out of scope, not used by the interface (guard test RG-6)

The productivity models belong to the original project scope. They remain in the repository because their removal needs a supervisor decision (Handoff section 9), and a guard test confirms that neither the live application nor the interface loads them.

5.3 Data model: the case file

A case is one JSON object (`docs/case_schema.md`). It is the only input to the engine, and it is what the interface saves and opens. Every numeric input other than a plain count is an object `{"value", "source", "evidence_ids", "note"}`, and time is in working days.

Table 5.5. Blocks of the case file. Source: `docs/case_schema.md`; `app/service.py` (`template_case`).

Block	Content	Notes
project	name, location, construction type, notes	descriptive only
activity	id, name, planned_duration_days {value, source}; optional deadline_days	impact class is measured against the planned duration
facts	key-value pairs: numbers, true/false	inputs to the 22 rules; planned_duration_days is added automatically
risks[]	id, name, category, status, p {value, source}, delay {kind, a, m, b, source}, evidence IDs; text-only owner, trigger, response type, action, review date	p is the probability that the risk occurs at least once during the activity; delay is the extra delay in days if it occurs
impact_bin_edges_fraction, impact_bin_edges_source	four ascending fractions of the planned duration, with a Source	user input, no default
use_literature_seed, seed_basis	switch for the literature tier; seed (default) or all	all also uses held-out studies, so no held-out comparison applies to it
cost, mitigations[], options[], constraints, simulation	optional decision layer	Section 5.12

Three design choices in the data model are worth stating.

1. **Source is part of the number.** A probability without a Source is rejected as a problem rather than defaulted, and a delay distribution carries its own Source. The six values of **Source** are those listed in the project's integrity rules.
2. **Delay is conditional.** The delay is the extra delay in working days given that the risk occurs, and p is the probability of occurrence during the activity. This is the same Bernoulli-times-delay structure used for risk events in the schedule-risk literature recorded in the research notes (for example the Beta-PERT times Bernoulli construction of Prins et al., 2022, a preprint that was not peer reviewed). The structure is a modelling choice of this tool, stated as such.
3. **Wrong is not the same as missing.** A value that is present but invalid (p outside 0 to 1, minimum above maximum, a number with no Source, a boolean or non-finite value where a number is needed) is a problem and the register keeps its last valid state. A value that is simply not entered leaves the risk in the register without a placement or with a literature tier.

A blank template is produced by `python -m app.cli template` and the ILLUSTRATIVE example by `python -m app.cli example`. The example has seven risks; Section 5.10 shows it in the interface and Chapter 6 recomputes it by hand.

5.4 The risk library

The library (`data/risk_library_masonry.json`) holds 46 risks. Thirty-eight were built on 7 October 2026 from the factors transcribed from the eleven studies of the literature seed, and eight were added on 8 October 2026 after the domain review (R-FRONT, R-VT, R-MORT, R-GPAY, R-OPEN, R-SCAF, R-FEST, R-HEAT). Each entry has an identifier, a name, a category, a description, the evidence IDs that show the

risk exists, an evidence note, and a scope (**activity** for 38, **project** for 8). Entries without a survey value carry a **seed_status** note.

Table 5.6. Library risks by category and by whether the literature seed has a value for them. Generated from the data by `report/scripts/gen_ch5_tables.py`. Source: `data/risk_library_masonry.json`, `data/literature_seed.json` (seed studies, direct mappings).

Category	Risks	With seed value	Without
Labour	13	9	4
Management	6	4	2
Material	6	4	2
Design	4	3	1
Equipment	3	1	2
External	3	2	1
Financial	3	2	1
Quality	2	1	1
Safety	2	2	0
Site	2	1	1
Weather	2	1	1
Total	46	30	16

The evidence IDs on a library entry support only the statement that the risk exists in published studies. They do not support a probability. The evidence note of R-MAT, for instance, records that M01 reports material shortage at RII 0.777 and that survey rankings show importance, not probability.

Sixteen library risks have no survey value in the seed studies. They are listed in Table 5.7. Eight were in the original 38 and have no matching survey item; eight were added later and are written in site language from the domain review. For the eight added risks the masonry-specific form of the risk is labelled Expert Judgment in the library and is to be confirmed by the expert survey, which has not been run.

Table 5.7. The 16 library risks without a seed value, with the rules that flag them. Generated by `report/scripts/gen_ch5_tables.py`. Source: `data/risk_library_masonry.json`, `data/rules_masonry.json`.

Risk	Name	Category	Scope	Added 2026-10-08	Rules that flag it
R-TRN	Inadequate training of workers	Labour	activity	no	none
R-PRD	Low labour productivity	Labour	activity	no	none
R-SUB	Subcontracting and contract-type problems	Management	activity	no	none
R-CLI	Client decision delay or pressure	Management	project	no	none
R-QLT	Inspection and quality-approval delays	Quality	activity	no	none
R-DEF	Defective or poor-quality material	Material	activity	no	none
R-GOV	Government policies or approvals	External	project	no	none
R-FIN	Contractor financial difficulties	Financial	project	no	none
R-FRONT	Work front not released (slab not de-shuttered, props or debris in place, frame not handed over)	Site	activity	yes	RL-FRONT
R-VT	Hoist or material lift unavailable or broken (bricks and mortar to upper floors)	Equipment	activity	yes	RL-VT
R-MORT	Mortar materials short or poor	Material	activity	yes	RL-MORT

Risk	Name	Category	Scope	Added 2026-10-08	Rules that flag it
	(cement, sand, or mixer down)				
R-GPAY	Gang wages or labour-contractor running bill overdue	Labour	activity	yes	RL-GPAY
R-OPEN	Openings, lintels, door/window frames and MEP sleeves not coordinated	Design	activity	yes	RL-OPEN
R-SCAF	Scaffolding not available, not erected or not passed safe	Equipment	activity	yes	RL-SCAF
R-FEST	Festival, harvest or election season exodus of the migrant gang	Labour	activity	yes	RL-FEST, RL-FEST2
R-HEAT	Peak-summer heat cuts effective working hours	Weather	activity	yes	RL-HEAT

Two limits follow from the table. Only 18 of the 46 library risks have any rule: the 10 original rule targets plus the 8 added risks. The other 28 enter the register by choice from the library, from an AI suggestion, or as custom risks. And the library is built from general building-construction surveys, with masonry-specific evidence mainly from M01 and A02, so it should not be read as a census of masonry delay causes on Indian sites. The risk of over-reading it is discussed in Chapter 7.

5.5 The rule engine

5.5.1 Purpose and constraints

The rule engine turns declared site facts into qualitative flags. A rule states a logical condition over facts, for example required workers exceed available workers. A rule flags one library risk as **elevated** or **normal** (relevant only). The module’s design constraints (`app/engine/rules.py`) are the following.

- A rule carries no probability and no numeric threshold invented by the software. Where a number appears, it is a fact the user enters (stock days, lead time) or the planned duration.
- A flag never produces a probability or a delay for a risk with entered numbers.
- A fact that is missing, or that has the wrong type for the rule, makes the rule “not evaluable”. It is reported as such and never guessed.
- Every rule has an identifier, a Source, evidence IDs, a rationale and a confidence, and the rule base is checked for consistency before use.

5.5.2 Evaluation

A rule has a list of **all_of** conditions and an optional list of **any_of** conditions. Each condition compares a fact with a literal or with another fact, using one of **lt**, **le**, **gt**, **ge**, **eq**, **ne**, **in**. The rule fires when every **all_of** condition is true and, if **any_of** is present, at least one **any_of** condition is true.

Typing is strict. Numeric comparisons need finite real numbers, and equality tests need the fact to have the same type as the value it is compared with. A string such as “10” or “yes” is not coerced. A rule is “not evaluable” if a needed fact is missing or has the wrong type and no other condition of the rule is already false.

When several rules fire on the same risk, the highest priority wins. If rules of equal priority disagree on the level, the outcome is reported as **conflict**. The rule base itself is checked for duplicate IDs, unknown facts, unknown levels, missing conditions, literature-sourced rules without evidence IDs, and identical conditions with contradictory levels (`check_rule_base`). Guard test RG-8 asserts that no pair of rules on one risk can give an equal-priority conflict.

5.5.3 The 22 rules

Table 5.8. The 22 site-fact rules. Generated from `data/rules_masonry.json` by `report/scripts/gen_ch5_tables.py`; the “condition in words” is the rule’s own description field.

Level **elevated** marks the risk as elevated; **normal** marks it as relevant only. Pri. is the priority used when several rules fire on one risk. Source: **data/rules_masonry.json**.

Rule	Condition (in words)	Formal condition	Risk flagged	Level	Pri.	Source	Conf.
RL-LAB	Required workers exceed workers available	required_workers > available_workers	R-LAB	elevated	0	Assumption	Moderate
RL-MAT	Stock on site runs out before a new delivery can arrive (for the material with the longest lead time)	material_stock_days <= material_lead_time_days	R-MAT	elevated	1	Assumption	Moderate
RL-MAT2	Stock on site covers fewer days than the planned duration	material_stock_days < planned_duration_days	R-MAT	normal	0	Assumption	Low
RL-TOOL	Tools or equipment shortage expected	tools_shortage = true	R-TOOL	elevated	0	Assumption	Low
RL-WX	Activity period overlaps the monsoon / rainy season	monsoon_overlap = true	R-WX	normal	0	Assumption	Moderate
RL-HGT	Masonry is done at height and scaffolding is not ready	work_at_height = true AND scaffolding_ready = false	R-SAFE	elevated	0	Assumption	Low
RL-PAY	Payment delays occurred previously or are expected	payment_delay_expected = true	R-PAY	elevated	0	Assumption	Low
RL-DES	Drawings or details are incomplete	design_incomplete = true	R-DES	elevated	0	Assumption	Low
RL-RWK	Rework was frequent in earlier work on this site or crew	prior_rework_history = true	R-RWK	elevated	0	Assumption	Low
RL-PLAN	Planner or site engineer judges the programme faster than the gang can achieve	schedule_compressed = true	R-PLAN	elevated	0	Assumption	Low
RL-SKILL	Skilled masons are fewer than needed	skilled_masons_short = true	R-SKILL	elevated	0	Assumption	Low
RL-WX2	Monsoon overlaps the activity and external walls are in scope	monsoon_overlap = true AND external_walls_in_scope = true	R-WX	elevated	1	Assumption	Low
RL-PACE	Planned daily output is higher than the output achieved last week	planned_daily_output > achieved_daily_output_last_week	R-PLAN	elevated	0	Assumption	Low
RL-FRONT	Floors or walls	work_front_re	R-FRONT	elevated	0	Assumption	Low

Rule	Condition (in words)	Formal condition	Risk flagged	Level	Pri.	Source	Conf.
	planned for the next 7 days are not de-propped, cleared and handed over	ady = false					
RL-VT	Masonry is above the ground floor and no hoist or material lift is available	masonry_floor_level >= 1 AND hoist_availability = false	R-VT	elevated	0	Assumption	Low
RL-SCAF	Scaffolding is not ready	scaffolding_ready = false	R-SCAF	elevated	0	Assumption	Low
RL-MORT	Sand or cement stock runs out before a new delivery can arrive	sand_cement_stock_days <= sand_cement_lead_time_days	R-MORT	elevated	0	Assumption	Low
RL-GPAY	The gang's weekly payment or the labour contractor's bill is overdue	gang_payment_overdue = true	R-GPAY	elevated	0	Assumption	Low
RL-OPEN	Door/window frames, MEP sleeve layout or lintel-level plan is not ready	frames_on_site = false OR mep_sleeve_layout_marked = false OR lintel_level_plan_issued = false	R-OPEN	elevated	0	Assumption	Low
RL-FEST	A major festival, harvest or election falls inside the activity window	festival_or_harvest_in_window = true	R-FEST	normal	0	Assumption	Low
RL-FEST2	A festival, harvest or election falls inside the window and the gang is mostly migrant	festival_or_harvest_in_window = true AND gang_mostly_migrant = true	R-FEST	elevated	1	Assumption	Low
RL-HEAT	The activity overlaps peak summer	summer_overlap = true	R-HEAT	normal	0	Assumption	Low

All 22 rules carry the Source label Assumption, 19 have confidence Low and 3 Moderate (RL-LAB, RL-MAT, RL-WX). The reason is that each rule states a logical condition that the team judged relevant, with evidence IDs showing that the risk exists. None of the rules is an empirical finding about how much a condition raises delay. The rationale text of each rule says this, and the interface shows the Source and the confidence beside each rule. Twenty-nine distinct fact names are used by the rules, one of which (the planned duration) is supplied automatically.

Several rules were changed after the review of 8 October 2026 (Audit_Corrections section 6). RL-MAT now compares stock days with lead time with a tie counted as a stock-out; RL-MAT2, RL-WX, RL-FEST and RL-HEAT are **normal** only, so that merely overlapping the monsoon does not mark weather as elevated; RL-HGT needs scaffolding not ready as well as work at height; and RL-WX2, RL-PACE and nine other rules were added. Priorities were set so that no pair of rules on one risk conflicts.

5.6 Register and matrix logic

5.6.1 The register

The register holds one card per risk. A card shows the name, category, status, the probability and the delay range with their Source badges, evidence chips, the rule flag if any, the matrix cell if any, and the expected delay if the risk occurs. The status is one of **literature-supported**, **expert/user-provided** and **AI-suggested-unverified**. The text-only fields (owner, early-warning trigger, response type, action, review date) follow the register practice described in the risk-response literature in the research notes (avoid, reduce, transfer, accept; Baker et al., 1999; Dey, 2011). They carry no numbers and take no part in any calculation.

5.6.2 Expected delay

For a risk with entered numbers the matrix uses the conditional expected delay $E[D]$, computed in closed form from the entered range (`DelayDist.mean`):

where a , m and b are the minimum, most likely and maximum delay in working days. With the default shape weight $\lambda = 4$ the PERT mean is $(a + 4m + b)/6$, the classical three-point mean (Herrerias-Velasco et al., 2011; Trietsch et al., 2012). The choice of $\lambda = 4$ is the conventional default and carries no site calibration. The notes record that no source supplies a masonry-specific value of λ .

5.6.3 Probability class, impact class, score and level

Let T_0 be the planned duration in working days, p the entered probability, and $f_1 < f_2 < f_3 < f_4$ the four impact bin edges entered by the user as fractions of T_0 . The probability class is

and the impact class is

Both classes run from 1 to 5. The indicator uses lower-edge-inclusive bins: a value exactly on an edge goes to the higher class. “Exactly on an edge” is judged with a relative tolerance of 10^{-9} , so that 1.2 days on a 12-day activity, which is 0.09999999999999999 in binary floating point, is still the 0.10 edge. The same tolerance applies on both axes (`EDGE_REL_TOL` in `app/engine/matrix.py`); it was added to the probability axis after the audit finding M4. The score and the level are

The probability edges and the score thresholds are equal-width choices of the tool. They are not taken from the literature, and every export prints them labelled Assumption. The impact edges have no default: they are the user’s input with a Source, and without them the matrix is not computed for entered numbers. Because the highest score in probability class 1 is $1 \times 5 = 5$, a risk in probability class 1 is always Low whatever its impact. This is a consequence of the product rule, and the matrix legend states it and marks the two affected cells. Risks in impact class 5 are also listed separately as a high-consequence filter, so that a severe but unlikely risk is not lost in a “Low” cell. The filter is not a score.

The matrix is an ordinal prioritisation aid. The literature on risk matrices (Cox, 2008; Duijm, 2015) warns that matrices have poor resolution, can rank a quantitatively smaller risk above a larger one, and depend on how the bins are drawn. The tool therefore keeps the continuous p and delay in the register next to the class, labels the thresholds as Assumptions, and calls the output “ordinal prioritisation only” on screen and in every export. The sensitivity of the levels to the thresholds and edges was analysed separately (Chapter 4); the matrix does not remove the limits the literature describes.

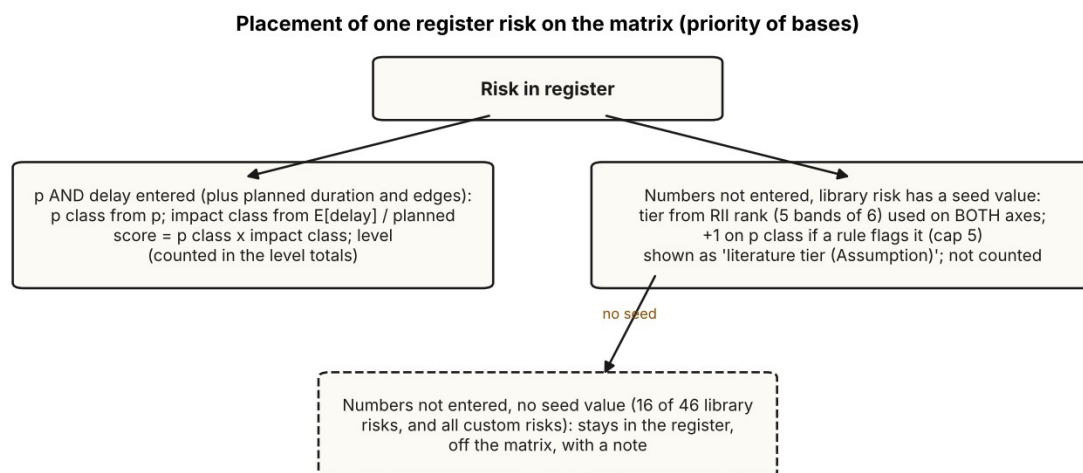


Figure 5.3. Placement of a register risk on the matrix. Drawn by `report/scripts/make_ch5_diagrams.py` from `run_case` in `app/service.py`.

5.7 Literature tier and source labels

5.7.1 What the tier is

A library risk whose probability and delay are not both entered still appears on the matrix if the literature seed has a value for it (30 of the 46 library risks). The placement is computed in `app/literature_seed.py`. For each risk, the direct-mapped RII values of each seed study are averaged first, so that one study counts once, and the study means are then averaged. The 30 risks are ranked by this mean, rank 1 being the highest, and cut into five equal-count bands of six. For rank p among $n = 30$ seeded risks,

The tier is used as both the probability class and the impact class. If a rule flags the risk as elevated, the probability class is raised by one, capped at 5: $c_p = \min(5, \text{tier} + 1)$. Entering both p and delay replaces the placement.

Both uses of the tier are Assumptions of this tool, and neither is a finding. An RII measures how important respondents rated a factor; it is not a probability and not a number of days. Using one rank on both axes squares it in the score, and the +1 step for a rule flag has no literature source. The seed values themselves match the printed papers (Audit_Corrections, row-level check of all 182 values), but the agreement of the seed with held-out surveys is weak and not statistically significant (Handoff section 8.4, reported in Chapter 4). The seed is therefore an ordinal starting position, to be replaced by expert-survey and site data.

5.7.2 How the tier is displayed

After the committee review of 8 October 2026, a seeded risk is shown differently from a risk with entered numbers, in all outputs.

- The chip on the grid is grey with a dashed outline and the label “literature tier (Assumption)”. The cell colour beneath it is the colour of the cell, not an assessed level for that risk.
- Seeded risks are left out of the level totals (`summary.levels`, the key-figure strip, and the High, Moderate, Low counts). The strip counts “on matrix (entered numbers)” separately from the seeded count.
- No probability, expected delay or monetary value is displayed or computed for a seeded risk.
- In `register.csv` the `level` cell is empty for a seeded row and the `tier` column holds the text “literature tier (Assumption): tier n”.
- The +1 step is labelled Assumption wherever it applies.

A fuller alternative, showing seeded risks in neutral colour with no level at all and dropping the +1 step, was considered in the review and left to a supervisor decision (Audit_Corrections section 6). The present display is the smaller fallback.

5.7.3 Source labels in the interface

Every numeric field has a Source picker, and the badge (for example ASM for Assumption) is shown beside the value on register cards, in the matrix table and in the exports. Typing a number into an empty field sets the Source to User Input; it never sets Literature automatically. The matrix legend prints the impact edges with their Source and the default probability edges and score thresholds as Assumption. If the impact edges have no Source, the tool gives a warning and the exports print “Source: not given”. A banner on every screen states “ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data” whenever the example is loaded (guard test RG-7 keeps the example labelled in every export).

5.8 AI evidence layer and guard rules

5.8.1 Purpose

The AI layer helps identify candidate risks. It does not estimate anything. The user types an activity and a query; the tool retrieves the best-matching records from the evidence store, sends only those records to a language model, and accepts back candidate risks that name a mechanism and cite record IDs. The layer works without a key in an offline mode that returns matching library risks and evidence records. With a key stored in Settings, either an Anthropic or a Google model can be called; default model identifiers are set in `app/ai_layer/clients.py` and can be overridden. The Gemini path was checked against the live service during development, and the Anthropic path was not checked because no key was available (`gui/README_GUI.md`). The key is stored in the user’s application-data folder, outside the project folder, with restricted file permissions, so that zipping the project for submission does not carry it.

5.8.2 Evidence store and retrieval

The store holds the curated records of `Literature_Evidence_Package.xlsx` and the structured records parsed from `Research_Notes/ (app/ai_layer/corpus.py)`. The running application reports 472 evidence records. Retrieval is lexical, by token overlap, and runs offline. The code comments and the GUI notes state that lexical retrieval is simple and that many harvested records have an abstract only. Retrieval grounding does not by itself prevent unsupported output, which is why the guard follows it.

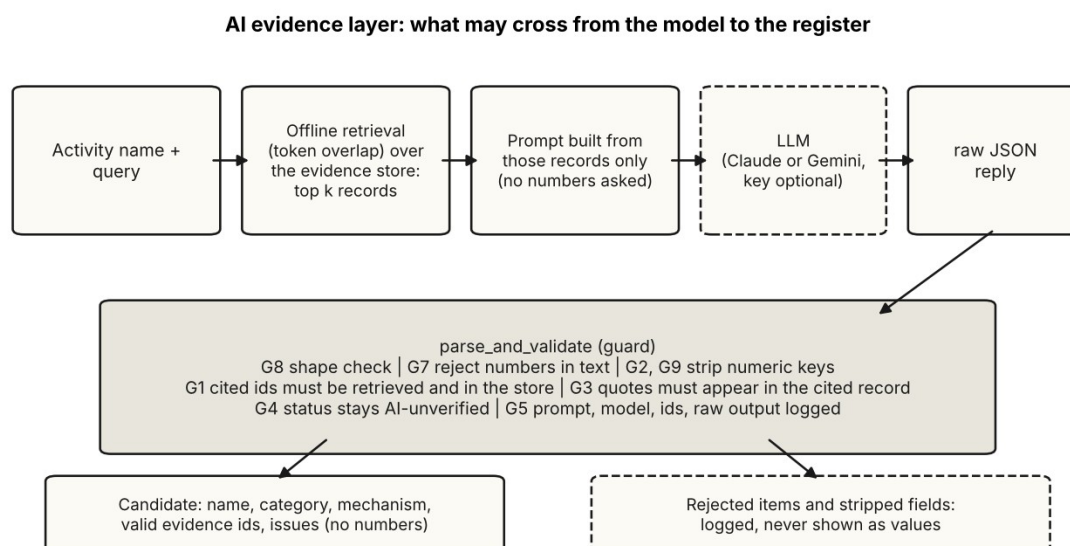


Figure 5.4. Path of an AI suggestion from query to register. Dashed boxes: optional or rejected paths. Drawn from `app/ai_layer/guard.py`.

5.8.3 Guard rules

The guard (`app/ai_layer/guard.py`) validates every reply before anything reaches the interface. Table 5.9 lists the rules. Each is enforced in code and exercised in `tests/test_ai_guard.py` or the relevance guard tests.

Table 5.9. Guard rules G1 to G9 of the AI layer. Source: `app/ai_layer/guard.py` header and code.

Rule	What it enforces
G1	Every cited ID must be in the retrieved set and in the store; otherwise it is dropped and the item flagged
G2	Any numeric field the model returns (probability, delay, cost, effect size and similar listed keys) is stripped and logged as rejected
G3	A quoted passage must appear in the cited record's stored text; otherwise the quote is flagged
G4	Output is always AI-suggested-unverified; only an explicit user confirmation changes the status
G5	The prompt, model identifier, retrieved IDs and raw output are kept verbatim in an audit log
G6	Belongs to the explanation module (<code>app/ai_layer/explain.py</code>), which checks numbers in a narrative against the result trace; that module is not exposed by the current interface
G7	A suggestion whose name, category or mechanism contains a probability, percentage, duration, day count or currency figure is rejected and logged
G8	Every field is type-checked; one malformed item is dropped and logged and the others survive
G9	Unknown numeric keys (likelihood, days, impact and similar) are dropped and logged as under G2

The consequence is that the AI path cannot place a number in the register. A candidate carries a name, a category, a mechanism and valid evidence IDs. **promote** turns a candidate into a risk only when the caller supplies the probability and delay with a Source, refuses a derived value as the source of either, and leaves the status at AI-unverified unless the user confirms. Guard test RG-10 checks that no path from the AI layer to the service can carry a number, and guard test RG-4 checks that the wording of the exports no longer claims “No number here was produced by AI”, a sentence that the audit (finding M7) showed was not enforced at the time. It was replaced by a statement of what the tool does: it does not generate probabilities, delays or costs, each number shown was entered with the Source printed next to it, and the AI feature can propose names, mechanisms and evidence IDs only.

5.8.4 Limits of the guard

The guard is pattern-based. G7 detects figures with regular expressions for percentages, probability phrases, durations and currency; a figure written in a form that none of the patterns matches could pass, and a suggestion with a harmless number in a name could be rejected without need. G3 compares text after whitespace and case normalisation, so a paraphrase is flagged but a quotation of the wrong sense is not. The guard does not judge whether a suggested risk is good: it judges whether it is traceable and number-free. Whether AI suggestions improve identification has not been tested; this would need a comparison with an expert-built list, which was not done.

5.9 Exports

All exports are produced from the result object, so they cannot disagree with the screen.

Table 5.10. Exports. Source: `app/reporting/*`, `gui/api.py`.

File	Content
register.md	Settings block (impact edges with Source; probability edges and thresholds as Assumption), the matrix grid, the register table, rule flags, warnings, and the statement of what the tool does and does not do
register.csv	One row per risk, 28 columns including basis (entered or literature tier), p and its Source, delay kind and range, class and level, tier, ILLUSTRATIVE note, and the text-only register fields
report.html	Printable page with the coloured matrix and the register (print to PDF from the browser)
matrix.svg, matrix.png	The 5 x 5 grid as a graphic

File	Content
case.JSON	The input case, for saving and reopening
analysis.md	Optional decision layer report (Section 5.12)

Several output-safety measures were added after the QA run (Chapter 6). The CSV neutralises cells that begin with =, +, -, @, tab or carriage return, so a risk named with a spreadsheet formula is written as text. The Markdown writer escapes the pipe character and replaces newlines. The HTML report escapes all text. The SVG writer escapes its inputs. The CSV is served with a byte-order mark so that Excel on Windows decodes non-ASCII names.

5.10 The interface

The interface has seven screens in the left rail (Table 5.11). A top bar holds the case name, an unsaved-changes indicator, an input-problem counter and the New, Open, Save and Load example buttons, and a theme switch. A strip under the bar summarises the register: risks, complete, on the matrix (entered numbers), levels, and rule-flagged. Shortcuts are Ctrl+S, Ctrl+O and Esc.

Table 5.11. Screens. Source: `gui/README_GUI.md`; `gui/static/js/app.js`.

#	Screen	Purpose
1	Case & Activity	project details, activity, planned duration with Source
2	Site facts & Rules	declare facts; see which rules fired, which are not evaluable
3	Risk register	cards with p, delay, Source, status, evidence; library and AI suggestions in a side panel
4	Risk matrix	impact edges with Source; the grid; the table of classes and levels; the high-consequence list
5	Cost, options & decision	optional decision layer (Section 5.12)
6	Evidence	curated records; search across the corpus
7	Export	downloads

The following figures show each screen with the ILLUSTRATIVE example loaded, at a viewport width of 1280 px in the light theme, captured with Playwright and Chromium on 8 October 2026 by `report/scripts/take_screenshots.py`. Long pages are cropped at the bottom; no other editing was done. Every number in these figures is an ILLUSTRATIVE placeholder labelled Assumption.

BRICKS-LY

ILLUSTRATIVE example

NO UNSAVED CHANGES

NEW

OPEN

SAVE

LOAD EXAMPLE

AI: OFFLINE MODE

DEFINE

Case & Activity

IDENTIFY

Site facts & Rules

Risk register

PRIORITISE

Risk matrix

DECIDE

Cost, options & decision

REFERENCE

Evidence

Export

Settings

Offline tool. The register and matrix come from the service; the GUI computes nothing.

V1.0.0 - 472 evidence records

RISKS IN REGISTER

COMPLETE

ON MATRIX (ENTERED NUMBERS)

HIGH

MODERATE

LOW

RULE-FLAGGED

7

7 / 7

7

1

5

1

6

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

[STEP 1]

Case & Activity

Name the project and the single brick-masonry activity assessed. All durations are in **working days**. Nothing is pre-filled with invented numbers.

[PROJECT]

Project name

ILLUSTRATIVE example

Location

n/a

Construction type

Residential building

Notes

Demonstration only. Every probability, delay and the planned duration is a placeholder labelled Assumption, not site data and not evidence. The evidence IDs (M01, M02, ...) point to real literature records supporting that each risk exists, not its size. Its site facts cover every input of the bundled rules and none of them points to a risk that is missing from the register. Replace all numbers with your own

ILLUSTRATIVE example loaded: every number is a placeholder, not evidence.

Activity ID

MAS-01

Activity name

Brick masonry, ground-floor walls

Planned duration of the activity

16

working days

ASM Assumption

ILLUSTRATIVE placeholder, not evidence

Needed for the risk matrix: each risk's impact is its expected delay as a fraction of this duration.

Also used by the material-stock rule.

Figure 5.5. Case & Activity screen with the ILLUSTRATIVE example. Planned duration (16 working days) and its Source are on this screen. The values are placeholders, not site data.

BRICKS-LY

ILLUSTRATIVE example

NO UNSAVED CHANGES

NEW

OPEN

SAVE

LOAD EXAMPLE

AI: OFFLINE MODE

DEFINE

[1] Case & Activity

IDENTIFY

[2] Site facts & Rules

[3] Risk register

PRIORITISE

[4] Risk matrix

DECIDE

[5] Cost, options & decision

REFERENCE

[6] Evidence

[7] Export

Settings

Offline tool. The register and matrix come from the service; the GUI computes nothing.
v1.0.0 - 472 evidence records

RISKS IN REGISTER

7

COMPLETE

7 / 7

ON MATRIX (ENTERED NUMBERS)

7

HIGH

1

MODERATE

5

LOW

1

RULE-FLAGGED

6

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

[STEP 2]

Site facts & Rules

Declare the site conditions the rule engine checks. Rules only **flag** risks as relevant or elevated; they never produce probabilities. A blank fact is *unknown*: rules that need it are reported as not evaluable instead of guessed.

[LABOUR]

site facts (User Input)

Workers required

6workers

required_workers · used by RL-LAB

Workers available (average daily attendance last week, or confirmed by the labour contractor)

5workers

available_workers · used by RL-LAB

Skilled masons fewer than needed

UNKNOWNYESNO

skilled_masons_short · used by RL-SKILL

Gang's weekly payment or labour contractor's bill is overdue

UNKNOWNYESNO

gang_payment_overdue · used by RL-GPAY

A major festival, harvest or election falls inside the activity window

UNKNOWNYESNO

festival_or_harvest_in_window · used by RL-FEST, RL-FEST2

Gang is mostly migrant (home state far from site)

UNKNOWNYESNO

gang_mostly_migrant · used by RL-FEST2

[MATERIAL & EQUIPMENT]

site facts (User Input)

Stock on site covers (same material)

2days

material_stock_days · used by RL-MAT, RL-MAT2

Supplier lead time (material with the longest lead time)

6days

material_lead_time_days · used by RL-MAT

Mixer, cutter, pump or hand tools short or broken

UNKNOWNYESNO

tools_shortage · used by RL-TOOL

Scaffolding is erected and passed safe

UNKNOWNYESNO

scaffolding_ready · used by RL-HGT, RL-SCAF

Hoist or material lift is available and working

UNKNOWNYESNO

hoist_available · used by RL-VT

Sand / cement stock on site covers

8days

sand_cement_stock_days · used by RL-MORT

Sand / cement delivery lead time

5days

sand_cement_lead_time_days · used by RL-MORT

[SITE & SCHEDULE]

site facts (User Input)

Activity overlaps monsoon / rainy season

UNKNOWNYESNO

monsoon_overlap · used by RL-WX, RL-WX2

Masonry is done at height

UNKNOWNYESNO

work_at_height · used by RL-HGT

Client payment delays occurred or expected

UNKNOWNYESNO

payment_delay_expected · used by RL-PAY

Good-for-construction drawings for wall layout, openings and lintel levels not issued

UNKNOWNYESNO

design_incomplete · used by RL-DES

Any wall dismantled or redone on this site / gang in the last month

UNKNOWNYESNO

prior_rework_history · used by RL-RWK

Programme judged faster than the gang can achieve

UNKNOWNYESNO

schedule_compressed · used by RL-PLAN

External walls are part of this activity

UNKNOWNYESNO

external_walls_in_scope · used by RL-WX2

Planned daily output (use one unit for both outputs)

10units/day

planned_daily_output · used by RL-PACE

Achieved daily output last week (same unit)

8units/day

achieved_daily_output_last_week · used by RL-PACE

Floors and walls planned for the next 7 days are de-propped, cleared and handed over

UNKNOWNYESNO

work_front_ready · used by RL-FRONT

Floor where masonry is done (0 = ground floor)

0floor

masonry_floor_level · used by RL-VT

Door and window frames are on site

UNKNOWNYESNO

frames_on_site · used by RL-OPEN

Electrical and plumbing sleeve / conduit positions are marked

UNKNOWNYESNO

mep_sleeve_layout_marked · used by RL-OPEN

Lintel and sill-band level plan is issued

UNKNOWNYESNO

lintel_level_plan_issued · used by RL-OPEN

Activity overlaps peak summer

UNKNOWNYESNO

summer_overlap · used by RL-HEAT

[AUTOMATIC FACT]

planned_duration_days is taken from the planned duration on step 1 (16 days) and is used by rule RL-MAT2 (relevant only, not elevated).

Every elevated flag is covered by the risk register

[RULE BASE]

data/rules_masonry.json · 22 rules · 9 fired, 0 not evaluable

RULE	CONDITION	FLAGS RISK	SOURCE	CONFIDENCE	EVIDENCE	RESULT
RL-LAB	Required workers exceed workers available required_workers > available_workers	R-LAB Labour absenteeism or shortage	ASM ASSUMPTION	Moderate	M01 M07	FIRE-ELEVATED facts used: required_workers = 6, available_workers = 5

Figure 5.6. Site facts & Rules screen (top part), with the *ILLUSTRATIVE* example. Each fact is a Yes / No / Unknown control or a number field with its unit, and names the rules that use it. The rule table below the facts is listed in Table 5.8. Cropped.

The facts screen avoids free text on purpose. Every input is a number with a unit, or a three-way Yes / No / Unknown control, so the user cannot type a value the rules cannot read. Unknown is the default and means “not evaluable”, which is different from No.

BRICKS-LY | ILLUSTRATIVE example

NO UNSAVED CHANGES | NEW | OPEN | SAVE | LOAD EXAMPLE | AI: OFFLINE MODE

DEFINE

[1] Case & Activity

IDENTIFY

[2] Site facts & Rules

[3] Risk register

PRIORITISE

[4] Risk matrix

DECIDE

[5] Cost, options & decision

REFERENCE

[6] Evidence

[7] Export

Settings

Offline tool. The register and matrix come from the service; the GUI computes nothing.
v1.0.0 - 472 evidence records

RISKS IN REGISTER: 7 | COMPLETE: 7/7 | ON MATRIX (ENTERED NUMBERS): 7 | HIGH: 1 | MODERATE: 5 | LOW: 1 | RULE-FLAGGED: 6

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

Risk register

Risks that can add delay to this activity. Each needs a probability and a delay range with their sources; a risk is placed on the matrix once both are entered (occurrence model).

+ CUSTOM RISK

[STEP 3]

R-MAT **Brick / material shortage** LITERATURE-SUPPORTED

Material p 0.5 ASM ASSUMPTION

delay 1 / 3 / 8 d PERT ASM ASSUMPTION M01 M06

matrix High (p class 3 × impact 5) expected delay if it occurs 3.5 d

rule flag elevated (RL-MAT, RL-MAT2)

R-LAB **Labour absenteeism / shortage** LITERATURE-SUPPORTED

Labour p 0.35 ASM ASSUMPTION

delay 1 / 2 / 6 d PERT ASM ASSUMPTION M01 M07 M15

matrix Moderate (p class 2 × impact 4) expected delay if it occurs 2.5 d

rule flag elevated (RL-LAB)

R-RWK **Rework due to workmanship** LITERATURE-SUPPORTED

Quality p 0.25 ASM ASSUMPTION

delay 1 / 2 / 5 d PERT ASM ASSUMPTION M03 M15

matrix Moderate (p class 2 × impact 4) expected delay if it occurs 2.33 d

rule flag elevated (RL-RWK)

R-WX **Rain / monsoon stoppage** LITERATURE-SUPPORTED

Weather p 0.2 ASM ASSUMPTION

delay 1 / 2 / 7 d PERT ASM ASSUMPTION R08

matrix Moderate (p class 2 × impact 4) expected delay if it occurs 2.67 d

rule flag elevated (RL-WX, RL-WX2)

R-PLAN **Poor planning or unrealistic scheduling** LITERATURE-SUPPORTED

Management p 0.1 ASM ASSUMPTION

delay 1 / 1 / 3 d PERT ASM ASSUMPTION M01

matrix Low (p class 1 × impact 3) expected delay if it occurs 1.33 d

rule flag elevated (RL-PLAN, RL-PACE)

R-SAFE **Unsafe conditions / work at height** LITERATURE-SUPPORTED

Safety p 0.3 ASM ASSUMPTION

delay 1 / 2 / 6 d PERT ASM ASSUMPTION M01

matrix Moderate (p class 2 × impact 4) expected delay if it occurs 2.5 d

R-SKILL **Unskilled or unqualified labour** LITERATURE-SUPPORTED

Labour p 0.22 ASM ASSUMPTION

delay 1 / 2 / 5 d PERT ASM ASSUMPTION M15 M14

matrix Moderate (p class 2 × impact 4) expected delay if it occurs 2.33 d

rule flag elevated (RL-SKILL)

LIBRARY (46) AI SUGGESTIONS

46 risks from the literature (data/risk_library_masonry.json). Evidence supports that a risk exists; it gives no probability. Risks not listed here can be added as custom or AI-suggested risks.

Filter the 46 + ADD ALL 39

Design (4) + ADD ALL 4

R-DES Design changes or incomplete documentation

Drawing changes, clarifications or missing details interrupt work.

M06 (India): frequent design changes and scope clarity among significant productivity factors; M13 (Saudi Arabia): design quality affects delay.

M06 M13 + ADD

R-SCP Unclear scope or inadequate project formulation PROJECT-LEVEL

The scope or the project definition is unclear at the start, causing changes and waiting.

M06 (India): scope clarity of the project R11 0.79; M06 (India): inadequate project formulation in the beginning R11 0.738; M06

Figure 5.7. Risk register screen with the *ILLUSTRATIVE* example: seven cards, each with p and delay and their Source badges, evidence chips, rule flag and matrix cell. The side panel offers the 46-risk library and AI suggestions.

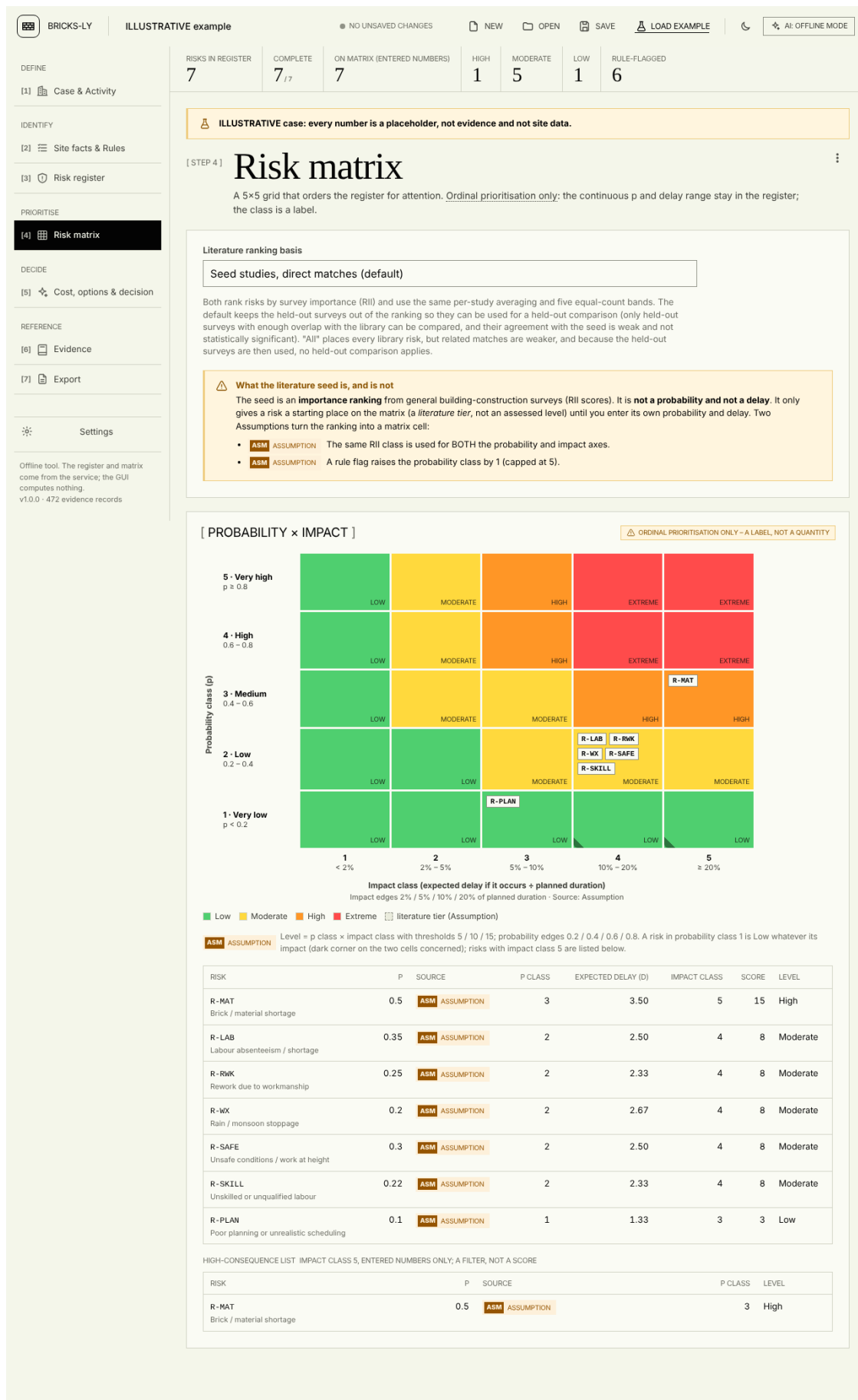


Figure 5.8. Risk matrix screen with the ILLUSTRATIVE example. The grey notice states what the literature seed is and is not. The grid shows the seven risks; the legend states the impact edges and their Source, the probability edges and score

thresholds as Assumption, and marks the two probability-class-1 cells where a risk is Low whatever its impact. Below the grid: the table of classes and levels and the high-consequence list.

In the example, R-MAT falls in probability class 3 and impact class 5 (score 15, High) and is the only entry in the high-consequence list; five risks are Moderate and R-PLAN is Low. The hand calculation in Chapter 6 reproduces these cells. The example has no risk placed from the literature tier, so no grey dashed chip appears in this figure; the dashed style is exercised in tests (Chapter 6) and in the legend.

BRICKS-LY | ILLUSTRATIVE example | NO UNSAVED CHANGES | NEW | OPEN | SAVE | LOAD EXAMPLE | AI: OFFLINE MODE

DEFINE

[1] Case & Activity

IDENTIFY

[2] Site facts & Rules

[3] Risk register

PRIORITISE

[4] Risk matrix

DECIDE

[5] Cost, options & decision

REFERENCE

[6] Evidence

[7] Export

Settings

Offline tool. The register and matrix come from the service; the GUI computes nothing.
v1.0.0 - 472 evidence records

RISKS IN REGISTER 7 | **COMPLETE** 7/7 | **ON MATRIX (ENTERED NUMBERS)** 7 | **HIGH** 1 | **MODERATE** 5 | **LOW** 1 | **RULE-FLAGGED** 6

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

[STEP 6]

Evidence

The literature records behind the evidence IDs. Curated records are listed; search reaches the full corpus, including the bulk literature harvest.

[EVIDENCE BROWSER] 472 records in the corpus

Search citation, context or notes (e.g. India labour) ☐ Only IDs cited in this case

472 curated records shown. Type to search the whole corpus, including the literature harvest.

- R01** Yaseen, Ali, Salih, Al-Ansari (2020). Prediction of Risk Delay in Construction Projects Using a Hybrid AI Model. Sustainability 12(4):1514 NC · depth: targeted read
- R02** Araujo-Rey, Sebastian (2021). Causes of Delays in Industrial Construction Projects through Planning and Statistical Computing. Sustainability 13(7):3975 Industrial construction (authors Spain; project data hypothetical) · depth: targeted read
- R04** Zorrilla, Seijo, Arenal, Mena (2026). AI-Driven Decision Support System for Proactive Risk Management in Construction Projects. Intelligent Infrastructure and Construction 2(2):4 (README name; prompt says 'International Infrastructure Construction' - verify) Spain (FCC Construcción dataset) · depth: targeted read
- R05** Ivanovic et al. (2022). Detection and In-Depth Analysis of Causes of Delay: ML and Expert Knowledge (DREAM). Sustainability 14(22):14927 Serbia (single case project) · depth: targeted read
- R06** Zagia, Kajewski, Motamedisedeh, Omrani, Rose (2026). Risk Maturity Level and Construction Delay Factors: ML Model. Buildings 16(16):3149 Australia · depth: targeted read
- R07** Alsugair et al. (2023). ANN Model to Predict Final Construction Contract Duration. Applied Sciences 13(14):8078 Saudi Arabia (135 projects) · depth: targeted read
- R08** Singh et al. (2025). Construction Delays Due to Weather in Cold Regions: SEM + ANN. Buildings 15(11):1916 India (cold regions) · depth: targeted read **CITED**
- R09** Plebankiewicz, Wlczorek (2020). Prediction of Cost Overrun Risk in Construction Projects. Sustainability 12(22):9341 Road-object example (country NC) · depth: targeted read
- R10** Ashtari, Ansari, Hassannayebi, Jeong (2022). Cost Overrun Risk Assessment: Bayesian Network Classifier. Buildings 12(10):1660

Figure 5.9. Evidence screen (top part, cropped). The search reaches the whole corpus; curated records are listed first.

BRICKS-LY

ILLUSTRATIVE example

NO UNSAVED CHANGES

NEW

OPEN

SAVE

LOAD EXAMPLE

AI: OFFLINE MODE

DEFINE

[1] Case & Activity

IDENTIFY

[2] Site facts & Rules

[3] Risk register

PRIORITISE

[4] Risk matrix

DECIDE

[5] Cost, options & decision

REFERENCE

[6] Evidence

[7] Export

Settings

Offline tool. The register and matrix come from the service; the GUI computes nothing.
v1.0.0 - 472 evidence records

RISKS IN REGISTER

7

COMPLETE

7 / 7

ON MATRIX (ENTERED NUMBERS)

7

HIGH

1

MODERATE

5

LOW

1

RULE-FLAGGED

6

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

[STEP 7]

Export

Download the colour-coded matrix as an image (PNG or SVG), a printable report with the matrix (HTML: open it and use Print to save as PDF), the register as Markdown or CSV, and the case file. Downloads always use the **current** inputs.

REPORT.HTML

MATRIX.PNG

MATRIX.SVG

REGISTER.MD

REGISTER.CSV

CASE.JSON

[PREVIEW]

register.md

Risk matrix

ILLUSTRATIVE example · Brick masonry, ground-floor walls

5 Very high	Low	Moderate	High	Extreme	Extreme
4 High	Low	Moderate	High	Extreme	Extreme
3 Medium	Low	Moderate	Moderate	High	High MAT
2 Low	Low	Moderate	Moderate LAB RWK WX SAFE SKILL	Moderate	Moderate
1 Very low	Low	Low	Low PLAN	Low	Low
	1	2	3	4	5

Impact class (expected delay if it occurs ÷ planned duration)

Low

Moderate

High

Extreme

Ordinal prioritisation only – a label, not a quantity.
Dark corner: probability class 1 is always Low whatever the impact (Assumption: level = p class x impact class, thresholds 5 / 10)

RISK REGISTER AND MATRIX: ILLUSTRATIVE EXAMPLE

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

n/a · Residential building

Activity: **Brick masonry, ground-floor walls**. Planned duration: **16 working days** (Source: Assumption).

> Demonstration only. Every probability, delay and the planned duration is a placeholder labelled Assumption, not site data and not evidence. The evidence IDs (M01, M02, ...) point to real literature records supporting that each risk exists, not its size. Its site facts cover every input of the bundled rules and none of them points to a risk that is missing from the register. Replace all numbers with your own before drawing any conclusion.

Summary

Figure 5.10. Export screen (top part, cropped).

Errors are handled in the same way on every screen. A value that is present but wrong is shown beside the field and in a problem counter in the top bar, with a button that jumps to the screen that holds it. The register and matrix keep their last valid state and are marked stale until the problem is fixed. Opening a saved case restores it exactly: a risk the user deleted is not re-added on open. This was defect 1 of the QA report and is now covered by a test (Chapter 6).

5.11 Desktop wrapper, build and continuous integration

5.11.1 Two ways to run

`python -m gui` starts the server and opens a browser tab at `http://127.0.0.1:8765/`; the next free port is used if the preferred one is busy. `python -m gui.qt_app` opens a PySide6 window that hosts the same interface in Qt WebEngine. The window starts the same Flask server on a free loopback port in a background thread, loads it, opens any external link in the normal browser, and replaces browser downloads with a Save dialog. The desktop window therefore adds no new logic: it is a different container for the same files, and a defect in the engine is the same defect in both.

The project description speaks of a PySide6 desktop application. The implementation is accurate to that description only in this sense: the window is a PySide6 application, but the interface inside it is web content. This is stated so that the wrapper is not mistaken for a native Qt interface.

5.11.2 Build and checks

`build_desktop_exe.bat` and `build_exe.bat` build Windows executables with PyInstaller from `fyp_desktop.spec` and `fyp_gui.spec`. A GitHub Actions workflow ([.github/workflows/build-exe.yml](https://github.com/workflows/build-exe.yml), run on demand or on a version tag) performs the following on a Windows runner: install the requirements, run the test suite, build the browser-version executable, start it and call `/api/health`, `/api/example` and `/api/run`, run the desktop window from source in a headless self-test (`--smoke`), build the desktop executable, run its self-test, and upload the zipped folders as artifacts. The self-test loads the page, checks that the page has buttons and text and that the server reports loaded evidence records, and writes a JSON result.

What was and was not checked. The workflow is in the repository. In the working session for this report it was not run, because the session is a Linux machine without Windows, PyInstaller or Qt WebEngine, and the QA report states that the Qt wrapper and `fyp_desktop.py` were not tested in its run. This report therefore makes no claim that the Windows executables were built or run successfully; the workflow is the means by which that is to be checked, and its log is the evidence. Python 3.12 is the version used on the build runner, and the development environment of this report used Python 3.13.

5.11.3 Dependencies and offline use

The runtime requirements are Flask, NumPy, SciPy, Matplotlib and openpyxl; the desktop window adds PySide6. No component of the core path needs a network. The only network use is an AI request, which happens only when the user has stored a key and presses the suggestion button, and the test route for the key. The server answers only requests addressed to `127.0.0.1`, `localhost` or the loopback IPv6 name, which blocks DNS-rebinding access to the settings routes (`gui/api.py`). The server is the Flask development server, which is adequate for one user on one machine and is not meant to be exposed on a network; the interface README says so.

5.12 Optional decision layer

5.12.1 Scope and status

On 8 October 2026 the team asked for a cost, options and decision screen to be added. It is implemented in `app/analysis.py` with the engine modules `simulation`, `emv`, `decision` and `options`, exposed on screen 5 and by `python -m app.cli analyze`. It is optional. The register and the matrix do not depend on it, and its result never feeds back into the register or the matrix level.

It is not the four-phase pipeline described in the original project description. In particular, it does not use a 27-state matrix to produce a delay probability, it does not produce a mitigation cost, and it does not compute the product of a P90 delay, a daily penalty and a delay probability. It computes the expected total cost of each response option from a simulated schedule, as set out below.

5.12.2 What goes in, and what does not

The layer uses only numbers the user enters, each with a Source: the cost of one delay day (no default), optional liquidated damages per day after a deadline, an optional deadline, per-response costs, and the effect of each response on a risk as a new probability and/or a new delay range. A response with no entered effect is reported and left out; the tool does not supply one. The mitigation catalogue (`data/mitigation_catalogue.json`, 13 entries) lists candidate actions and what must be elicited for each, and contains no effect sizes and no costs, because the evidence review found no transferable risk-reduction factor for a masonry-specific response (Research_Notes/D_mitigation.md, section 7, as recorded in the catalogue header). The layer never uses a literature-tier (RII) placement as a probability. If a risk has no entered `p` and delay, it is left out of the simulation with a note.

5.12.3 Method

For each risk i with entered probability p_i and delay distribution F_i , a simulated run draws an occurrence $I_i \sim \text{Bernoulli}(p_i)$ and, if the risk occurs, a delay $D_i \sim F_i$. The duration is

with independent, additive risks unless correlations are entered. The delay distributions are Beta-PERT with shape weight $\lambda = 4$, triangular, uniform or fixed. The draws for each risk come from a random stream keyed on the seed and the risk ID, so a response option that changes p_i or F_i reuses the same underlying draws as the baseline (common random numbers). Differences between options are then driven by the entered changes and not by sampling noise. The default is 10,000 runs and seed 12345, both user-editable (up to 200,000 runs).

The cost of a run is

where C_d is the cost of a delay day, ℓ the liquidated damages per day after the deadline T_{dl} , and d_i an optional direct cost if risk i occurs. Early finishes earn no credit: negative delay is clipped at zero in the cost. The event expected monetary value of one risk is analytic,

and its sum equals the simulated expected delay cost in expectation for independent additive risks. It does not include liquidated damages, whose term is non-linear in T and cannot be recovered from event EMVs.

For response options, each option is a set of at most one response per risk. The tool evaluates each response alone and combinations of responses that target different risks (at most 40 options), plus the option “accept”. For option j with mitigation cost M_j the total expected cost is

where the residual cost is simulated with the changed p and F . The default criterion is the minimum TEC; minimum probability of exceeding the deadline and minimum P90 duration are also available, with optional constraints on the deadline probability and the mitigation budget. Infeasible responses and options that violate a constraint are removed before ranking. The result is one of the commands AUTHORIZE MITIGATION or ACCEPT RISK, with the figures behind it. The command text says the option is “preferred under the stated criterion among the options evaluated”, never “optimal”, and a test enforces that wording. Two further checks accompany every command: the preferred option under three different seeds, and the preferred option as the cost of a delay day is scaled from 0.25 to 4 times.

5.12.4 Illustrative output

The ILLUSTRATIVE analysis example (`python -m app.cli example-analysis`) adds to the register example a deadline of 20 working days, a delay-day cost of INR 8,000, liquidated damages of INR 5,000 per day, and six responses with placeholder costs and effects, one of them marked infeasible. All of these numbers are Assumptions of the example and are not evidence, site data or an expert’s estimate. Figure 5.11 shows the entry fields and Figure 5.12 the decision part of the screen after the analysis was run.

BRICKS-LY

ILLUSTRATIVE analysis example

NO UNSAVED CHANGES

NEW

OPEN

SAVE

LOAD EXAMPLE

AI: OFFLINE MODE

DEFINE

[1] Case & Activity

IDENTIFY

[2] Site facts & Rules

[3] Risk register

PRIORITISE

[4] Risk matrix

DECIDE

[5] Cost, options & decision...

REFERENCE

[6] Evidence

[7] Export

Settings

Offline tool. The register and matrix come from the service; the GUI computes nothing.
v1.0.0 - 472 evidence records

RISKS IN REGISTER

7

COMPLETE

7 / 7

ON MATRIX (ENTERED NUMBERS)

7

HIGH

1

MODERATE

5

LOW

1

RULE-FLAGGED

6

ILLUSTRATIVE case: every number is a placeholder, not evidence and not site data.

LOAD ILLUSTRATIVE EXAMPLE

RUN ANALYSIS

ANALYSIS.MD

[STEP 7]

Cost, options & decision

Turns the register's delay risks into a simulated schedule and cost, compares candidate responses on the same random draws, and states which option is preferred under the chosen criterion. Every cost and effect is your input with a Source; the tool never supplies one. Not a claim of optimality.

[COST MODEL]

every number needs a Source

Cost of one delay day

8000 INR ASM Assumption

ILLUSTRATIVE placeholder, not evidence

Liquidated damages per day after the deadline

5000 INR ASM Assumption

ILLUSTRATIVE placeholder, not evidence

Deadline for this activity

working days ASM Assumption

ILLUSTRATIVE placeholder, not evidence

[CANDIDATE RESPONSES]

6 entered

RESPONSE	TARGETS RISK	COST	PROBABILITY AFTER	DELAY AFTER
M-BUF Buffer stock of bricks and mortar materials	R-MAT	60 INR ASM Assumption ILLUSTRATIVE placeholder, not evidence	0.2 ASM Assumption ILLUSTRATIVE placeholder, not evidence	not set
M-SUP Second approved brick supplier	R-MAT	30 INR ASM Assumption ILLUSTRATIVE placeholder, not evidence	0.35 ASM Assumption ILLUSTRATIVE placeholder, not evidence	not set
M-GANG Extra mason gang for the critical weeks	R-LAB	90 INR ASM Assumption ILLUSTRATIVE placeholder, not evidence	not set	PERT 0 / 1 / 3 d
M-COV Covers and sheltered work sequence for the rain period	R-WX	25 INR ASM Assumption ILLUSTRATIVE placeholder, not evidence	0.12 ASM Assumption ILLUSTRATIVE placeholder, not evidence	not set
M-QC Inspection hold point before each course	R-RWK	15 INR ASM Assumption ILLUSTRATIVE placeholder, not evidence	0.15 ASM Assumption ILLUSTRATIVE placeholder, not evidence	not set
M-MECH Switch to larger blocks	R-SKILL	40 INR ASM Assumption ILLUSTRATIVE placeholder, not evidence	0.1 ASM Assumption ILLUSTRATIVE placeholder, not evidence	not set

Figure 5.11. Cost, options & decision screen, inputs (cost model and candidate responses), ILLUSTRATIVE analysis example. Cropped. In this capture the narrow number fields clip the entered values: the cost of M-BUF is stored as 6000 but the field shows “60”, and the deadline field (20 days) shows no digits. The data are intact; the display is a defect recorded in Chapter 6.

[✖ DECISION]							criteron: min_expected_total_cost
AUTHORIZE MITIGATION							
Preferred option: O-M-BUF+M-QC							
Under the criterion 'minimum total expected cost $TEC = C_{mitigation} + E[C_{residual}]$ ', 'Buffer stock of bricks and mortar materials + Inspection hold point before each course' is preferred among the 22 admissible options evaluated: it costs 7,500 INR and changes total expected cost by -7,047 INR compared with accepting the risk.							
Mitigation cost 7,500 INR; expected total cost saving 7,047 INR; expected delay avoided 1.30 days. Closest alternative is only 130 INR (0.3%) more expensive.							
Seed check (12345, 12346, 12347): the choice is stable .							
Result depends on user/expert inputs marked as assumptions; it is not a claim of optimality.							
[OPTIONS COMPARED]							same random draws for every option
OPTION	MITIGATION COST (INR)	RESIDUAL EXPECTED COST	TOTAL EXPECTED COST	P90 (D)	P(T > DEADLINE)	ADMISSIBLE	
Accept (no response)	0	50,821	50,821	25.6	59.8%	yes	
Buffer stock of bricks and mortar materials	6,000	38,929	44,929	24.3	45.7%	yes	
Second approved brick supplier	3,000	45,046	48,046	25.1	52.5%	yes	
Extra mason gang for the critical weeks	9,000	48,737	57,737	25.2	58.7%	yes	
Covers and sheltered work sequence for the rain period	2,500	48,270	50,770	25.4	57.5%	yes	
Inspection hold point before each course	1,500	48,018	49,518	25.4	56.8%	yes	
Buffer stock of bricks and mortar materials + Extra mason gang for the critical weeks	15,000	36,933	51,933	23.9	43.5%	yes	
Buffer stock of bricks and mortar materials + Covers and sheltered work sequence for the rain period	8,500	36,523	45,023	23.9	42.8%	yes	
Buffer stock of bricks and mortar materials + Inspection hold point before each course (preferred)	7,500	36,274	43,774	24.0	42.4%	yes	
Second approved brick supplier + Extra mason gang for the critical weeks	12,000	43,003	55,003	24.6	51.0%	yes	
Second approved brick supplier + Covers and sheltered work sequence for the rain period	5,500	42,573	48,073	24.8	49.9%	yes	
Second approved brick supplier + Inspection hold point before each course	4,500	42,305	46,805	24.8	49.4%	yes	
Extra mason gang for the critical weeks + Covers and sheltered work sequence for the rain period	11,500	46,200	57,700	24.9	56.2%	yes	
Extra mason gang for the critical weeks + Inspection hold point before each course	10,500	45,945	56,445	24.9	55.6%	yes	
Covers and sheltered work sequence for the rain period + Inspection hold point before each course	4,000	45,491	49,491	25.1	54.4%	yes	
Buffer stock of bricks and mortar materials + Extra mason gang for the critical weeks + Covers and sheltered work sequence	17,500	34,544	52,044	23.6	40.4%	yes	

Figure 5.12. Decision part of the same screen after “Run analysis”: the command, the criterion, and the first rows of the table of options compared. Cropped. Every number is derived from ILLUSTRATIVE placeholders.

With those placeholders the tool reports an expected delay of 5.13 days, a probability of 59.8 % of finishing after the deadline, and the command AUTHORIZE MITIGATION for the option “buffer stock plus inspection hold point” (mitigation cost INR 7,500; change in total expected cost INR -7,047 against accepting). The closest alternative is INR 130 more expensive, a gap of 0.3 %, which the tool itself describes as close. The preferred option is the same under three seeds and changes to “accept” when the delay-day cost is scaled to a quarter. This shows how sensitive the command is to its inputs, which are invented. It is not a result about any real project. Section 6.5 compares the first of these figures with an independent calculation.

5.12.5 Limits

The layer is a calculator over user inputs. It is only as good as the probabilities, delay ranges, costs and response effects entered, none of which this project has measured. Independence between risks is the default, though correlations can be entered. A pre-implementation review noted that the evidence base offers no masonry-specific effect sizes for any response, and the layer was therefore built so that it refuses to invent them. The distribution shape (PERT with $\lambda = 4$) has no site calibration. Whether a site manager would use the screen, or would trust its command, has not been tested.

5.13 Summary

The tool is a deterministic, offline register-and-matrix aid with a transparent rule base, a labelled literature tier, a guarded AI suggestion path that cannot supply numbers, and an optional cost layer that uses only the user's own numbers. Its main design decisions are to keep the engine separate from the interface, to attach a Source to every number, to prefer “not evaluable” or “not placed” over a guess, and to word outputs as ordinal or conditional. Its main design weaknesses, carried forward to Chapter 6 and Chapter 7, are the Assumption-laden class edges and thresholds, the use of one RII band on two axes, the thin evidential basis of the rules, and the absence of any evaluation with practitioners.

Chapter 6. Verification and Testing

This chapter reports what was checked, how, and with what result, and then states what was not checked. The checks establish that the software does what Chapter 5 says it does: that it computes the matrix cells it claims to, rejects the input it claims to reject, keeps its wording within limits, and gives the same output for the same input. They do not establish that the tool is useful to a site manager, that its probabilities or thresholds are right, or that it improves any decision. No expert evaluation, no usability study and no site trial were carried out, and Section 6.9 lists what that leaves open.

All counts in this chapter were produced by commands run on 8 October 2026 against commit 9f265f9 of the repository (working tree with the report files untracked), on Linux with Python 3.13.16. Commands and scripts are named so that the reader can repeat them.

6.1 Verification approach

Four kinds of evidence are used, in order of strength for the claim they support.

1. **Automated tests** (Section 6.2 to 6.3) show that specified behaviour holds for the inputs the tests choose. They are only as good as those inputs.
2. **Independent recomputation** (Sections 6.4 to 6.6) shows that the numbers the tool prints for the ILLUSTRATIVE example equal numbers obtained by a separate calculation, by hand in the text and by a second implementation of the arithmetic in `report/scripts/hand_check.py`.
3. **Robustness probing** (Section 6.7) shows how the tool behaves on malformed and hostile input. It was done by an independent QA pass and repeated for a sample of its findings in this session.
4. **Audit and review** (Section 6.8) shows what independent reviewers found wrong, and what was changed.

The ILLUSTRATIVE example is used for the worked checks because it is the only case with complete numbers. Its probabilities, delays, costs and planned duration are placeholders labelled Assumption. A match between hand calculation and tool therefore shows that the arithmetic is right. It says nothing about whether the placeholders resemble any site.

6.2 The automated test suite

6.2.1 How it was run

pytest is not installed in the system Python of the environment used for this report. It was run from the isolated environment of the `pytest` tool that was already present, with the project's dependencies on the path:

```
PYTHONPATH=/usr/local/lib/python3.13/dist-packages \
/root/.local/share/uv/tools/pytest/bin/python -m pytest -q -rA
PYTHONPATH=... python -m pytest --collect-only -q
```

The run took 29 seconds. The result was **513 passed, 3 skipped, 0 failed** from **516 collected tests** in 17 files. The earlier project documents quote 137 passed and 3 skipped on 8 October before the committee changes (Handoff section 8), and 133 before that; the larger number now is mostly parametrised cases of the hardening tests (one test function can generate many cases) and the tests added with the committee upgrades, not 380 new behaviours.

Two further test files, `tests/test_explain.py` (7 test functions) and `tests/test_india_norms.py` (6 test functions), are not collected. `tests/conftest.py` excludes them with `collect_ignore` because they import service functions (`explain_result`, `india_norms`) of the earlier simulation, cost and decision features that the lean application no longer exposes. They are kept on disk until their removal is confirmed. These 13 test functions did not run and are not counted as passed.

The three skipped tests are in `tests/test_corpus.py` and skip on a missing optional resource: the bulk literature harvest (`data/openalex_corpus.jsonl`, not in git) and the SQLite database `data/literature.db` (not in git, built by `scripts/build_literature_db.py`). The tests that check the database against its screening log and the 3,000-record work target therefore did not run in this environment, and the corpus-size discrepancy noted in the audit (finding F08) is not tested.

6.2.2 Tests per file

Table 6.1. Tests per file: collected, passed, skipped (all failed counts are zero). Collected counts from `pytest --collect-only -q`; passed and skipped from `pytest -rA`, both run for this report. A parametrised test counts once per case. Source: command output saved under the session scratch directory; commands as in Section 6.2.1.

File	Collected	Passed	Skipped	What it covers
<code>test_validation_hardening.py</code>	140	140	0	Input validation: non-finite numbers, booleans as numbers, bad shapes, missing Source, probability and impact edge rules, string facts
<code>test_export_api_hardenin g.py</code>	129	129	0	Exports and HTTP layer: CSV injection, Markdown escaping, BOM, odd bodies on every POST route, host check, guard rules G7 to G9, settings masking
<code>test_service.py</code>	42	42	0	<code>run_case</code> : example reproducibility, matrix rows, literature tier, rules in the service, register fields, exports
<code>test_engine.py</code>	32	32	0	Distribution means and variances, Monte Carlo against analytic moments, reproducibility, EMV, residual risk, constraints, audit hash
<code>test_analysis.py</code>	29	29	0	Decision layer: sections of the result, wording, input errors, deadline criterion, API routes
<code>test_legacy_hardening.py</code>	26	26	0	Earlier simulation, EMV and productivity code: bad values rejected, EMV against hand calculation, Monte Carlo mean within 3 standard errors
<code>test_options.py</code>	17	17	0	Response options: break-even targets, option

File	Collected	Passed	Skipped	What it covers
				generation, net benefit, command wording, seed stability, catalogue has no numbers
test_settings.py	16	16	0	AI key storage: outside the project, masked, never echoed, retry behaviour
test_relevance_guard.py	14	14	0	Guards that tie documents, data and wording to the facts (Section 6.3)
test_gui_api.py	14	14	0	HTTP routes of the interface
test_corpus.py	13	10	3	Evidence store and corpus loading; the 3 skips need data not in git
test_productivity.py	13	13	0	Earlier productivity models (out of scope)
test_gui_settings_api.py	12	12	0	Settings routes
test_ai_guard.py	10	10	0	Guard rules G1 to G4, audit log, promotion rules
test_rules.py	5	5	0	Rule firing, missing facts, priority and conflicts, consistency checker
test_gui_static.py	3	3	0	Static files are UTF-8 without garbled characters; opening a case does not auto-add risks; honest wording
test_report.py	1	1	0	Earlier Monte Carlo report
Total	516	513	3	

Two things stand out. First, the two largest files are hardening tests added after audit and QA findings, so more than half of the suite is a regression guard for defects that were found and fixed (Section 6.8). Second, 40 tests (`test_legacy_hardening`, `test_productivity` and `test_report`) cover earlier code that is out of scope and not used by the interface; they pass but support no claim about the current tool. The same holds in part for `test_engine.py` and `test_options.py`, which test the simulation and decision modules; those modules are live only in the optional decision layer.

6.2.3 Test categories

The suite can be read as six kinds of test.

Unit tests of the logic (`test_engine`, `test_rules`, parts of `test_validation_hardening`).

Examples: the closed-form mean of each delay distribution equals the mean from a simulation within sampling error; a value exactly on a class edge goes to the higher class; a brute-force check over a one-decimal grid of ratios agrees with exact arithmetic for the impact class (`test_m4_brute_force_one_decimal_grid_matches_exact_arithmetic`); a rule with a missing fact is not evaluable and never fires; two rules of equal priority and different levels give **conflict**.

Integration tests of the service (`test_service`, `test_analysis`). They run whole cases through `run_case` or `run_analysis` and check the result: that the example is reproducible, that every matrix row equals the engine's classification of the register risk, that an entered number replaces the literature tier, that a rule flag raises the probability class by one for a seeded risk only, that a risk with partial numbers stays in the register, and that every evidence ID cited by the example exists in the corpus.

API and export tests (`test_gui_api`, `test_export_api_hardening`, `test_gui_settings_api`). They call the Flask routes with valid, malformed and hostile bodies, and parse the exports (CSV with the CSV module, HTML for unescaped markup, SVG as XML).

Guard tests (`test_ai_guard`, `test_relevance_guard`, parts of `test_export_api_hardening`). They check the AI guard rules and the document and wording guards of Section 6.3.

Regression tests for named defects. The hardening files name the audit item they guard: `test_h1_*` (non-finite numbers), `test_h2_*` (delay without Source), `test_m1_*` to `test_m9_*`, `test_l1_*` to `test_l5_*` and `test_qa14_*`. Each test was written after the defect was reproduced, so each one is a record that the defect existed.

What the suite does not include. There are no automated tests of the JavaScript interface code: `test_gui_static.py` checks encoding, wording and one behaviour of the load function by reading the source, but no browser test drives the screens. The browser behaviour was checked manually and with a Playwright pass (Section 6.7), whose scripts are not in the repository. The Qt desktop wrapper has no test in the suite; it has a self-test in the build workflow that was not run here (Section 5.11.2). There is no test of a real LLM reply for the Anthropic path (no key was available), and no coverage measurement was made, so the proportion of lines exercised is not known.

6.3 The relevance guard tests

`tests/test_relevance_guard.py` holds ten guards added on 8 October 2026 after the committee review. They protect the report and the interface from drifting away from the data. In this report they are cited as RG-1 to RG-10, matching the function names `test_g1_*` to `test_g10_*` in the file; they are distinct from the AI guard rules G1 to G9 of Chapter 5.

Table 6.2. The relevance guard tests. Source: `tests/test_relevance_guard.py`; all 14 test cases (RG-5 has two parametrised cases; RG-4, RG-6 and RG-9 have two functions each) passed in the run of 8 October 2026.

Guard	What it asserts	Why it matters
RG-1	Counts read from the data are consistent with each other: seeded plus unseeded equals the number of library risks (46), seed plus held-out studies equals the number of studies (11), the unseeded risks are exactly those with a <code>seed_status</code> note, IDs are unique	These counts are quoted throughout the report; they are not typed into the test
RG-2	README.md, docs/case_schema.md and gui/README_GUI.md state the rule, row, study and risk counts found in the data	Documents cannot silently go stale
RG-3	The 182 seed rows are frozen: a checksum of the study, factor, rii and rank fields equals the stored value	A change to a transcribed survey value fails the suite
RG-4	Withdrawn wording (“validated”, “predicts”, “built from every factor” and others listed in Handoff section 8) does not occur in live text; the handoff mentions withdrawn wording only on lines that say it is withdrawn	Keeps the claims within what the work supports
RG-5	Every number in the exports has its Source printed next to it, for the example and for a case with the whole library added	Integrity rule 2
RG-6	Legacy modules are not imported by the live application, and the interface does not load the legacy screens	Out-of-scope code cannot influence outputs
RG-7	The example case stays labelled ILLUSTRATIVE in every export	Placeholders cannot be mistaken for data
RG-8	Rule-base sanity: every rule is Assumption-sourced or carries evidence IDs; no pair of rules on one risk has different levels at equal priority; every fact used by a rule has a label on the facts screen; every rule targets a library risk	Rule base stays consistent with the interface
RG-9	Unseeded library risks carry the <code>seed_status</code> note and have no seed rows; evidence IDs of the eight added risks exist in the corpus	The 16 unseeded risks cannot acquire a literature tier by accident
RG-10	promote refuses a derived (AI) value as the source of p or delay; the response of <code>/api/suggest-risks</code> contains no numeric-looking key	NFR3

These guards test consistency, not truth. RG-3 shows that the seed rows are unchanged since the row-level check against the printed papers; it does not re-check them. RG-4 shows that listed phrases are absent; a new over-claim in different words would pass.

6.4 Hand calculation against the tool: the matrix

6.4.1 The example

The ILLUSTRATIVE example (`app.service.example_case`, also `examples/example_case.json`) has a planned duration $T_0 = 16$ working days, impact edges (0.02, 0.05, 0.10, 0.20), and seven risks with PERT delays. All values are Assumptions. The calculation uses the formulas of Section 5.6: $E[D] = (a + 4m + b)/6$, $r = E[D]/T_0$, c_p from the edges 0.2, 0.4, 0.6, 0.8, c_i from the impact edges, $s = c_p c_i$, and the level thresholds 5, 10, 15.

6.4.2 By hand, for R-MAT

R-MAT has $p = 0.50$ and a delay range $a = 1$, $m = 3$, $b = 8$ days.

- $E[D] = (1 + 4 \times 3 + 8)/6 = 21/6 = 3.5$ days.
- $r = 3.5/16 = 0.21875$, which is at least the fourth edge 0.20, so the impact class is $c_i = 5$.
- $p = 0.50$ is at least 0.4 and below 0.6, so the probability class is $c_p = 3$.
- $s = 3 \times 5 = 15$, and $11 \leq 15 \leq 15$ gives the level High.

For R-LAB ($p = 0.35$, $a/m/b = 1/2/6$): $E[D] = (1 + 8 + 6)/6 = 2.5$; $r = 0.15625$ lies in $[0.10, 0.20)$, so $c_i = 4$; $p = 0.35$ lies in $[0.2, 0.4)$, so $c_p = 2$; $s = 8$, Moderate. R-PLAN ($p = 0.10$, $a/m/b = 1/1/3$): $E[D] = (1 + 4 + 3)/6 = 1.3333$; $r = 0.08333$ lies in $[0.05, 0.10)$, so $c_i = 3$; $p = 0.10$ is below 0.2, so $c_p = 1$; $s = 3$, Low. The remaining four risks are computed in the same way in Table 6.3.

6.4.3 All seven risks

Table 6.3. Hand calculation and tool output for the seven risks of the ILLUSTRATIVE example. All inputs are placeholders labelled Assumption. “Hand” columns come from the formulas of Section 5.6 applied with a separate script (`report/scripts/hand_check.py`, which does not call `app.engine`); “Tool” columns come from `app.service.run_case`. Source: `report/tables/hand_check_output.txt`, produced by running the script.

Risk	p	a / m / b (d)	E[D] hand (d)	r = E[D]/16	c_p	c_i	s	Level (hand)	Tool: E[D], c_p, c_i, s, level
R-MAT	0.50	1 / 3 / 8	3.5000	0.2188	3	5	15	High	3.5000, 3, 5, 15, High
R-LAB	0.35	1 / 2 / 6	2.5000	0.1562	2	4	8	Moderate	2.5000, 2, 4, 8, Moderate
R-RWK	0.25	1 / 2 / 5	2.3333	0.1458	2	4	8	Moderate	2.3333, 2, 4, 8, Moderate
R-WX	0.20	1 / 2 / 7	2.6667	0.1667	2	4	8	Moderate	2.6667, 2, 4, 8, Moderate
R-PLAN	0.10	1 / 1 / 3	1.3333	0.0833	1	3	3	Low	1.3333, 1, 3, 3, Low
R-SAFE	0.30	1 / 2 / 6	2.5000	0.1562	2	4	8	Moderate	2.5000, 2, 4, 8, Moderate
R-SKILL	0.22	1 / 2 / 5	2.3333	0.1458	2	4	8	Moderate	2.3333, 2, 4, 8, Moderate

All seven rows agree: expected delay to within 10^{-9} days, and class, score and level exactly. The tool’s summary reports one High, five Moderate and one Low. The $p = 0.20$ of R-WX is exactly on the edge 0.2 and goes to class 2, which is the lower-edge-inclusive convention of Section 5.6.3; this case is a deliberate check of

the convention. The test `test_m4_exact_edge_goes_to_the_higher_class` guards the same behaviour.

What this check shows is limited. It confirms the arithmetic and the edge convention. The classes and levels depend on the impact edges (0.02 to 0.20), the probability edges and the score thresholds, which are Assumptions; with other edges the same seven risks would be classed differently. The sensitivity of the levels to those choices was examined in Chapter 4 and is not repeated here.

6.4.4 Rules for the example, by hand

The example's 28 facts (Chapter 5, Figure 5.6) fire nine rules. By hand, with the facts of the example (required workers 6, available 5, material stock 2 days, lead time 6 days, planned duration 16, monsoon overlap Yes, external walls Yes, schedule compressed Yes, planned daily output 10, achieved 8, prior rework Yes, skilled masons short Yes):

- RL-LAB fires ($6 > 5$), RL-MAT fires ($2 \leq 6$) at priority 1, RL-MAT2 fires ($2 < 16$) as **normal**, RL-WX fires as **normal**, RL-RWK fires, RL-PLAN fires, RL-SKILL fires, RL-WX2 fires (monsoon and external walls) at priority 1, RL-PACE fires ($10 > 8$).
- Thirteen rules do not fire: RL-TOOL (no tools shortage), RL-HGT (work at height but scaffolding is ready), RL-PAY, RL-DES, RL-FRONT (work front ready), RL-VT (floor 0 is below 1), RL-SCAF (scaffolding ready), RL-MORT ($8 > 5$ days), RL-GPAY, RL-OPEN (all three items ready), RL-FEST, RL-FEST2, RL-HEAT.

On R-MAT, RL-MAT (elevated, priority 1) and RL-MAT2 (normal, priority 0) both fire, and the higher priority wins, so R-MAT is **elevated**, not **conflict**. On R-WX, RL-WX (normal, 0) and RL-WX2 (elevated, 1) give **elevated**. The tool reports the same nine fired rules, no rule not evaluable, and six elevated risks (R-LAB, R-MAT, R-WX, R-RWK, R-PLAN, R-SKILL), matching the “rule-flagged 6” in the key-figure strip of Figure 5.8. R-SAFE has no flag, as its scaffolding is ready.

6.4.5 Literature tier, by hand

The tier is not used by the example, whose risks all have entered numbers. To check the tier arithmetic, a case with R-MAT entered and R-TOOL and R-WX left without numbers was run, with `tools_shortage` set to true and the other facts as in the example. The seed table (`app.literature_seed.seed_table()`) gives R-TOOL a mean RII of 0.7528 (study means M01 0.7455 and A02 0.760, the mean of the two study means) at rank 12 of 30, and R-WX a mean RII of 0.6934 (A14 0.6528, M01 0.764, A02 0.700, M06 0.6567) at rank 24.

- R-TOOL: tier = $5 - \lfloor 5 \times 11/30 \rfloor = 5 - 1 = 4$. It is flagged elevated by RL-TOOL, so $c_p = \min(5, 4 + 1) = 5$ and $c_i = 4$, giving score 20.
- R-WX: tier = $5 - \lfloor 5 \times 23/30 \rfloor = 5 - 3 = 2$. RL-WX2 (elevated, priority 1) beats RL-WX (normal), so $c_p = 3$ and $c_i = 2$, giving score 6.

The tool reports exactly these cells and marks both as “literature tier (Assumption)” with `raised_by_rule` true. Its level totals count only R-MAT (High), as the specification requires: the seeded risks do not enter the High, Moderate or Low counts even though their cells carry a colour. Band membership also holds in the data: the 30 seeded risks fall six into each tier (Counter of tiers 1 to 5: 6, 6, 6, 6, 6). Both the 20 and the 6 are cell positions produced by two Assumptions (one RII band on both axes, +1 for a flag), and not assessments of those risks.

6.5 Independent check of the decision layer

The decision layer is checked on the ILLUSTRATIVE analysis example, again with placeholders. Two kinds of check are possible: closed-form quantities that can be computed by hand, and simulated quantities that need an independent simulation.

Closed-form: expected delay and event EMV. The expected total delay of independent additive risks is the sum of $p_i E[D_i]$:

$0.50 \times 3.5 + 0.35 \times 2.5 + 0.25 \times 2.3333 + 0.20 \times 2.6667 + 0.10 \times 1.3333 + 0.30 \times 2.5 + 0.22 \times 2.3333 = 1.75 + 0.875 + 0.5833 + 0.5333 + 0.1333 + 0.75 + 0.5133 = \mathbf{5.1383 \text{ days}}$.

At INR 8,000 per delay day the sum of event EMVs is $5.1383 \times 8,000 = \mathbf{INR 41,106.67}$. The tool's event EMV rows sum to INR 41,106.67, equal to the cent.

Simulated: expected delay. The tool's 10,000-run simulation with the default seed gives an expected delay of 5.1323 days with a standard error of 0.0334, which is 0.18 standard errors below the hand value. A second implementation written for this check (`hand_check.py`, own beta sampler, own random generator) with 4,000,000 runs gives 5.1401 days (standard error 0.0016), 1.1 standard errors above the hand value and consistent with it. The tool with 200,000 runs gives 5.1521.

Simulated: probability of any delay. Every risk, if it occurs, adds at least one day, so the probability that the activity finishes late is $1 - \prod(1 - p_i) = 1 - 0.5 \times 0.65 \times 0.75 \times 0.8 \times 0.9 \times 0.7 \times 0.78 = \mathbf{0.9042}$. The tool's default run reports 0.8975, 0.0067 below, which is 2.3 standard errors of a proportion at 10,000 runs and therefore looks like a sampling fluctuation but needs a check. Six other seeds give 0.9084, 0.9037, 0.9039, 0.9028, 0.9037 and 0.9092, and 200,000 runs give 0.9036, all in agreement with 0.9042. The seed 12345 simply happens to be a low draw for this quantity. This is the reason the tool shows the number of runs, the seed and standard errors, and why a result should not be read to more digits than the sampling error allows.

Deadline and cost. The probability of exceeding the 20-day deadline and the expected cost (which includes the liquidated-damages term) have no closed form here, because they depend on the distribution of the sum. The tool reports $P(T > 20) = 0.598$ and an expected cost of INR 50,821 at 10,000 runs. The independent 200,000-run simulation gives 0.6025 and INR 50,929. The 200,000-run tool run gives 0.6041. The two simulators agree to within about 0.5 % on the cost and 0.5 percentage points on the probability, which is within the sampling error of the 10,000-run result. As a consistency check on the liquidated-damages term, the expected cost minus $8,000 \times 5.1323 = \mathbf{INR 41,058}$ leaves about INR 9,763, which is the expected liquidated damages $5,000 \times E[\max(T - 20, 0)]$, so $E[\max(T - 20, 0)]$ is about 1.95 days; this is a derived consistency figure and was not computed independently.

Decision command. The command (AUTHORIZE MITIGATION with “buffer stock plus inspection hold point”) rests on the simulated cost of 22 admissible options and was not recomputed by hand. What was verified is that the preferred option does not change with the seed (three seeds), that the repository's tests compare the net benefit of a response with the simulated comparison when the cost is linear (`test_net_benefit_agrees_with_simulated_comparison_when_cost_is_linear`), and that the break-even targets agree with hand checks (`test_break_even_targets_hand_checked`). The gap between the first and second option is 0.3 % of total expected cost, smaller than a difference that the sampling error of a 10,000-run simulation would make dependable by itself; the tool reports the gap for that reason. The result is a demonstration of the arithmetic on placeholders and not a recommendation about any activity.

6.6 Reproducibility and determinism

Repeating the same case gives the same output. In the QA run, six cases were each sent 96 times concurrently, in forward and reversed order, and the hashes of `/api/run` and of all four exports were identical across repetitions; no server state carried between requests. The test `test_example_case_builds_the_register_and_matrix_and_is_reproducible` and `test_result_is_reproducible_and_input_is_not_modified` give the same check in the suite. The Monte Carlo layer is reproducible for a given seed and number of runs, as the hand check above shows by the stable 12345-seed values, and the result of a change of seed is shown, not hidden.

6.7 Robustness and fuzz testing (QA report)

6.7.1 What the QA pass did

An independent QA pass on 8 October 2026 tested the interface and service with scripted fuzzing and a real Chromium run. The report is at `/tmp/claude-0/-home- claude- fyp/b6e485db-2d31-5ab7-a4a5-10d8b255e54b/scratchpad/qa/qa_report.md`, and the scripts it names (`fuzz.py`,

fuzz2.py, probe*.py, det.py, b1.py to b4.py) are in the same session directory, not in the repository. The pass covered API fuzzing of every route with empty, malformed, wrongly typed and oversized bodies; export checks; determinism; the settings store; the command line; a Chromium run through all screens, themes and widths with XSS payloads in every text field; and a static review. The suite at that time had 137 passed and 3 skipped.

What it found to be sound is recorded in the report: all POST routes returned 422 or `ok:false` for empty bodies, bad JSON, wrong content type and list or null bodies (except two routes, defect 6 below); a 30 MB body returned 413; 5,000 risks validated in 0.4 seconds and 1.2 MB names were handled by all five exports; probabilities outside [0, 1] in object form, reversed delay ranges, negative delays, bad planned durations, duplicate IDs and bad impact edges were all rejected with clear messages; CSV parsed with 17 columns on every row at that time; the SVG was well-formed; the HTML report contained no raw script or image tags; path-traversal strings returned 404; the register was identical after save and reopen of the unmodified example; and none of the XSS payloads executed on any screen or in the problem list.

6.7.2 Defects found, and their status

The QA report listed 17 defects. They were reported at a commit before the committee upgrades; the table gives the status found when selected items were re-probed in this session on the current code, with the test that now guards each.

Table 6.4. QA defects and their status. “Re-probed” means the reproduction of the QA report was repeated in this session against the current code (command output not saved to a file); “test” names the guard in the suite. Source: QA report, section “Defects”; probes run on 8 October 2026.

#	Severity	Defect (QA report)	Status on current code
1	medium	Opening a saved case re-added a rule-flagged risk the user had deleted	Fixed: <code>test_load_case_does_not_auto_add_risks</code> in <code>test_gui_static.py</code> (checks the source of the load function)
2	medium	Non-string names and non-object activity gave HTTP 500	Re-probed: now <code>ok:false</code> with “activity.name must be text, not int” and “activity: expected an object”; <code>test_m6_non_string_names</code> , <code>test_validate_never_returns_html_for_odd_case_shapes</code>
3	medium	CSV formula injection	Re-probed: cell now written as text with a leading apostrophe; <code>test_csv_cells_starting_with_formula_chars_are_neutralised</code>
4	medium	Invalid bare p (-1, 1.5, “abc”) silently treated as not entered	Re-probed: <code>p = -1</code> gives “expected an object with value and source”; <code>test_m2_bare_probability_value_rejected</code>
5	low-med	register.md table broken by \ or newline in names	Re-probed: pipe escaped, newline replaced; <code>test_markdown_table_escapes_pipes_and_newlines</code>
6	low-med	/api/suggest-risks and /api/settings/test returned HTML 500 for non-object bodies	Re-probed: HTTP 422; <code>test_settings_test_and_suggest_reject_wrong_types</code>
7	low	Garbled characters (“Â”, “â€”) in app.js	Fixed: <code>test_static_files_are_utf8_without_mojibake</code>
8	low	NaN, Infinity and booleans accepted as numbers	Re-probed: “must be a finite number”; boolean gives “must be a number, not true”; <code>test_h1_*</code> , <code>test_m1_booleans_are_not_numbers</code>
9	low	Case with bad types left the screen at “Checking the case...” with the cause hidden	Same validation as 2; screen behaviour not re-probed in a browser
10	low	Export screen overflowed at 800 px	Handoff section 9 records a screenshot check at 800 px with

#	Severity	Defect (QA report)	Status on current code
			no overflow; not re-probed here
11	low	Deeply nested JSON gave HTML 500	Re-probed: HTTP 422 with a JSON body; test_unhandled_exception_becomes_json_500, test_every_post_route_returns_clean_json_error_for_odd_bodies
12	low	CLI printed tracebacks for bad files	Re-probed for a missing file: one-line message; test_cli_friendly_errors
13	low	CSV had no BOM	Fixed: the HTTP route and the CLI encode as utf-8-sig; test_register_csv_endpoint_has_bom_and_is_parseable, test_cli_writes_csv_with_bom
14	low	Unknown evidence IDs accepted without warning	Re-probed: a warning is given; test_qa14_unknown_evidence_ids_are_warnings_not_errors
15	info	Masked key revealed short keys; any Host header accepted	Fixed: test_masked_key_never_reveals_a_short_key, test_host_header_is_checked
16	info	Dead JavaScript screens shipped; dark-theme level colours differ from the specified values	Dead-code part guarded by RG-6; dark-theme colours not re-checked here
17	low	Label for attributes missing; one select text clipped; toast uses innerHTML for a few server strings	Not re-checked; treat as open

Fourteen of the seventeen were re-probed or have a named guard test that passed; items 9, 10, 16 (dark-theme part) and 17 were not re-checked in this session. A new display defect was found while capturing the screenshots for Chapter 5 and is not in the QA report.

6.7.3 A defect found in this verification

On the “Cost, options & decision” screen at a viewport width of 1280 px, the number fields for the response costs and the deadline are too narrow for their values. A measurement in the page shows the deadline field (value 20) with a client width of 20 px and a content width of 41 px, and the cost fields (value 6000) 37 px wide against 56 px of content, so the digits are cut off (Figure 5.11 shows “60” for 6,000 and an empty-looking deadline). The stored values are correct and are the ones used in the analysis; the display misleads the user about what is entered. This is a usability defect in a part of the tool that decides on money. It is not fixed, and it is listed with the known defects below.

6.8 Audit findings and what was corrected

6.8.1 The audit

Before the committee upgrades, a research-methods audit of the repository (30 findings, F01 to F30), a row-level check of the literature seed against the printed papers (17 findings, V-prefixed), and an engine audit (3 high, 10 medium and 14 low findings, H, M and L) were carried out.

`docs/Audit_Corrections_2026-10-08.md` lists each with its status. The status tags in the file (a finding can carry two) show that, of the 30 research findings, 19 were fixed in documents, 5 in data, 12 needed a code change when the file was written, and 10 were left as decisions for the supervisor.

6.8.2 The main result: the data were right, the claims were not

The row-level check compared all 182 RII values and ranks in `data/literature_seed.json` with the printed papers and found them all to match. No row transcribed from M01 (28 of 28 table rows), M06 (40 of 40), A14 (as entered) or M10 (33 of 33 Table 7 rows) was missing. What was wrong was metadata, notes and wording. This finding supports the reliability of the transcription; it does not extend to the mapping of survey items to library risks, which is a judgement (finding F12), and for which no second coder or agreement statistic exists. Two further entries, A11 and P1-23 (listed under “other indices” in the seed file, not used in the seed), were not verified against PDFs because the files were not on disk.

6.8.3 Statistical findings that were corrected

The audit found the following problems in what had been said about the literature seed. Each was corrected in the documents; the scripts that generate some of the old figures were not all changed (Audit_Corrections section 5).

- **Circular agreement figures** (F01). The agreement of the pooled ranking with the seed (ρ 0.73) included the seed studies themselves. The held-out-only values are ρ 0.08 ($n = 22$, $p = 0.73$) for the pooled rank against the seed and 0.09 ($p = 0.70$) for the standardised pooled score. The circular values are withdrawn.
- **Over-claim of validation** (F05). “Validation”, “out-of-sample test” and “predicts the order of study X” were withdrawn. The leave-one-study-out comparisons against the seed give ρ 0.09 (M01, p 0.76), 0.37 (A02, p 0.29), 0.64 (M06, permutation p 0.066) and 0.60 (A14, n 4); these are seed studies, not held-out ones. For the three held-out studies M15, M10 and M14 the Holm-adjusted p are 0.13, 0.48 and 0.55. The proper description is an agreement check that is weak, low-powered and inconclusive.
- **Same band on both axes** (F02). Using one class on both axes squares the rank: 12 of the 30 seeded risks fall in the Extreme level and none in High. The use was kept as a labelled Assumption and the seeded cells were demoted to a “literature tier” display (Section 5.7).
- **Subset claims** (F03). Only 5 of the 39 factors of A14 were entered, so “built from every factor” was withdrawn and the subset stated.
- **Post-hoc finding** (F09). “Weather is rated lower than the rest” ($p = 0.049$) was item-level and post hoc; the study-level Kruskal-Wallis p was 0.25. It was withdrawn.
- **Kendall W** (F10). $W = 0.400$, permutation $p = 0.153$ (chi-square approximation 0.156; four studies, six risks) is no evidence of agreement and was reworded.

These are findings about the evidence behind the tier, discussed in Chapter 4. They are included here because the tier is a feature of the tool and its description in the interface changed as a result.

6.8.4 Engine and QA findings fixed in code

After the committee review the code changes in Audit_Corrections section 6 were made and tests added. Re-probing on the current code in this session (Section 6.7 and the checks below) shows that most items listed as “needs code change” in sections 4 and 5 of the audit file are now rejected or handled, although those sections of the file were not updated. The statuses found were:

- H1 non-finite numbers: a NaN planned duration is rejected with “must be a finite number”. The audit’s remaining point, the command line writing invalid JSON for NaN, was not re-probed.
- H2 delay without Source: rejected (“source is required”). Previously it was silently labelled Expert Judgment.
- M1 booleans, M2 out-of-range probability without a delay, M5 string facts, M6 malformed shapes, M9 edge count (six edges now rejected, “exactly four numbers”), L1 (a conflicting planned-duration fact is rejected), L4 (a string `use_literature_seed` is rejected): re-probed, all handled.
- M3, partly entered numbers: a risk with only p entered stays in the register with the entered p kept, is placed from the literature seed if it has a seed value, and carries a note that the p is not used for placement until the delay is entered. This is a documented design choice and not a defect.
- L5, a uniform distribution without m , is accepted, as intended: uniform uses only a and b .
- M4, rounding of values exactly on an edge: fixed with the 10^{-9} relative tolerance on both axes, with the brute-force grid test.
- M7, the unenforced sentence “No number here was produced by AI”: replaced (Section 5.8.3).

The deferred items, which remain open, are in Section 6.9.

6.9 Known defects and deferred items

This section lists what is known to be wrong or unfinished, so that nothing in the earlier sections is read as completeness.

Known defects

1. The number fields of the “Cost, options & decision” screen clip their values at 1280 px (Section 6.7.3).
2. QA items 9, 10, 16 (dark-theme colours) and 17 (label associations, one clipped select, `innerHTML` in a few toasts) were not re-checked, and item 17 is treated as open.
3. The stale sections 4 and 5 of `docs/Audit_Corrections_2026-10-08.md` still list items as needing code changes that the code now handles. The document describes a state before the committee changes and was only extended in its section 6. The report does not rely on those sections for status.
4. The scripts that generate corpus statistics (`scripts/corpus_statistics.py`, `scripts/stats_corpus.py`) still print the old margin-of-error and N sentences, so re-running them overwrites the corrected documents (F19, F20). The statistics scripts still compute the circular agreement figures (F01, F21), and still use the t-based p-value for two studies where a permutation p is recommended (F11).
5. The three corpus sizes (21,902 / 5,214; 17,788 / 3,975; 11,599 / 3,550) have not been reconciled (F08), and the test that would compare the database with its screening log is skipped in this environment.
6. The Windows executables and the Qt desktop window were not built or run in this verification (Section 5.11.2).
7. Retrieval is lexical; no test measures how good the retrieved records are, and the guard is pattern-based (Section 5.8.4).
8. The `explain` module with guard rule G6, and 13 test functions of the earlier decision features, are not exposed or run.

Deferred decisions (Audit_Corrections section 6 and Handoff section 9): an event-or-condition tag on library risks; merging and removing risks (R-TRN into R-SKILL, R-OVT with R-FAT, removal of R-PRD), which would change the seed mapping and every band; renaming risks into site language; the uncoloured-tier display and dropping the +1 step; entering the 34 omitted A14 factors; a second-coder check of the mappings (F12); handling of the truncated M01 table (F13); updating the stale `existence_evidence` of core risks (F16); recomputing M14 on its stated scale (V-M14); the sensitivity table of thresholds in the interface; the title-versus-scope decision (F07); the real title of HAS25 and the details of A15 (V-A15, V-HAS25).

6.10 What was not done

The verification stops where the evidence stops. The following were not done, and no statement in this report depends on them.

- **No expert evaluation.** No expert survey was run and no expert reviewed the library, the rules, the thresholds or the response catalogue. The statements that a rule or risk is “relevant” rest on literature records and on the team’s judgement, not on practitioner confirmation. The rules are labelled Assumption with Low or Moderate confidence for this reason.
- **No usability study.** No site manager or engineer used the tool. The claims about the interface (no free-text fact fields, plain error messages, Source badges) describe its design; whether they work for users is untested. The defect in Section 6.7.3 would probably have been found by even a short session.
- **No site data and no calibration.** No project data were collected, so nothing in this chapter shows that the tool’s classes or its decision layer agree with what happened on any project. No accuracy figure for any real site exists or is claimed.
- **No validation of the thresholds or the tier.** The probability edges, score thresholds and tier construction are Assumptions. Their effect was explored by sensitivity analysis (Chapter 4), which shows how much the output depends on them and not that any setting is correct.

- **No inter-rater check of the risk mapping**, no second coder, no kappa.
- **No test of AI usefulness or of the Anthropic path.** The guard blocks numbers; whether suggestions improve identification was not tested, and the Anthropic client was not run against the live service.
- **No code coverage measurement and no browser test suite in the repository.**
- **No penetration test or security review beyond the QA probes.** The tool is for a single user on one machine and should not be exposed on a network.

6.11 Summary

The automated suite passes (513 passed, 3 skipped for missing optional data, 0 failed, 516 collected), but more than half of it guards defects that were found and fixed, and under a tenth covers out-of-scope code. For the ILLUSTRATIVE example, an independent calculation reproduces every matrix cell, the nine fired rules, the tier arithmetic and the closed-form delay and EMV, and a second simulator agrees with the Monte Carlo layer within sampling error. Fuzz and QA probing found 17 defects, most of which are now fixed and guarded by tests, and this verification found one more, a display defect in the decision screen. The audit confirmed that the transcribed survey values match the papers and corrected a set of over-claims about what the seed shows. The tool has not been evaluated by experts or users, has not been run on any real project, and its Windows builds were not exercised here; those gaps limit what can be concluded about its worth, and they are the main limitations carried into Chapter 7.

Chapter 7. Discussion

This chapter says what the results of Chapters 4 to 6 mean, how far they can be pushed, how the tool compares with the other ways a site engineer could keep a masonry risk register, and what could make the conclusions wrong. It repeats results only where the argument needs them. All numbers carry a Source label; those labelled Derived Calculation come from `analysis/out/` or from the scripts in `report/scripts/`, and those labelled Literature are values printed in the surveys.

What was found, and how strongly

Four findings carry the report. Table 7.1 states each with the evidence behind it and the strength with which it can be said.

Table 7.1. Principal findings and the strength of the evidence. Source: Chapters 4 to 6; `analysis/out/` (Derived Calculation); `docs/Audit_Corrections_2026-10-08.md`; Appendix E.

Finding	Evidence	Strength
The transcribed RII values and ranks match the printed papers; errors found were in metadata and notes	Row-by-row check of 161 of the 182 rows against the PDFs on 2026-10-08, listed in the audit file; the 21 rows from A02, A15 and HAS25 were taken from research notes or an abstract and were not checked against a PDF	Strong for the 161 rows checked. The values of A11 and P1-23 were not re-checked, and the seed is a subset of the factors printed in several papers (A14 5 of 39, A15 5 of 32, A02 14 of 35)
Published RII rankings agree only weakly with one another on the risks they share	Held-out comparison: M10 rho 0.28 (n = 19), M15 0.63 (n = 11), M14 -0.18 (n = 13). Leave-one-study-out cross-prediction among seed studies: M01 0.09 (n = 13), A02 0.37 (n = 10), M06 0.64 (n = 9). Kendall's W 0.40, permutation p = 0.15 (4 studies, 6 risks)	Weak, low-powered and inconclusive: no comparison is significant after correction, and the overlaps are small enough that only large correlations could be detected
The class a seeded risk receives depends on banding and study choices; the level a risk receives from entered numbers depends more on probability class and thresholds than on the delay range	Seed variants: 6.7% to 80.0% of the 30 classes change across the pre-specified variants; example: no delay multiplier from 0.5 to 1.5 changes any of 7 levels, whereas halving every probability changes 6 of 7 (analyses S3 and S1)	Descriptive of the rules; says nothing about real risks, because the example inputs are placeholders
The software does what the documents say	513 tests passed, 3 skipped, 0 failed (Appendix E); worked example in Chapter 6	Verification only. Not validation, and not a guarantee for the defects still listed as open in the audit file

What was not found matters as much. No site data were collected, no expert was asked, and no user has been observed using the tool. The report therefore contains no evidence that the tool improves a decision, that its levels correspond to delay on a real site, or that a site engineer finds it usable. These are the open questions that Chapter 8 turns into future work.

Reading the weak agreement between surveys

The agreement analysis was built to ask how far the literature can stand in for project data. The honest answer is that it cannot be said, and the reasons deserve to be separated, because they lead to different remedies.

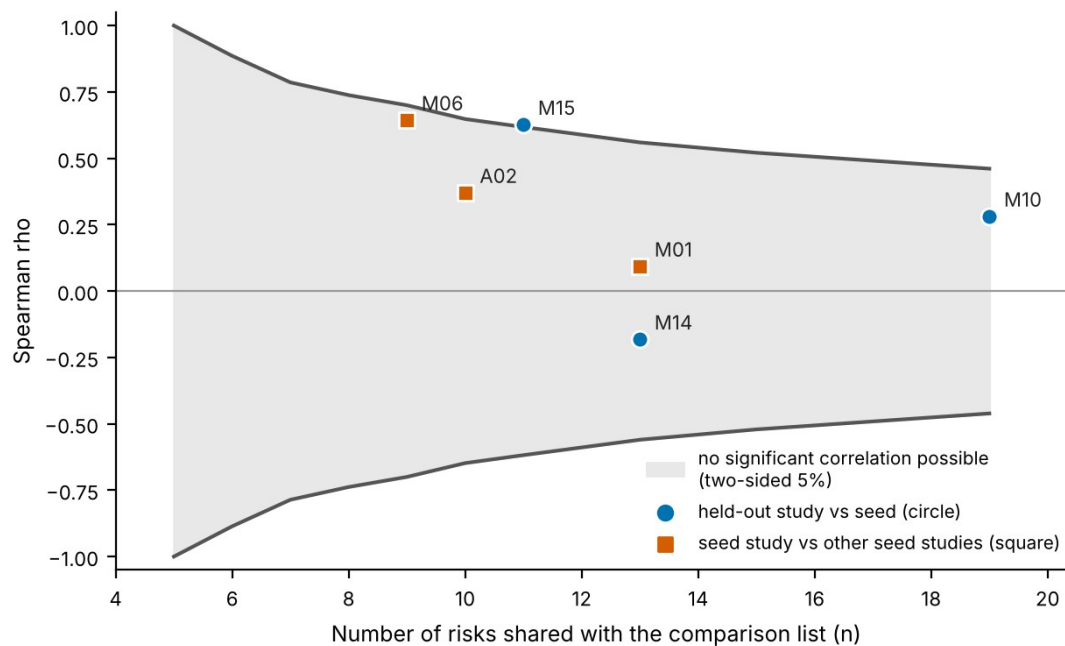
Real heterogeneity. The surveys differ in country, in the kind of project, in who answered and in what was asked. Two are Sri Lankan, one pools ten Middle Eastern studies, one is Brazilian, one Indonesian. M14 is the only held-out study to correlate negatively with the seed (ρ -0.18), and its authors rated factors with workers only (the managers were consulted on mitigation, not on the ratings); why the order differs is a hypothesis that this project cannot test. If the surveys really rank risks differently because the sites differ, then a pooled ordering describes no particular site, and the case for replacing it with local data is stronger.

Sampling noise. The RII of one item carries sampling error. For M01 ($N = 44$) an interval of about plus or minus 0.06 around a single RII is reasonable (Derived Calculation, docs/Project_Handoff_Summary.md section 8.12), while neighbouring seed risks differ by far less: the top six risks lie within 0.019 of each other, the top 24 within 0.096, and ranks 12 and 13 (R-TOOL and R-INC) differ by 0.0008 (Derived Calculation, audit file F30 and F02). The order of most risks is therefore not resolved by the data, and the sampling error would produce disagreement between surveys even if the true order were identical.

Different things being rated. An RII measures how important respondents rated a factor on a Likert scale. One survey may ask about causes of delay, another about causes of lost labour productivity (M01, A02, M06, A15 and M14 are productivity surveys; A14, A10, A21 and M15 are about delay). The same word, for example “material shortage”, can then carry different weight in the two lists. Mixing 1 to 4 and 1 to 5 scales also moves values, though rescaling changed only 2 of 30 seeded classes (6.7%, analysis S3), so scale mixing is the smallest of these concerns.

Selection of what is printed. M01 prints only factors with RII of 0.700 or more (28 of 38). A14 and A15 are represented by their top factors (5 of 39 and 5 of 32 entered). A risk whose only evidence comes from a top-factors list is pulled upwards by construction. Dropping A14 changes 6.7% of the classes and dropping A15 10.0% (analysis S3), which is small, but A02 is influential: dropping it changes 13 of the 27 classes of the risks that remain (48.1%).

Low power. With few shared risks, only a large correlation can be distinguished from chance. Figure 7.1 shows the smallest correlation that would be two-sided significant at 5% for each overlap (Derived Calculation, report/scripts/critical_rho.py: 0.62 at $n = 11$, 0.56 at $n = 13$, 0.46 at $n = 19$) together with the observed values. M15’s ρ of 0.63 at $n = 11$ sits just beyond the line (raw permutation $p = 0.045$), while the others lie well inside. After Holm correction over the three held-out comparisons (Holm, 1979), the adjusted p -values are 0.13 (M15), 0.48 (M10) and 0.55 (M14); over all six testable leave-one-study-out comparisons of analysis S4 the adjusted value for M15 is 0.27. None is significant. A non-significant result with this power is not evidence of disagreement either: it leaves the question open.



Derived Calculation; 6 comparisons with $n \geq 5$ (S4), 200,000 permutations ($n > 8$), seed 20261008

Figure 7.1. Smallest absolute Spearman correlation that could be called significant (two-sided, 5%) at each overlap of shared risks, with the observed leave-one-study-out correlations. Derived Calculation from `analysis/out/s4_leave_one_study_out.csv` and `report/scripts/out/critical_rho.csv`.

Taken together, the results support a narrow statement: the literature gives a coarse order, with a few risks (material availability, skilled labour) near the top in several lists and a few (social environment, training, client decisions) clearly lower, but it does not give a stable ranking of the middle. The bootstrap over the five seed studies agrees: three risks reach the top band in at least 80% of the 2,000 resamples (R-MAT, R-SAFE and R-SKILL), and R-SAFE rests on a single study (Derived Calculation, analysis S3). Resamples that omit a study drop its risks, so these shares are conditional on the risk being present. This is the reason the tool shows seeded risks as a labelled, uncounted “literature tier” and asks for entered numbers, instead of treating the seed as an estimate.

The disagreement also has a use. Because the surveys do not converge, the report can offer a methods point that does not depend on any site: rankings of delay factors from a handful of South Asian RII surveys are not stable enough to be copied into a risk register as if they were probabilities. This is a modest, checkable observation and the one the audit trail best supports.

What the matrix level means

The tool multiplies a probability class by an impact class and labels the product Low, Moderate, High or Extreme. This is a conventional device and it has known weaknesses. Cox (2008) lists poor resolution, errors in ranking, suboptimal resource allocation and ambiguous inputs and outputs as problems of risk matrices, and Duijm (2015) gives recommendations on their design; both are cited here for the general critique, and only the abstract of Cox was read in this project. Three of those concerns are visible in the tool.

First, **resolution is coarse**. Only five classes per axis exist, so risks with quite different entered numbers share a cell. In the ILLUSTRATIVE example, five of seven risks share the cell probability class 2 by impact class 4 (Chapter 6), although their probabilities range from 0.20 to 0.35 and their expected delays from 2.3 to 2.7 days.

Second, **the score is symmetric and class 1 on either axis always gives Low**. A risk with a probability below 0.2 is Low whatever its consequence, since $1 \times 5 = 5$. The tool prints this as an Assumption and also lists impact-class-5 risks separately, as a filter and not a score, so that a rare but severe risk is not lost.

Third, **the result depends on the edges**. In the example, replacing the probability edges with alternative sets changes between 2 and 7 of the 7 levels, and changing the level thresholds over 2,024 admissible triples

changes a median of 5 of 7 (IQR 1 to 6; analysis S2). By contrast, scaling the impact edges between 0.5 and 2 changes the impact class of between 1 and 7 of the 7 risks but never a level, because the probability class caps the score. Two things follow. The level is less sensitive to the entered delay than a reader might expect; and the one number that moves the example most is a probability sitting exactly on an edge (R-WX, $p = 0.20$, which the tool places in class 2 under its lower-edge-inclusive convention). A reader should therefore treat the level as a prompt for discussion, and the probability class edges and thresholds as parameters that deserve a stated reason, which for this tool is that they are Assumptions (Appendix H).

For risks that carry only a literature tier, the weakness is larger. Using one band as both axes squares the rank (diagonal scores 1, 4, 9, 16 and 25), so a risk at the top fifth of an importance list scores 25. That is not a statement about probability or impact, and the earlier version of the tool coloured such placements Extreme, High or Low as if it were. After the committee review these placements are grey, dashed and excluded from the level totals. The more conservative option, removing level colours and the one-class step for rule flags altogether, remains a decision for the supervisor (audit file, F02).

Comparison with other ways of keeping a register

Three alternatives are realistic for a site engineer: a spreadsheet register, a schedule-risk package used by the planning office, and no formal register. The comparison below is limited to what could be established without testing commercial products. The project did not install or evaluate Primavera Risk Analysis or any other package, and the account of schedule-risk software rests on how it is described in the case studies of the corpus: Gheewala et al. (2025) used Primavera P6 with Risk Analyzer and Ichsan et al. (2025) used @RISK, in both cases on project schedules with distributions supplied by the study team. Anything not stated in those papers or visible in the repository is marked “not assessed”.

Table 7.2. Features of the tool compared with a spreadsheet register and with schedule-risk software as used in two cited case studies. Source: tool column from the repository (`README.md`, `docs/case_schema.md`, Appendices B to D); software column from Gheewala et al. (2025) and Ichsan et al. (2025) as summarised in Chapter 2; spreadsheet column describes what a plain sheet is, not any particular template. No commercial product was tested.

Capability	This tool	Spreadsheet register	Schedule-risk software (as used in the cited studies)
Masonry-specific risk library with evidence IDs	Yes: 46 risks, 30 with a survey seed value, 16 without	Only if the author builds one	Not assessed; the studies built their risk lists from surveys or focus groups
Source label required on every number	Yes: six labels; a number without one is rejected	Only if the author adds a column and keeps it	Not reported in the papers read
Site facts flag risks	Yes: 22 rules, Source Assumption, no numbers	Possible with formulas written by the author	Not assessed
Probability-impact matrix	Yes, with edges and thresholds printed as Assumptions	Possible; edges are whatever the author chose	Not assessed
Schedule simulation	Optional layer, entered numbers only, no defaults	Not in a plain sheet	Yes in both studies; distributions and probabilities supplied by the user (Chapter 2)
Whole-project schedule network	No: one activity	No	Yes: project-level in both studies
Works offline on one computer	Yes (local server or desktop window)	Yes	Not assessed
Evidence of use on real projects	None: verified by tests only; no user evaluation	Not assessed	Published case studies exist (two cited)

The differences are practical, not methodological. This tool contains no calculation that a competent engineer could not build in a spreadsheet, and it does not contain the project-level scheduling that the packages offer. What it adds is a particular discipline: a referenced list of masonry risks to start from, a prompt from site conditions, and an output that cannot show a number without its source. Whether that discipline helps a site engineer is not known. The closest published prior art for the optional quantitative layer is activity-level residual-risk simulation after mitigation (Ichsan et al., 2025, who took response effects from a five-person focus group); the response types in the register follow the avoid, reduce, transfer, accept classification used in the risk-response literature (Dey, 2011; Baker et al., 1999). No claim of novelty is made for any of these mechanics.

Implications for site engineers

The findings have practical consequences even though the tool has not been evaluated with users.

Use the library and the rules as a checklist, not as a forecast. The 46 risks and the rule prompts are the part most likely to help. A flag such as “stock on site runs out before a new delivery can arrive” asks a question the engineer can answer in a minute. It costs nothing to check and it is derived from facts the engineer entered. The flag says that the condition holds, not that a delay will follow.

Enter your own numbers, and say where they come from. A literature tier is a starting point for a conversation about which risks to look at first. As soon as the engineer has an estimate of a probability and a delay range, the estimate replaces the tier and carries its own Source label. An estimate from the engineer’s own experience is Expert Judgment; a count from the site diary is Historical Data. Both are better than a survey rank, and both are visible as such in the register and its exports.

Do not read the level as a measurement. The example shows that the level can change when a probability moves across an edge, and that the delay range has little influence on it. The ordering inside a level is not meaningful. A high-consequence risk of low probability appears in the separate impact-class-5 list for that reason.

Use the response fields. The register asks for an owner, an early-warning sign, a response type and a review date. These are text fields and carry no numbers, and they are what turns a ranked list into something a site team can act on. Several of the committee’s recommendations concerned exactly these fields, and the report treats them as at least as valuable as the matrix.

Treat the optional cost layer with caution. It compares responses using the engineer’s own costs and effects. If those effects are guesses, the output is a guess presented with a command word. The wording “preferred under the stated criterion” is deliberate, and a stability check across random seeds is built in, but neither guards against poor inputs.

Time and effort were not measured. The tool needs entry of facts and numbers for each activity. How long this takes, and whether a busy engineer would do it daily, is a question for the pilot in Chapter 8.

Threats to validity

Table 7.3. Threats to validity, their likely effect and what has been done or remains to do. Source: methods review of 2026-10-08, audit file, and the authors’ assessment.

Type	Threat	Likely effect	Done so far	Still needed
Construct	An RII is importance, not likelihood or days; the seed puts one band on both axes	A seeded placement has no quantitative meaning	Labelled Assumption everywhere; shown as an uncounted, grey literature tier; wording “starting position”	Expert likelihood and impact ratings on separate axes
Construct	Multiplicative matrix compresses range and depends on edges	Levels differ under other edges or thresholds	Edges and thresholds printed as Assumptions; sensitivity reported (analyses S1, S2)	Practitioner view on edges; option of dropping level colours for seeded risks
Internal	Single coder for the factor-to-risk mapping; some decisions inconsistent (F12)	The seed depends on one person’s judgment	Mapping rule documented; Direct versus Related distinguished	Second coder and Cohen’s kappa (Appendix G)
Internal	Analysis choices made after seeing the 10-risk data; the 38-risk run was a second look	Forking paths	One pre-specified family of variants; all variants reported	Re-run once on a reconciled mapping with the plan fixed in advance
Statistical conclusion	Five seed studies; 14 of 30 seeded risks rest on one; overlaps of 5 to 19 risks	Low power; unstable ranks; intervals understate uncertainty	Exact or permutation p-values; Holm correction; power shown (Figure 7.1)	More studies; complete the omitted A14, A10 and A21 factors and re-run once
Statistical conclusion	Surveys are not independent (M10 pools ten studies); M14 printed RII on a basis that differs from its stated scale	Agreement may be over- or understated	Noted; values kept as printed	Recompute M14 on its stated basis if the supervisor agrees
External	General building-construction surveys, two	Seed may not describe Indian masonry	Stated in every table note	Masonry-specific expert survey

Type	Threat	Likely effect	Done so far	Still needed
	from Sri Lanka, one road-heavy (A14 26.4% road respondents), manager-dominated			
Reliability of the tool	Defects listed in the engine audit (for example NaN accepted by the command-line writer, booleans accepted as numbers)	Wrong output on malformed input	Several fixed and tested after the review; 513 tests pass	Close the remaining items before any release
Ecological	No site data; example is ILLUSTRATIVE; no user evaluation	No evidence that levels match what happens on site	Claims limited accordingly	Site pilot and usability sessions (Chapter 8)
Corpus	Three corpus sizes appear in the project records (F08)	Reported search coverage uncertain	The three versions are listed with their source files in the audit file	Re-run the corpus statistics once against the database and keep one figure

One threat deserves a sentence of its own. The report was written after the tool, and several of its checks, such as the audit of the seed, the held-out comparison and the guard tests, were produced by reviewers who were asked to find faults. That is a strength for integrity and a weakness for independence: the same team that built the tool prepared the corrections. An outside reader should treat the audit trail as evidence of care, not as an independent evaluation.

What can and cannot be claimed

Table 7.4. Claims, whether the evidence supports them, and the wording to use. Source: docs/Project_Handoff_Summary.md section 8.11 and 8.12 (withdrawn and replacement wording), extended by this report.

Claim	Supported?	Wording to use
The tool predicts delay or labour productivity	No	“orders risks”; “starting position”
The literature seed has been validated	No	“agreement check with held-out surveys: weak, low-powered and inconclusive”
Published rankings agree	Not shown	“weak agreement, not distinguishable from chance”
The seed values are probabilities	No	“an importance rank used as an ordinal tier, an Assumption”
Every number in an export has a source	Yes, by design and by test G-5	“each number is printed with its Source”
The AI supplies no numbers	By design and guard; fixed after the audit	“the AI can propose names, mechanisms and evidence IDs only; numeric fields are rejected”
The tool is accurate on real sites	No evidence	not claimed
The tool is useful to site engineers	Not evaluated	“intended to help; to be evaluated”
A first or new framework	Not claimed	“an evidence-traceable tool; no new method”
Published agreement shows masonry-specific importance	No; surveys are general	“general building-construction surveys”

The conclusion of this chapter is that the project delivers a transparent, offline, verified tool and an honest account of how little published rankings can carry, and that it stops short of everything that would need data from people or from sites.

Chapter 8. Conclusions, Limitations and Future Work

Conclusions

The project set out to build and verify an offline tool that helps a site engineer identify, register and prioritise delay risks in a brick or block masonry activity, with every value traceable to its source and with published survey evidence used only as a labelled ordinal starting point. Table 8.1 states what was concluded for each objective of Chapter 1, and how far.

Table 8.1. Conclusions mapped to the objectives. Source: Chapters 3 to 7; docs/Audit_Corrections_2026-10-08.md; Appendix E (Derived Calculation and Literature, as labelled in those chapters).

Objective	Conclusion	How far it goes
1. Compile and verify a masonry delay-risk library with a documented, inter-coder-checked mapping	A library of 46 risks (38 original, 8 added on 2026-10-08) was built, 30 of them with a survey seed value and 16 without. 182 RII rows (161 checked against the PDFs) from 11 studies (5 seed, 6 held-out) were transcribed; every one of the 161 rows checked matches the printed papers; the other 21 were not checked against a PDF. The mapping rule and its Direct / Related distinction are documented	Partly met. One coder made all mappings, and some decisions are inconsistent (audit finding F12). The second-coder check and kappa have not been done (Appendix G). The seed is a subset of the printed factors (A14 5 of 39, A15 5 of 32, A02 14 of 35), and the surveys are general building-construction surveys, not masonry surveys
2. Implement site-condition rules, a source-labelled register and a 5 x 5 matrix with assumptions stated	22 site-fact rules, a register in which each number carries one of six Source labels (and in which text fields record owner, early-warning sign, response type, action and review date), a 5 x 5 matrix, and exports (Markdown, CSV, HTML, SVG, case file) were implemented. The assumptions that shape the output (equal-count bands, one band on both axes, probability edges, level thresholds, the one-class step for a rule flag) are listed in Appendix H and printed on every export	Met as software. The rule conditions and the matrix edges are Assumptions that no practitioner has yet endorsed
3. Analyse how consistent published RII rankings are, to establish how far literature can stand in for project data	Agreement between surveys is weak and not distinguishable from chance: held-out correlations 0.28, 0.63 and -0.18 on 19, 11 and 13 shared risks, none significant after Holm correction; leave-one-study-out cross-prediction 0.09 to 0.64; Kendall's W 0.40 (permutation p = 0.15). Band and study choices change 6.7% to 80.0% of seeded classes. Only a coarse order can be read from the literature	Met, with a negative result. The check is low-powered, so it cannot show that surveys disagree either. It supports the limited use of the seed as a starting position
4. Verify the software and evaluate it with practitioners; define site validation as future work	The suite passes 513 tests (3 skipped because the literature database is not in the repository) in 17 files; a worked example is given in Chapter 6. Site validation is defined below	Verification met; evaluation not done. No expert rating, usability session or site case exists

Four general conclusions follow.

1. **The most defensible contribution is the traceability.** Every number in a register carries a Source, every assumption is named, and the claims that the audit withdrew are recorded with the reason. A reader can see which numbers are evidence and which are placeholders.
2. **A survey importance index cannot stand in for a probability.** The seed is an ordinal starting position and nothing more. The weak agreement among surveys shows that even the order of the middle of the list is not settled by the data in hand.
3. **The matrix is a prompt for discussion.** Its levels depend on edges and thresholds that are Assumptions and, in the example, depend more on the probability class than on the delay entered. It helps a team look at the same list of risks at the same time; it does not measure them.
4. **The registered title promised more than was done.** Productivity prediction, a rules matrix giving probability and cost, and the financial decision command were not built as described, because no site data existed to feed them. Chapter 1 gives the term-by-term account. The optional quantitative layer added on 2026-10-08 runs only on numbers the user enters and has been verified only as software.

Limitations

The limitations that bear on every result are these.

- **No primary data.** No site was visited, no site record collected, and no practitioner consulted. No number in this report is Historical Data, and no number is an expert rating.
- **The seed is secondary and indirect.** The RII surveys are general building-construction surveys from India and Sri Lanka (plus held-out studies from other countries); none is masonry-specific. The transcribed factors are a subset. M01 prints only RII of 0.700 or more. Scales differ between studies and the scale of two is not recorded. Fourteen of the 30 seeded risks rest on one study.
- **One coder.** The factor-to-risk mapping has no second coder and no agreement statistic.

- **The agreement check is low-powered.** With 5 to 19 shared risks only a correlation of about 0.46 (19 shared risks) to 1.00 (5 shared risks) could be detected (Figure 7.1). The analysis was first run on a 10-risk set and then on the 38-risk library, so the second run is a second look, and the library has since grown to 46 risks, the 8 new ones having no survey rows.
- **Assumptions drive the placement.** Equal-count bands, one band on both axes, equal study weights, averaging within a study first, and alphabetical tie-breaking are choices of the team (Appendix H).
- **The tool has not been used by anyone but its authors.** Usability, time to complete a case and acceptability are unknown. The Anthropic path of the optional AI feature was never run against the live service; the Google path was.
- **Open defects.** The engine audit lists defects that were fixed and tested after the committee review, and others that still need a code change (Audit Corrections, sections 4 and 6; for example NaN in the command-line JSON writer and booleans accepted as numbers). The corpus-size figure has three versions that are not reconciled (F08). Legacy productivity code with a known norms defect remains in the repository, unused.
- **The registered title has not been changed.** The decision belongs to the project guide.

Recommendations

For a site engineer who wants to try the tool. Use it on one masonry activity at a time. Start with the site facts and the rule prompts; treat the literature tier as a list of things to think about, not as a ranking of danger; replace it with your own estimates as soon as you have them, with the right Source label; fill in owner, early-warning sign and review date; and keep the exported register with its stated assumptions. Do not use the optional cost layer to justify spending unless the costs and effects entered have a source you can show.

For the project guide (decisions recorded as open). (i) Retitle the work to match what was done, or keep the registered title with the scope-revision table of Chapter 1. (ii) Decide whether to drop the one-class step for rule flags and to show seeded risks with no level colour (F02, F18). (iii) Decide whether to enter the omitted A14, A10 and A21 factors and re-run the statistics once, with the analysis plan fixed in advance (F03); the basis of the seed must not be chosen by which agreement figure looks better. (iv) Decide how to treat the truncated M01 list (F13) and whether to recompute M14 on its stated basis. (v) Approve the second-coder check before any mapping is changed (F12). (vi) Reconcile the corpus count with one re-run (F08).

For the maintainers of the code. Close the open items of the engine audit that touch integrity (the items listed as open in Audit Corrections; NaN, booleans, missing delay Source and malformed input shapes were fixed on 2026-10-08); move the unused productivity, norms and legacy simulation modules to a folder marked as legacy so that no reader mistakes them for the deliverable; and keep the guard tests of Appendix E, which tie the counts and the wording rules in the documents to the data files.

Future work

The next steps are ordered by how much each adds for the effort. The first three need no site access.

1. Expert survey

Administer the 15-item instrument of Appendix F (frequency and severity rated separately, with N/A and Don't know, neutral wording, randomised order, profile questions) to practitioners who have supervised masonry, with the analysis plan dated before the first reply. The target is at least 30 respondents, because 10 to 15 would be a pilot: with a planning standard deviation of one Likert point (Assumption AS-18) the half-width of one item's index is about 0.10 at $n = 15$ and 0.07 at $n = 30$, while neighbouring seed risks differ by 0.001 to 0.02. Results would be labelled Expert Judgment with n , response rate and respondent profile, never converted to probabilities or days, and compared with the seed by Spearman correlation on the items with a one-to-one mapping, as an exploratory check with the detectable correlation stated in advance. The same survey can ask respondents to endorse the rule conditions and to name causes the library lacks. Institutional ethics approval and consent text must be settled first; neither has been sought.

2. Second coder

Carry out the protocol of Appendix G: freeze the coding manual, have a coder who did not build the mapping code the blinded rows, compute Cohen's kappa for the risk assignment and a weighted kappa for the strength, apply the pre-committed rule that kappa below 0.60 triggers a revision of the manual (Cohen, 1960; Landis & Koch, 1977), log every change, and re-run the seed table and the statistics once, reporting the original and reconciled results side by side.

3. Face validity and usability session

Ask 5 to 8 site engineers to complete a case on the example activity and then on one of their own, observe where they hesitate, and collect a standard usability questionnaire such as the System Usability Scale (Brooke, 1996). Report the real number of participants and no synthetic ones. This is the only form of tool evaluation possible before site data exist, and it would show whether the rule prompts, the Source labels and the register fields are understood.

4. Site pilot

A site pilot would collect a daily diary for each masonry activity, with cause codes equal to the library risk identifiers: date, masons and helpers present, quantity laid (in one unit chosen at the start), hours lost and cause, rain and heat, hoist, scaffold and mixer status, material received, hold points passed; plus a start-of-activity snapshot of every rule fact. The project records plan three to five completed activities for a first phase. With so few activities only descriptive comparisons are possible, and the report should say so. The four comparisons planned in the project records are: (a) did the tool's flags include the risks that happened (hit rate); (b) does the order of the matrix match the days lost (Spearman correlation); (c) do entered probabilities match how often risks occurred, which needs many activities; and (d) only if the quantitative layer is used, did the actual delay fall inside the simulated range. All diary numbers would be Historical Data. Site access and permission are the main obstacles.

5. Prerequisites for the parts of the registered title that were not built

Table 8.2. What each deferred part of the registered title would need before it could be built or switched on honestly. Source: docs/Project_Handoff_Summary.md sections 6 and 8; Research_Notes/ (productivity and norms notes); committee reviews of 2026-10-08. Nothing in the third column exists yet.

Part	Needs	Why the present data cannot supply it
Labour-productivity model	Measured output per crew per day on masonry activities (unit chosen in advance), crew size and mason-to-helper ratio, wall height and number of openings, weather and working hours, supervision time	The seed surveys rate importance; they contain no output measurements. CPWD and BIS labour norms are planning norms, not measured productivity, and the legacy norms file has a documented defect
Prediction from managerial factors	Many activities with managerial factors recorded (supervision hours, material staging, crew ratio) and an observed outcome for each, in enough number to fit and test a model out of sample	There are no projects and no outcomes. Fitting a model to survey importance values would produce a number without meaning
Monte Carlo schedule simulation on real risks	For each risk, a frequency from several activities or an elicited probability, and a delay distribution from days lost per event, each with a Source	RII is not a probability or days; the tool refuses to use seed placements in the simulation. The layer already runs on entered numbers, but nothing entered so far is evidence
Expected-monetary-value command	Cost of one delay day from the contract (liquidated damages, site overheads), cost quotations for each response, and an observed or elicited effect of each response on probability or delay	No cost or effect size exists in the project; the tool deliberately supplies none
Rules matrix of 27 states	A source for every cell, or a replacement by rules with practitioner endorsement	The original cells had no source; the present 22 rules carry no numbers

None of these is a coding task. Each is a data-collection task, and the pilot in item 4 is its first step.

6. A single clean re-run of the seed statistics

When the second coder has reported and the omitted factors of A14, A10 and A21 have been entered, re-run the secondary-data analysis once with the family of tests fixed in advance, permutation p-values, Holm correction and held-out studies kept out of any reference they are compared with. Correct the generator text of the corpus

scripts (F19, F20) and the circular figures that the older statistics scripts still print (F01), so that a re-run does not reintroduce withdrawn wording.

7. Routes to a paper

The work supports two modest routes, and not a claim of a validated predictive framework. The first is a paper that combines the expert survey (item 1) with a face-validity and usability evaluation of the tool (item 3). The second is a methods note on how unstable RII-based rankings of delay factors are across South Asian surveys, and what that means for seeding a risk register, built on the secondary-data analysis already done and the single clean re-run of item 6. Either needs the corresponding primary data or the re-run first. Publication in a particular indexed journal is not claimed here.

Closing statement

What has been built is a small, careful instrument: a library of masonry delay risks tied to the surveys that support them, a set of site-fact prompts, a register in which no number appears without its Source, and a record of what was checked and what was withdrawn. What it has not done is show that any of this changes a decision on a site. The most useful next step is to put it in front of site engineers and to record, with their permission and with real numbers, what happens on their next masonry activities.

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References are in APA 7th edition author-date style, assembled from the lists kept by each chapter writer. Only works that appear in the repository evidence corpus or in the project's methods notes are listed. Where a bibliographic detail (author initials, volume, pages, DOI, edition) is not recorded in the repository it has been left out and not guessed; such entries are marked with a dagger (†) and itemised in `report/references_unverified.txt`. Standards are listed without an edition year where the year is not recorded.

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Appendix A. Literature seed table: all 182 rows

This appendix reproduces every row of `data/literature_seed.json` in file order (Source: Literature; transcribed from the printed papers and checked row by row against the PDFs on 2026-10-08, see `docs/Audit_Corrections_2026-10-08.md` section 3). Generated by `report/scripts/gen_appendices.py`. No value has been altered. The RII is a survey importance index; it is not a probability and not a number of days. The rank is the rank printed in the source study, within that study's own list, and is not comparable between studies. "Mapped risk" is the library risk the team assigned (Source: Expert Judgment, single coder; the second-coder check is not done, see Appendix G). Mapping type "direct" means the factor is the same thing as the library risk; "related" means a looser match, and only "direct" rows of seed studies feed the literature seed. 135 rows are direct, 43 related and 4 unmapped (three are group-level rows or are noted as such in the file, one is the M14 item "No consumption"). The seed studies are a subset of the factors printed in several papers (for A14 only 5 of 39 factors were entered, for A15 5 of 32, for A02 14 of 35); M01 prints only factors with RII of 0.700 or more. These selection effects are discussed in Chapters 4 and 7.

Table A.1. The 11 studies in the seed file (5 seed, 6 held-out) and the number of rows each contributes. Source: `data/literature_seed.json`, field `studies`; row counts are a Derived Calculation from the file.

Study	Role	Year	Country / context	N	Scale	Rows
M01	seed	2019	India (Telangana)	44	1-5	28
A02	seed	2018	India (Chennai)	Not reported	1-4	14
M06	seed	2019	India	201	1-5	40
A14	seed	2022	Sri Lanka	163	1-5 (very low to very high significance)	5
A15	seed	2022	Sri Lanka	90 firms	Not recorded	5
M10	held-out	2023	Middle East (10 studies: Malaysia, Palestine, Kuwait, Qatar, Saudi Arabia, Egypt, Bahrain, Oman, Yemen, Vietnam)	10 studies	Pooled RII	33
M15	held-out	2021	Brazil (Brasilia)	47	1-4	24
M14	held-out	2026	Indonesia (East Java)	83	1-4	23
A10	held-out	2025	India	72	1-5	6
A21	held-out	2024	Thailand	30 projects; 380 respondents	1-5 importance scale	2
HAS25	held-out	2025	Iraq	Not recorded	Not recorded	2

Table A.2. The 182 seed rows. "Not printed" means the paper does not print a rank for that row. Source: Literature (`data/literature_seed.json`, field `rows`); mapped risk and mapping type: Expert Judgment.

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
1	M01 (seed)	Unsafe working conditions	0.809	1	R-SAFE	direct
2	M01 (seed)	Poor work planning	0.791	2	R-PLAN	direct
3	M01 (seed)	Unrealistic scheduling	0.791	3	R-PLAN	direct
4	M01 (seed)	Little or no financial rewards	0.777	4	R-INC	direct
5	M01 (seed)	Material shortages	0.777	5	R-MAT	direct
6	M01 (seed)	Payment delay	0.773	6	R-PAY	direct
7	M01 (seed)	Working at heights	0.764	7	R-SAFE	direct
8	M01 (seed)	Weather conditions	0.764	8	R-WX	direct

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
9	M01 (seed)	Lack of skill and experience of worker	0.759	9	R-SKILL	direct
10	M01 (seed)	Poor or no supervision method	0.759	10	R-SUP	direct
11	M01 (seed)	Incompetent supervisors	0.759	11	R-SUP	direct
12	M01 (seed)	Accidents	0.759	12	R-ACC	direct
13	M01 (seed)	Poor relation between labour and superintendents	0.755	13	R-COM	related
14	M01 (seed)	Poor condition of equipment and tools	0.750	14	R-TOOL	direct
15	M01 (seed)	Bad leadership skill	0.750	15	R-DIS	direct
16	M01 (seed)	Working 7 days per week	0.741	16	R-OVT	direct
17	M01 (seed)	Equipment and tools shortages	0.741	17	R-TOOL	direct
18	M01 (seed)	Low labour morale/commitment	0.736	18	R-MOT	direct
19	M01 (seed)	Rework	0.736	19	R-RWK	direct
20	M01 (seed)	Lack of empowerment	0.732	20	R-MOT	related
21	M01 (seed)	Change order during execution	0.732	21	R-DES	direct
22	M01 (seed)	Low amount of pay	0.727	22	R-INC	direct
23	M01 (seed)	High workforce absenteeism	0.723	23	R-LAB	direct
24	M01 (seed)	Lack of labour surveillance	0.714	24	R-SUP	direct
25	M01 (seed)	Inadequate safety plan	0.714	25	R-SAFE	related
26	M01 (seed)	Frequency of working overtime	0.709	26	R-OVT	direct
27	M01 (seed)	Effectiveness of management system	0.709	27	R-MGT	direct
28	M01 (seed)	Method of construction	0.700	28	R-MTH	direct
29	A02 (seed)	Lack of experience (manpower)	0.787	1	R-SKILL	direct
30	A02 (seed)	Absenteeism	0.751	2	R-LAB	direct
31	A02 (seed)	Age	0.730	not printed	R-FAT	related
32	A02 (seed)	Misunderstanding among labourers	0.680	not printed	R-COM	direct
33	A02 (seed)	Accidents	0.650	not printed	R-ACC	direct
34	A02 (seed)	Personal problems	0.610	not printed	R-FAT	direct
35	A02 (seed)	Lack of required construction material	0.780	1	R-MAT	direct
36	A02 (seed)	Lack of tools/equipment	0.760	not printed	R-TOOL	direct
37	A02 (seed)	Poor site condition	0.751	not printed	R-SITE	direct
38	A02 (seed)	Material storage location	0.700	not printed	R-STO	direct
39	A02 (seed)	Increase in price of material	0.550	9	R-PRC	direct
40	A02 (seed)	Payment delays	0.776	not printed	R-PAY	direct

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
41	A02 (seed)	Weather condition	0.700	3	R-WX	direct
42	A02 (seed)	Shortage of water	0.650	5	R-WAT	direct
43	M06 (seed)	Availability of resources	0.802	1	R-MAT	related
44	M06 (seed)	Contractual disputes	0.799	2	R-CTR	direct
45	M06 (seed)	Scope clarity of the project	0.790	3	R-SCP	direct
46	M06 (seed)	Design capability and frequent design changes	0.785	4	R-DES	direct
47	M06 (seed)	Rework	0.776	5	R-RWK	direct
48	M06 (seed)	Coordination between all stakeholders	0.774	6	R-MGT	direct
49	M06 (seed)	Project manager's authority to take financial decisions and select key team members	0.772	7	R-MGT	related
50	M06 (seed)	Timely payment of completed works	0.771	8	R-PAY	direct
51	M06 (seed)	Site clearance/availability	0.771	9	R-SITE	direct
52	M06 (seed)	Ability to handle the crisis by the PM	0.771	10	R-MGT	related
53	M06 (seed)	Use of inappropriate planning tools and techniques	0.763	11	R-PLAN	direct
54	M06 (seed)	Quality	0.754	12	R-QLT	related
55	M06 (seed)	Leadership qualities	0.752	13	R-DIS	related
56	M06 (seed)	Top management support to PM	0.752	14	R-MGT	related
57	M06 (seed)	Human resource and labour strike	0.749	15	R-STR	direct
58	M06 (seed)	Cost	0.744	16	R-CST	direct
59	M06 (seed)	Project coordination meetings	0.740	17	R-MGT	direct
60	M06 (seed)	Inadequate project formulation in the beginning	0.738	18	R-SCP	direct
61	M06 (seed)	Obsolete construction equipment, methods and technology	0.738	19	R-TOOL	related
62	M06 (seed)	Conflict of interest among team members	0.737	20	R-CTR	related
63	M06 (seed)	Customer/client satisfaction	0.730	21	R-CLI	related
64	M06 (seed)	Conflict of interest among team members (second entry)	0.730	22	R-CTR	related
65	M06 (seed)	Availability of training and development for enhancing skills	0.727	23	R-SKILL	related

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
66	M06 (seed)	Selection of PM with proven track record	0.724	24	R-MGT	related
67	M06 (seed)	Supply chain	0.700	25	R-MAT	related
68	M06 (seed)	Willingness to adopt change	0.700	26	R-MGT	related
69	M06 (seed)	Claim genuineness	0.695	27	R-CTR	direct
70	M06 (seed)	Regular budget update	0.693	28	R-CST	direct
71	M06 (seed)	Social skills of key team managers	0.692	29	R-COM	related
72	M06 (seed)	Working hours	0.688	30	R-OVT	direct
73	M06 (seed)	Developing and maintaining a short and informal line of communication	0.679	31	R-COM	direct
74	M06 (seed)	Urgency emphasized by the owner while issuing tender	0.674	32	R-CLI	related
75	M06 (seed)	Ability to delegate authority	0.670	33	R-MGT	related
76	M06 (seed)	Political and economic environment	0.666	34	R-ECO	direct
77	M06 (seed)	Exceptional difference between client and architect	0.666	35	R-CTR	related
78	M06 (seed)	Interpersonal skills	0.663	36	R-COM	related
79	M06 (seed)	Climate conditions	0.657	37	R-WX	direct
80	M06 (seed)	Interest and inflation rates	0.657	38	R-ECO	direct
81	M06 (seed)	Availability of accurate historical information	0.642	39	R-SCP	related
82	M06 (seed)	Social environment	0.594	40	R-SOC	direct
83	A14 (seed)	Shortage of skilled subcontractors/suppliers	0.829	1	R-MAT	related
84	A14 (seed)	Shortage of labourers (skilled, semi-skilled, unskilled)	0.825	2	R-LAB	direct
85	A14 (seed)	Delay of delivering materials to the site	0.810	4	R-MAT	direct
86	A14 (seed)	Fluctuation of material prices in the market	0.789	6	R-PRC	direct
87	A14 (seed)	Bad weather conditions	0.653	30	R-WX	direct
88	A15 (seed)	Skills shortage	0.820	1	R-SKILL	direct
89	A15 (seed)	Lack of thinking abilities	0.810	2	R-SKILL	related
90	A15 (seed)	Lack of work experience	0.800	3	R-SKILL	direct
91	A15 (seed)	Lack of knowledge in	0.790	4	R-SKILL	direct

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
		construction works				
92	A15 (seed)	Poor labour discipline	0.790	5	R-DIS	direct
93	M10 (held-out)	Inadequate workers' supervision	0.816	2	R-SUP	direct
94	M10 (held-out)	A shortage of skilled labour	0.801	3	R-SKILL	direct
95	M10 (held-out)	Extent of change orders	0.793	4	R-DES	direct
96	M10 (held-out)	Clarity of technical specifications	0.791	5	R-DES	related
97	M10 (held-out)	Coordination among design disciplines	0.785	6	R-DES	related
98	M10 (held-out)	Lack of incentive schemes	0.783	7	R-INC	direct
99	M10 (held-out)	Economic conditions	0.778	8	R-ECO	direct
100	M10 (held-out)	Material shortage at project site	0.775	9	R-MAT	direct
101	M10 (held-out)	Proportion of work subcontracted	0.761	10	R-SUB	direct
102	M10 (held-out)	Rework	0.761	10	R-RWK	direct
103	M10 (held-out)	Quality and availability of drawings	0.758	12	R-DES	direct
104	M10 (held-out)	Payment delay	0.751	13	R-PAY	direct
105	M10 (held-out)	Remuneration scale	0.738	14	R-INC	direct
106	M10 (held-out)	Stringent inspections	0.732	15	R-QLT	direct
107	M10 (held-out)	Physical fatigue	0.723	16	R-FAT	direct
108	M10 (held-out)	Inclement weather conditions	0.715	17	R-WX	direct
109	M10 (held-out)	Poor workers' motivation	0.709	18	R-MOT	direct
110	M10 (held-out)	Unrealistic deadline for project completion	0.698	19	R-PLAN	direct
111	M10 (held-out)	Government policies	0.698	19	R-GOV	direct
112	M10 (held-out)	Poor site management	0.696	21	R-PLAN	related
113	M10 (held-out)	Working overtime	0.696	21	R-OVT	direct
114	M10 (held-out)	Unavailability of suitable tools and equipment	0.691	23	R-TOOL	direct
115	M10 (held-out)	Problem of communication	0.682	24	R-COM	direct
116	M10 (held-out)	Unsuitability of storage location	0.680	25	R-STO	direct
117	M10 (held-out)	Design complexity	0.668	26	R-MTH	direct
118	M10 (held-out)	Construction method	0.666	27	R-MTH	direct
119	M10 (held-out)	Absenteeism	0.660	28	R-LAB	direct
120	M10 (held-out)	Working within a confined space	0.659	29	R-SITE	related
121	M10 (held-out)	Inspection delay	0.638	30	R-QLT	direct
122	M10 (held-out)	Accident	0.632	31	R-ACC	direct
123	M10 (held-out)	Lack of labour recognition	0.555	33	R-MOT	related

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
		programmes				
124	M10 (held-out)	Project location	0.555	33	R-SITE	related
125	M10 (held-out)	Inadequate training	0.459	35	R-TRN	direct
126	M15 (held-out)	Rework	0.750	1	R-RWK	direct
127	M15 (held-out)	Low labor productivity	0.730	2	R-PRD	direct
128	M15 (held-out)	Lack of skilled labor	0.725	3	R-SKILL	direct
129	M15 (held-out)	Low qualification of labor	0.705	4	R-SKILL	direct
130	M15 (held-out)	Modifications to the project	0.695	5	R-DES	direct
131	M15 (held-out)	Failure to manage deadlines	0.690	6	R-PLAN	related
132	M15 (held-out)	Poor specification or lack of definition of the project	0.690	6	R-DES	direct
133	M15 (held-out)	Incompatibility of projects	0.670	8	R-DES	related
134	M15 (held-out)	Poor qualification in the contractor's technical staff	0.650	9	R-MGT	related
135	M15 (held-out)	Delay in delivery of material	0.650	9	R-MAT	direct
136	M15 (held-out)	Unrealistic schedule	0.640	11	R-PLAN	direct
137	M15 (held-out)	Timing failures due to overestimation of productivity	0.630	12	R-PLAN	related
138	M15 (held-out)	Delay in ordering materials	0.615	13	R-MAT	direct
139	M15 (held-out)	Conflict between parties (contractor and contractor)	0.610	14	R-CTR	direct
140	M15 (held-out)	Delay in delivery of material at the front of service (site handling)	0.590	15	R-MAT	related
141	M15 (held-out)	Delay in the mobilization of the work	0.520	16	R-MGT	direct
142	M15 (held-out)	Low efficiency and productivity of equipment	0.510	17	R-TOOL	related
143	M15 (held-out)	Delay for defective material	0.485	18	R-DEF	direct
144	M15 (held-out)	Effects of climate on construction (heat, rain)	0.475	19	R-WX	direct
145	M15 (held-out)	Equipment shortage	0.455	20	R-TOOL	direct
146	M15 (held-out)	Accident during construction	0.455	20	R-ACC	direct
147	M15 (held-out)	Complexity of the project	0.444	22	R-MTH	direct
148	M15 (held-out)	Scarcity of building materials on the market	0.435	23	R-MAT	direct
149	M15 (held-out)	Equipment	0.415	24	R-TOOL	direct

No.	Study (role)	Factor as printed	RII	Rank	Mapped risk	Mapping
		defects				
150	M14 (held-out)	Low salary	0.735	1	R-INC	direct
151	M14 (held-out)	Lack of supervision	0.720	2	R-SUP	direct
152	M14 (held-out)	No incentives	0.716	3	R-INC	direct
153	M14 (held-out)	Late payment	0.716	4	R-PAY	direct
154	M14 (held-out)	Personal problems	0.711	5	R-FAT	direct
155	M14 (held-out)	Changing jobs	0.701	6	R-MOT	related
156	M14 (held-out)	Inadequate tools	0.689	7	R-TOOL	direct
157	M14 (held-out)	Project scale	0.660	8	R-MTH	related
158	M14 (held-out)	Types of work contracts	0.660	9	R-SUB	direct
159	M14 (held-out)	Weather	0.643	10	R-WX	direct
160	M14 (held-out)	Fatigue	0.612	11	R-FAT	direct
161	M14 (held-out)	No consumption	0.598	12	none (unmapped)	none
162	M14 (held-out)	Lack of skills	0.523	13	R-SKILL	direct
163	M14 (held-out)	Poor communication	0.518	14	R-COM	direct
164	M14 (held-out)	Late material	0.508	15	R-MAT	direct
165	M14 (held-out)	Project location	0.508	16	R-SITE	related
166	M14 (held-out)	Job scheduling	0.504	17	R-PLAN	related
167	M14 (held-out)	Work method	0.504	18	R-MTH	direct
168	M14 (held-out)	Compressed time	0.475	19	R-PLAN	direct
169	M14 (held-out)	Worker dissatisfaction	0.460	20	R-MOT	direct
170	M14 (held-out)	Inadequate safety	0.460	21	R-SAFE	direct
171	M14 (held-out)	Lack of motivation	0.458	22	R-MOT	direct
172	M14 (held-out)	The owner/supervisor cannot decide	0.412	23	R-CLI	direct
173	A10 (held-out)	Late material deliveries	0.652	not printed	R-MAT	direct
174	A10 (held-out)	Material shortages	0.619	not printed	R-MAT	direct
175	A10 (held-out)	Weather conditions (external)	0.616	not printed	R-WX	direct
176	A10 (held-out)	Material-related issues (group)	0.562	1	none (unmapped)	none
177	A10 (held-out)	Construction-site factors (group)	0.555	not printed	none (unmapped)	none
178	A10 (held-out)	Contractor-related (group)	0.506	not printed	none (unmapped)	none
179	A21 (held-out)	Contractor financial issues	0.777	1	R-FIN	direct
180	A21 (held-out)	Labour issues (shortage during harvest season or festivals)	0.733	2	R-LAB	direct
181	HAS25 (held-out)	Planning	0.874	not printed	R-PLAN	direct
182	HAS25 (held-out)	Team size	0.856	not printed	R-LAB	related

Appendix B. Risk library: all 46 risks

The risk library (`data/risk_library_masonry.json`) holds 46 candidate delay risks for one brick or block masonry activity: 38 from the original compilation and 8 added on 2026-10-08 (R-FRONT, R-VT, R-MORT, R-GPAY, R-OPEN, R-SCAF, R-FEST, R-HEAT). 38 are scoped to the activity and 8 to the project as a whole. “Seeded” means at least one direct survey row of a seed study maps to the risk, so the literature seed can place it at an ordinal tier; rank and class are those of `app.literature_seed.seed_table()` on the five seed studies (Derived Calculation, Assumption: five equal-count bands). “Unseeded” risks have no

survey value; they stay off the matrix until probability and delay are entered, and carry the **seed_status** text from the library. Evidence IDs point to records in the evidence corpus (**Literature_Evidence_Package.xlsx, Research_Notes/**) and support that the risk exists, never its size. The masonry-specific form of the 8 added risks is labelled Expert Judgment, to be confirmed by the expert survey (Appendix F).

Table B.1. Library risks by category. Source: Derived Calculation from **data/risk_library_masonry.json**.

Category	Risks
Labour	13
Management	6
Material	6
Design	4
Equipment	3
External	3
Financial	3
Quality	2
Safety	2
Site	2
Weather	2

Table B.2. The 46 library risks. Source: **data/risk_library_masonry.json** (name, category, scope, evidence IDs: Literature); seed status: Derived Calculation from **data/literature_seed.json**.

ID	Name	Category	Scope	Seed status	Evidence IDs
R-MAT	Brick / block / mortar material shortage or late delivery	Material	activity	seeded: rank 1 of 30, class 5, 3 studies	M01, M06
R-TOOL	Tools and equipment delay	Equipment	activity	seeded: rank 12 of 30, class 4, 2 studies	M01
R-LAB	Labour absenteeism or shortage	Labour	activity	seeded: rank 7 of 30, class 4, 3 studies	M01, M07, M15
R-SKILL	Unskilled or unqualified labour	Labour	activity	seeded: rank 3 of 30, class 5, 3 studies	M15, M14
R-PLAN	Poor planning or unrealistic scheduling	Management	activity	seeded: rank 4 of 30, class 5, 2 studies	M01
R-SAFE	Unsafe conditions or work at height	Safety	activity	seeded: rank 2 of 30, class 5, 1 study	M01
R-WX	Rain / monsoon stoppage	Weather	activity	seeded: rank 24 of 30, class 2, 4 studies	R08, M01
R-RWK	Rework due to workmanship or defects	Quality	activity	seeded: rank 11 of 30, class 4, 2 studies	M03, M15
R-PAY	Payment delay	Financial	activity	seeded: rank 5 of 30, class 5, 3 studies	M01
R-DES	Design changes or incomplete documentation	Design	activity	seeded: rank 10 of 30, class 4, 2 studies	M06, M13
R-SUP	Inadequate supervision or monitoring of workers	Labour	activity	seeded: rank 16 of 30, class 3, 1 study	M01, M10, M14
R-MOT	Low motivation, morale or labour turnover	Labour	activity	seeded: rank 17 of 30, class 3, 1 study	M01, M10, M14
R-INC	Inadequate wages or incentives	Labour	activity	seeded: rank 13 of 30, class 3, 1 study	M01, M10, M14
R-OVT	Overtime and long working hours	Labour	activity	seeded: rank 20 of 30, class 2, 2 studies	M01, M06, M10
R-FAT	Worker fatigue, age or personal problems	Labour	activity	seeded: rank 29 of 30, class 1, 1 study	A02, M10, M14
R-COM	Poor communication or misunderstanding on site	Labour	activity	seeded: rank 25 of 30, class 1, 2 studies	A02, M01, M06, M10, M14
R-DIS	Poor labour	Management	activity	seeded: rank 6 of 30,	A15, M01, M06

ID	Name	Category	Scope	Seed status	Evidence IDs
	discipline or weak site leadership			class 5, 2 studies	
R-TRN	Inadequate training of workers	Labour	activity	unseeded: no survey RII value; enter probability and delay	M10
R-PRD	Low labour productivity	Labour	activity	unseeded: no survey RII value; enter probability and delay	M15
R-STR	Labour strikes or disputes	Labour	activity	seeded: rank 14 of 30, class 3, 1 study	M06
R-ACC	Accidents and injuries on site	Safety	activity	seeded: rank 21 of 30, class 2, 2 studies	A02, M01, M10, M15
R-MGT	Poor site management and coordination	Management	activity	seeded: rank 18 of 30, class 3, 2 studies	M01, M06, M15
R-SUB	Subcontracting and contract-type problems	Management	activity	unseeded: no survey RII value; enter probability and delay	M10, M14
R-CTR	Contractual disputes and claims	Management	project	seeded: rank 15 of 30, class 3, 1 study	M06, M15
R-SCP	Unclear scope or inadequate project formulation	Design	project	seeded: rank 8 of 30, class 4, 1 study	M06
R-CLI	Client decision delay or pressure	Management	project	unseeded: no survey RII value; enter probability and delay	M06, M14
R-CST	Cost control and budget updating problems	Financial	project	seeded: rank 19 of 30, class 2, 1 study	M06
R-QLT	Inspection and quality-approval delays	Quality	activity	unseeded: no survey RII value; enter probability and delay	M06, M10
R-DEF	Defective or poor-quality material	Material	activity	unseeded: no survey RII value; enter probability and delay	M15
R-SITE	Poor site access, clearance or conditions	Site	activity	seeded: rank 9 of 30, class 4, 2 studies	A02, M06, M10, M14
R-STO	Poor material storage or handling	Material	activity	seeded: rank 23 of 30, class 2, 1 study	A02, M10
R-PRC	Material price escalation	Material	activity	seeded: rank 26 of 30, class 1, 2 studies	A02, A14
R-WAT	Water shortage	Material	activity	seeded: rank 28 of 30, class 1, 1 study	A02
R-MTH	Construction method or design complexity	Design	activity	seeded: rank 22 of 30, class 2, 1 study	M01, M10, M14, M15
R-ECO	Economic conditions, inflation or interest rates	External	project	seeded: rank 27 of 30, class 1, 1 study	M06, M10
R-GOV	Government policies or approvals	External	project	unseeded: no survey RII value; enter probability and delay	M10
R-SOC	Social or community environment	External	project	seeded: rank 30 of 30, class 1, 1 study	M06
R-FIN	Contractor financial difficulties	Financial	project	unseeded: no survey RII value; enter probability and delay	A21
R-FRONT	Work front not released (slab not de-shuttered, props or debris in place, frame not handed over)	Site	activity	unseeded: no survey RII value; enter probability and delay	D01, P1-05

ID	Name	Category	Scope	Seed status	Evidence IDs
R-VT	Hoist or material lift unavailable or broken (bricks and mortar to upper floors)	Equipment	activity	unseeded: no survey RII value; enter probability and delay	P1-02, P3-02
R-MORT	Mortar materials short or poor (cement, sand, or mixer down)	Material	activity	unseeded: no survey RII value; enter probability and delay	A02, D01
R-GPAY	Gang wages or labour-contractor running bill overdue	Labour	activity	unseeded: no survey RII value; enter probability and delay	A08, D21
R-OPEN	Openings, lintels, door/window frames and MEP sleeves not coordinated	Design	activity	unseeded: no survey RII value; enter probability and delay	M10, M06
R-SCAF	Scaffolding not available, not erected or not passed safe	Equipment	activity	unseeded: no survey RII value; enter probability and delay	P1-05, P3-01
R-FEST	Festival, harvest or election season exodus of the migrant gang	Labour	activity	unseeded: no survey RII value; enter probability and delay	A21, A03, A08
R-HEAT	Peak-summer heat cuts effective working hours	Weather	activity	unseeded: no survey RII value; enter probability and delay	P2-08, M15, P2-06

Unseeded risks (16): R-TRN, R-PRD, R-SUB, R-CLI, R-QLT, R-DEF, R-GOV, R-FIN, R-FRONT, R-VT, R-MORT, R-GPAY, R-OPEN, R-SCAF, R-FEST, R-HEAT.

Appendix C. The 22 site-fact rules

The rule base (`data/rules_masonry.json`) turns site facts the engineer already knows (Yes / No / Unknown, or a number entered by the user) into a flag on a library risk. A rule is a logical condition on facts. It contains no probability, no delay and no numeric threshold other than comparisons between two entered facts or a Yes/No fact. All 22 rules carry Source = Assumption (a logical condition proposed by the team); the evidence IDs support that the risk exists, not the rule's size or trigger. Level "elevated" marks an observed shortfall; level "normal" marks the risk as relevant only. Where several rules fire on one risk the highest priority wins; equal priority with different levels is reported as a conflict (guard test G-8). A rule whose facts are missing is shown as "not evaluable", never guessed. For a risk placed from the literature seed, an "elevated" flag raises its probability class by one (cap 5); that step is an Assumption of the tool (register item AS-11, Appendix H). Rule levels: 18 elevated, 4 normal.

Table C.1. The 22 rules: condition, flagged risk, level, priority and evidence. Source: `data/rules_masonry.json`. Conditions are rendered from the rule's `all_of` / `any_of` fields; "yes" is a Boolean fact set to true.

Rule	Flags risk	Condition	Level	Priority	Source	Evidence IDs	Confidence
RL-LAB	R-LAB	required_workers > available_workers	elevated	0	Assumption	M01, M07	Moderate
RL-MAT	R-MAT	material_stock_days <= material_lead_time_days	elevated	1	Assumption	M01, M06	Moderate
RL-MAT2	R-MAT	material_stock_days < planned_duration_days	normal	0	Assumption	M01	Low
RL-TOOL	R-TOOL	tools_shortage = yes	elevated	0	Assumption	M01	Low
RL-WX	R-WX	monsoon_overlap = yes	normal	0	Assumption	R08, M01	Moderate

Rule	Flags risk	Condition	Level	Priority	Source	Evidence IDs	Confidence
RL-HGT	R-SAFE	work_at_height = yes AND scaffolding_ready = no	elevated	0	Assumption	M01	Low
RL-PAY	R-PAY	payment_delay_expected = yes	elevated	0	Assumption	M01	Low
RL-DES	R-DES	design_incomplete = yes	elevated	0	Assumption	M06, M13	Low
RL-RWK	R-RWK	prior_rework_history = yes	elevated	0	Assumption	M03, M15	Low
RL-PLAN	R-PLAN	schedule_compressed = yes	elevated	0	Assumption	M01	Low
RL-SKILL	R-SKILL	skilled_masons_short = yes	elevated	0	Assumption	M15	Low
RL-WX2	R-WX	monsoon_overlap = yes AND external_walls_in_scope = yes	elevated	1	Assumption	A09, P1-05	Low
RL-PACE	R-PLAN	planned_daily_output > achieved_daily_output_last_week	elevated	0	Assumption	M01	Low
RL-FRONT	R-FRONT	work_front_ready = no	elevated	0	Assumption	D01, P1-05	Low
RL-VT	R-VT	masonry_floor_level >= 1 AND hoist_available = no	elevated	0	Assumption	P1-02, P3-02	Low
RL-SCAF	R-SCAF	scaffolding_ready = no	elevated	0	Assumption	P1-05, P3-01	Low
RL-MORT	R-MORT	sand_cement_stock_days <= sand_cement_lead_time_days	elevated	0	Assumption	A02, D01	Low
RL-GPAY	R-GPAY	gang_payment_overdue = yes	elevated	0	Assumption	A08, D21	Low
RL-OPEN	R-OPEN	any of: frames_on_site = no OR mep_sleeve_layout_marked = no OR lintel_level_plan_issued = no	elevated	0	Assumption	M10, M06	Low
RL-FEST	R-FEST	festival_or_harvest_in_wind_row = yes	normal	0	Assumption	A21, A03, A08	Low
RL-FEST2	R-FEST	festival_or_harvest_in_wind_row = yes AND gang_mostly_migrant = yes	elevated	1	Assumption	A21, A03, A08	Low
RL-HEAT	R-HEAT	summer_overlap = yes	normal	0	Assumption	P2-08, M15	Low

Table C.2. Description and rationale text of each rule as stored in the file. Source: data/rules_masonry.json.

Rule	Description	Rationale (as stored)
RL-LAB	Required workers exceed workers available	Logical shortfall condition; carries no numeric threshold. Risk existence: M01, M07.
RL-MAT	Stock on site runs out before a new delivery can arrive (for the material with the longest	Logical stock-out condition: stock days at or below the lead time leave no margin (a tie is

Rule	Description	Rationale (as stored)
	lead time)	a stock-out risk). Carries no numeric threshold. Risk existence: M01, M06.
RL-MAT2	Stock on site covers fewer days than the planned duration	Restocking is required during the activity, so the risk is relevant; it is not raised to elevated by this fact alone. Whether it fails depends on lead time (see RL-MAT, which wins when both fire).
RL-TOOL	Tools or equipment shortage expected	User-declared condition.
RL-WX	Activity period overlaps the monsoon / rainy season	Calendar overlap declared by the user. Internal walls can continue in rain, so monsoon alone marks the risk as relevant only (see RL-WX2 for external walls).
RL-HGT	Masonry is done at height and scaffolding is not ready	Work at height alone applies to almost every superstructure wall, so it is raised only when scaffolding is not ready. M01 ranks working at heights among the listed factors.
RL-PAY	Payment delays occurred previously or are expected	User-declared condition.
RL-DES	Drawings or details are incomplete	User-declared condition.
RL-RWK	Rework was frequent in earlier work on this site or crew	User-declared condition.
RL-PLAN	Planner or site engineer judges the programme faster than the gang can achieve	User-declared (subjective) condition. See RL-PACE for the checkable form.
RL-SKILL	Skilled masons are fewer than needed	User-declared condition.
RL-WX2	Monsoon overlaps the activity and external walls are in scope	Logical condition declared by the user; external walls are the work that rain stops. Risk existence: A09 (Kolkata monsoon), P1-05.
RL-PACE	Planned daily output is higher than the output achieved last week	Checkable form of RL-PLAN: both facts are User Input in the same unit (the user chooses it once); no norm is used.
RL-FRONT	Floors or walls planned for the next 7 days are not de-propped, cleared and handed over	User-declared condition. Risk existence: D01, P1-05. Masonry-specific form: Expert Judgment, to be confirmed by the expert survey.
RL-VT	Masonry is above the ground floor and no hoist or material lift is available	Floor 0 is the ground floor, so the only number is the floor index 1. Risk existence (related): P1-02, P3-02. Expert Judgment, to be confirmed by the expert survey.
RL-SCAF	Scaffolding is not ready	User-declared condition. Risk existence: P1-05, P3-01. Expert Judgment, to be confirmed by the expert survey.
RL-MORT	Sand or cement stock runs out before a new delivery can arrive	Same logical stock-out condition as RL-MAT, for mortar materials. Risk existence: A02, D01.
RL-GPAY	The gang's weekly payment or the labour contractor's bill is overdue	User-declared condition. Risk existence: A08 (abstract only), D21.
RL-OPEN	Door/window frames, MEP sleeve layout or lintel-level plan is not ready	Any one of the three being 'No' fires the rule. If one is left blank and the others are 'Yes', the rule is reported as not evaluable. Risk existence (related): M10, M06.
RL-FEST	A major festival, harvest or election falls inside the activity window	Calendar fact declared by the user: the risk is relevant only. Expert Judgment (to be confirmed by the expert survey) for the Indian form.
RL-FEST2	A festival, harvest or election falls inside the window and the gang is mostly migrant	Raised above RL-FEST (priority 1) when the gang is mostly migrant. Expert Judgment, to be confirmed by the expert survey.
RL-HEAT	The activity overlaps peak summer	Calendar fact declared by the user: the risk is relevant only. Heat figures from brick-making (P2-06) are not used.

Appendix D. Case file schema and interface endpoints

D.1 Case JSON schema, summary

A case is one JSON object that describes one masonry activity. `python -m app.cli template` prints a blank case and `python -m app.cli example` prints the ILLUSTRATIVE one. Every numeric input that is not a plain count is an object `{"value": ..., "source": ..., "evidence_ids":`

[...], "note": "..."}]. The **source** must be one of **Literature**, **Historical Data**, **User Input**, **Expert Judgment**, **Derived Calculation** or **Assumption**; a number without a source is rejected. Time is in working days. The full definition is `docs/case_schema.md`; this appendix summarises it.

Table D.1. Top-level blocks of the case file. Source: `docs/case_schema.md` (summarised).

Block	Fields	Notes
project	name, location, construction_type, notes	Descriptive text.
activity	id, name, planned_duration_days {value, source}	The planned duration is the denominator of the impact fraction for risks with entered numbers and an input to rule RL-MAT2. Risks placed from the literature seed do not need it.
facts	free key / value pairs	Inputs to the rules, for example required_workers, available_workers, material_stock_days, material_lead_time_days, scaffolding_ready, monsoon_overlap. Every fact used by the 22 rules is listed in <code>data/rules_masonry.json</code> . A missing fact makes a rule “not evaluable”.
rules	optional list	Default is <code>data/rules_masonry.json</code> . A rule flags a library risk as relevant or elevated; a flag never produces a probability or a delay for a risk with entered numbers.
risks	id, name, category, description, status, p {value, source, note}, delay {kind, a, m, b, lam, source, note}, evidence_ids; optional text fields owner, trigger, response_type (avoid, reduce, transfer, accept), response_action, review_date (YYYY-MM-DD)	p is the probability that the risk occurs at least once during the activity. delay is the extra delay in days if it occurs, as PERT, triangular, uniform or fixed. A risk with p or delay missing stays in the register; if the literature seed has a value for it, it is placed at a labelled literature tier, otherwise it is left off the matrix with a note. The register text fields carry no numbers.
use_literature_seed, seed_basis	optional	use_literature_seed is true by default. seed_basis is seed by default (the 5 seed studies, direct mappings); all also uses held-out studies and related mappings, so the held-out agreement check no longer applies to it.
impact_bin_edges_fraction	four ascending fractions of the planned duration	User input, no default. A value exactly on an edge goes to the higher class (relative tolerance 1e-9, both axes).
impact_bin_edges_source	optional {source, note}	Source of the four edges. If missing, a warning is issued and the exports print “Source: not given”.

Results returned by `run_case` (Appendix I shows an example) are: **project**, **activity**, **risks** (one row per register entry with a completeness flag and any rule flag), **matrix** (probability class, impact class, score, level and expected delay if it occurs, for every placed risk; literature-tier rows carry **basis: literature-seed** and no expected delay), **summary** (counts and level totals; literature-tier rows are not counted in the level totals), **rules** (fired, not evaluable, flags), **matrix_settings** (impact edges with Source; probability edges and level thresholds, both Assumption), **high_consequence** (entered-number risks in impact class 5; a filter, not a score), **warnings** and **report_markdown**. The command line writes `register.md`, `register.csv` and `result.json`. The register CSV appends **delay_note**, **expected_delay_source**, **tier**, **owner**, **trigger**, **response_type**, **response_action** and **review_date**; its **level** column is empty for literature-tier rows.

Table D.2. Optional analysis blocks of the decision layer (`app/analysis.py`, `python -m app.cli analyze`). Every number is a {"value", "source"} object and nothing has a default value except the simulation settings. Source: `docs/case_schema.md` (summarised).

Block	Content
activity.deadline_days	Optional deadline, needed for liquidated damages and the probability of exceeding the deadline.
risks[].direct_cost_if_occurs	Optional money lost if the risk occurs, on top of the cost of its delay days.
cost	cost_per_delay_day (required), ld_per_day_after_deadline (optional),

Block	Content
	currency (default INR).
mitigations[]	id, risk_id, action, strategy, cost (required), p_after and / or delay_after (entered, with a source, normally Expert Judgment), secondary risks, feasibility. A response with no modelled effect is reported and left out.
options[], constraints	Optional option sets (blank = generated: each response alone, combinations that target different risks, and Accept); optional maximum probability of exceeding the deadline and maximum budget.
simulation	n (default 10,000, at most 200,000), seed (default 12345), criterion, correlations.

The command is one of **AUTHORIZE MITIGATION**, **ACCEPT RISK**, **NO ADMISSIBLE OPTION** or **NO RESPONSE EVALUATED**. It is worded as “preferred under the stated criterion among the options evaluated”, never as optimal, and it is not produced from literature-tier placements.

D.2 Interface endpoints

The local interface (`python -m gui`) is a Flask application that serves a single page on `127.0.0.1` only. `gui/api.py` registers 23 routes, 21 of them under `/api/`. The table below is parsed from the source file by `report/scripts/gen_appendices.py`; the purposes are summarised from `gui/README_GUI.md` and `docs/case_schema.md`. The interface computes nothing itself: every figure on screen comes from `app.service.run_case` (or `run_analysis`), which runs on every edit.

Table D.3. Routes registered in `gui/api.py`. Source: `gui/api.py` (route paths); purposes summarised from the repository documents.

Method	Path	Purpose
GET	/	Serves the single-page interface
GET	/favicon.ico	Icon
GET	/api/health	Liveness check; reports whether AI is offline and the number of curated evidence records
GET	/api/meta	Version, counts and run settings for the interface
GET	/api/template	Blank case JSON
GET	/api/example	The ILLUSTRATIVE example case (placeholders, labelled Assumption)
GET	/api/risk-library	The risk library (Appendix B) with seed placement
GET	/api/rules	The site-fact rules (Appendix C)
GET	/api/evidence	Evidence records (curated list, by ids, or search with ?q=)
GET	/api/settings	Whether an AI key is configured (never returns the key)
POST	/api/settings	Store the AI provider and key outside the project folder
POST	/api/settings/test	One small real request to the provider to check the key
POST	/api/validate	Check a case; always HTTP 200 with ok true/false and a problem list
POST	/api/run	Run the register, rules and matrix on a case (HTTP 422 on problems)
POST	/api/analyze	Optional decision layer: Monte Carlo schedule, delay cost / EMV, response comparison on user-entered numbers
POST	/api/analysis-report	Markdown report of the optional decision layer
GET	/api/mitigation-catalogue	Catalogue of candidate responses (text only, no effect sizes)
GET	/api/example-analysis	ILLUSTRATIVE case with cost model, deadline and responses
POST	/api/suggest-risks	Guarded AI suggestions: risk names, mechanisms and evidence IDs only
POST	/api/report	register.md
POST	/api/register-csv	register.csv
POST	/api/matrix-svg	matrix.svg download

Method	Path	Purpose
POST	/api/report-html	report.html download

Errors on `/api/` paths return JSON. `POST /api/validate` always answers HTTP 200 with `{"ok": true, "result": ...}` or `{"ok": false, "problems": [...]}`; `POST /api/run` answers 422 when the case has problems. The desktop version (`python -m gui.qt_app`) hosts the same interface in a Qt WebEngine window; it is a wrapper, not a separate implementation.

Appendix E. Software test log

The automated test suite was run for this appendix on 2026-10-08 13:10 UTC with `pytest -v` (Python 3.13.16, pytest 9.1.1) from the repository root, using `report/scripts/gen_appendices.py`. The raw output is saved as `report/scripts/out/pytest_log.txt`. Result line: **513 passed, 3 skipped in 30.21 s**. That is 516 tests in 17 files: 513 passed, 3 skipped, 0 failed or errored.

The tests verify software behaviour (that the code does what the documents say it does). They do not validate the tool against real sites and say nothing about the accuracy of any risk level (Chapter 6). The counts are a Derived Calculation from the log; the module docstring column is the first line of each test module's own docstring, where one exists.

Two further files, `tests/test_explain.py` and `tests/test_india_norms.py`, are excluded from collection by `tests/conftest.py` (they exercise legacy modules); they are not part of the counts below.

Table E.1. Test results by file. Source: `pytest -v` run of 2026-10-08 13:10 UTC; docstrings from the test modules.

Test file	Tests	Passed	Skipped	Failed	Module docstring (first line)
<code>test_ai_guard.py</code>	10	10	0	0	
<code>test_analysis.py</code>	29	29	0	0	Tests for the quantitative analysis flow (<code>app/analysis.py</code>), its service wrappers, report and HTTP routes.
<code>test_corpus.py</code>	13	10	3	0	
<code>test_engine.py</code>	32	32	0	0	
<code>test_export_api_harden.py</code>	129	129	0	0	Regression tests for export / HTTP API / CLI / AI-guard hardening.
<code>test_gui_api.py</code>	14	14	0	0	Tests for the GUI's HTTP layer (<code>gui/api.py</code>) using Flask's built-in test client.
<code>test_gui_settings_api.py</code>	12	12	0	0	Tests for the <code>/api/settings</code> routes. Isolated from the real machine: <code>APPDATA/XDG_CONFIG_HOME</code> are
<code>test_gui_static.py</code>	3	3	0	0	Static-content checks for the GUI front end (no browser needed).
<code>test_legacy_harden.py</code>	26	26	0	0	Regression tests for the legacy quantitative modules (simulation, EMV, productivity, India norms).
<code>test_options.py</code>	17	17	0	0	Tests for option generation and the decision command

Test file	Tests	Passed	Skipped	Failed	Module docstring (first line)
					(app/engine/options.py) and the memoised Beta quantile.
test_productivity.py	13	13	0	0	
test_relevance_guard.py	14	14	0	0	Relevance / consistency guards G-1 .. G-10 (committee 2026-10-08).
test_report.py	1	1	0	0	
test_rules.py	5	5	0	0	
test_service.py	42	42	0	0	
test_settings.py	16	16	0	0	
test_validation_harden.py	140	140	0	0	Regression tests for the validation hardening of app.service / app.engine (audit 2026-10-08).
Total	516	513	3	0	

Table E.2. Skipped tests and the reason printed by pytest. Source: `pytest -rs` section of the same run.

Reason location	Count	Reason
tests/test_corpus.py:118	1	bulk harvest not run yet
tests/test_corpus.py:128	1	literature.db not built
tests/test_corpus.py:134	1	literature.db not built (scripts/build_literature_db.py)

E.1 Relevance and consistency guards

`tests/test_relevance_guard.py` pins the counts and the wording rules used in this report. It reads the numbers from the data files and compares them with the documents, so a change to the library, the rules or the seed that is not reflected in the documents fails the suite. Its 14 tests in this run are listed below; the guard numbering G-1 to G-10 follows `docs/Project_Handoff_Summary.md` section 9.

- `test_g1_counts_are_consistent_with_each_other`
- `test_g2_current_docs_state_the_counts_in_the_data`
- `test_g3_seed_rows_are_frozen`
- `test_g4_withdrawn_wording_does_not_occur`
- `test_g4_handoff_only_mentions_withdrawn_wording_on_lines_that_say _so`
- `test_g5_every_exported_number_has_a_source[example_case]`
- `test_g5_every_exported_number_has_a_source[_full_library_case]`
- `test_g6_legacy_modules_are_not_imported_by_the_live_app`
- `test_g6_front_end_does_not_load_legacy_screens`
- `test_g7_the_example_stays_illustrative_everywhere`
- `test_g8_rule_base_sanity`
- `test_g9_unseeded_library_risks_are_labelled_and_have_no_seed_rows`
- `test_g9_new_risk_evidence_ids_exist_in_the_corpus`
- `test_g10_ai_path_cannot_carry_numbers`

E.2 Known defects not covered by the suite

The engine audit of 2026-10-08 ([docs/Audit_Corrections_2026-10-08.md](#), section 4) lists defects that were reported and, for the items marked FIXED in section 6 of that file, repaired and tested after the committee review. Items still marked “needs code change” there (for example H1, the CLI JSON writer accepting NaN; M1, booleans accepted as numbers; M2, M3, M5, M6, M8, M9 and L1 to L14 except those listed as fixed) are not guaranteed by this test log. Several of the test files named above (for example [test_validation_hardening.py](#)) were written after the audit and pin the repaired behaviour. This list is the one to re-check before any release of the tool.

Appendix F. Proposed expert-survey instrument

PROPOSED, NOT ADMINISTERED. This instrument has not been sent to anyone. No response exists, no result in this report comes from it, and no number in this report should be read as survey evidence. It is the instrument the team proposes to use in the next phase (Chapter 8). Item wording and scales follow the committee’s domain-expert review (2026-10-08, “Expert survey and site data form”); the statistical plan follows the methods review of the same date. Both are the team’s own working documents. Ethics approval and the institution’s consent requirements must be confirmed before any distribution (not yet sought).

F.1 Purpose and design

The aim is to obtain the team’s own, masonry-specific evidence on how often each of 15 delay causes slows brick or block masonry on Indian sites and how severe the slip is when it happens. Frequency and severity are asked separately because a published RII mixes them, and neither is a probability or a number of days. The results would be labelled Expert Judgment, with n stated, and would sit beside the literature seed; they would not be converted into probabilities or days. The 15 items were chosen to cover the library’s activity-level risks that a site engineer can recognise and that the committee judged most likely to stop a masonry gang. A 46-item survey would invite fatigue and was rejected.

Table F.1. The 15 items, in the order of the working document (the order is randomised for each respondent). The last two columns show the nearest library risk; 6 of the 15 map to a risk that has a literature seed value and 9 to a risk without one. Source: Expert Judgment (committee review, proposed); library names from [data/risk_library_masonry.json](#).

Item	Wording shown to respondents	Nearest library risk	Library name
1	Bricks or blocks not on site when needed.	R-MAT	Brick / block / mortar material shortage or late delivery
2	Sand or cement short, or mortar mixer not working.	R-MORT	Mortar materials short or poor (cement, sand, or mixer down)
3	Masons or helpers absent or fewer than planned on the day.	R-LAB	Labour absenteeism or shortage
4	Skilled masons short, helpers doing mason work.	R-SKILL	Unskilled or unqualified labour
5	Masons leave around festivals, harvest or elections.	R-FEST	Festival, harvest or election season exodus of the migrant gang
6	Gang’s weekly payment or labour contractor’s bill paid late.	R-GPAY	Gang wages or labour-contractor running bill overdue
7	Work front not released by the RCC/shuttering team (props, debris, slab not handed over).	R-FRONT	Work front not released (slab not de-shuttered, props or debris in place, frame not handed over)
8	Hoist or material lift not available or broken.	R-VT	Hoist or material lift unavailable or broken (bricks and mortar to upper floors)
9	Scaffolding not available or not safe.	R-SCAF	Scaffolding not available, not erected or not passed safe
10	Door/window frames, lintel casting or electrical/plumbing positions not ready when masonry reaches that level.	R-OPEN	Openings, lintels, door/window frames and MEP sleeves not coordinated
11	Rain stopping external masonry.	R-WX	Rain / monsoon stoppage
12	Extreme summer heat reducing	R-HEAT	Peak-summer heat cuts effective

Item	Wording shown to respondents	Nearest library risk	Library name
	working hours.		working hours
13	Walls redone because of line, level, plumb, bond or opening errors.	R-RWK	Rework due to workmanship or defects
14	Inspection or approval (consultant, QA) delayed before masonry can continue or be plastered.	R-QLT	Inspection and quality-approval delays
15	Drawings for wall layout/openings not final or changed after start.	R-DES	Design changes or incomplete documentation

F.2 Introduction shown to respondents (proposed text)

“This survey is part of a student final-year project at [institution] on delay risks in brick and block masonry. It takes about 10 minutes. Participation is voluntary and anonymous: we do not ask for your name, your employer’s name or any project name, and you may stop at any time. Answers are used only in the project report and any paper derived from it, and are reported in groups. Please think of the masonry (brick or block wall) activities that you personally supervised or managed in the last 3 years. There are no right or wrong answers.”

F.3 Section A: respondent profile (all choices from a list; no free text except where stated)

1. Current role: site / project engineer; project manager; contractor or proprietor; labour or masonry sub-contractor; QA/QC; client or project-management consultant; faculty with site experience.
2. Total years in construction (bands).
3. Years directly supervising masonry (bands). Respondents with less than 1 year of masonry supervision are excluded from the analysis (pre-specified).
4. Number of building projects with masonry supervised (bands).
5. Building types (multiple choice): low-rise residential; high-rise residential; commercial; institutional or government.
6. Unit types used (multiple choice): burnt clay brick; fly-ash brick; AAC; concrete block.
7. State(s) or region of the sites.
8. Employer type: developer; general contractor; labour contractor; government / PWD; consultant.
9. How masons were mostly employed: through a labour contractor / piece-rate; daily wage directly.

F.4 Section B: the 15 items, two ratings each

For every item the respondent sees the same two questions. The frame above the section reads: “Think of the masonry activities you personally supervised or managed in the last 3 years.”

Frequency (how often it caused delay to masonry):

Code	Label
1	Never or almost never
2	Occasionally
3	Sometimes
4	Often
5	On almost every masonry activity
N/A	Not applicable / no experience

Severity (when it occurs, its effect on the masonry finish date):

Code	Label
1	No noticeable slip
2	Absorbed within the week
3	Finish date slips a little
4	Finish date slips noticeably

Code	Label
5	Holds up following trades (plaster, flooring) as well
DK	Don't know

The anchor wording is the team's own scale and is an Assumption. It is never converted to days or to a probability in the report.

F.5 Section C: closing items

- (a) "From the list, pick the three causes that most often delayed masonry on your sites." This gives a ranking check that is independent of the Likert items.
- (b) Open text: "Any other cause of masonry delay you have seen?" This feeds the completeness of the library and is the only free-text field.
- (c) Optional, for items 1, 3, 7 and 8 only: "Of the last 10 masonry activities you supervised, in about how many did this happen?" A count may later be entered for a risk as Expert Judgment with n stated; it is never labelled Literature.
- (d) One instructed-response (attention-check) item placed inside Section B: "For this row, please select 'Occasionally' for frequency." Respondents who fail it are excluded (pre-specified).

F.6 Bias controls

- Literature ranks, the tool and its matrix are not shown before or during the survey.
- Item order is randomised per respondent; frequency and severity are separate questions.
- "Not applicable" and "Don't know" are offered; they are never imputed (pairwise deletion).
- Item wording is neutral (no blame words such as "indiscipline"); recall is anchored to a time window and to activities the respondent personally supervised.
- Contractor-side and client / consultant-side respondents are both recruited and reported separately (each side tends to rate the other side's causes higher; this is a hypothesis, Expert Judgment).
- Anonymity; team members do not answer; the wording is piloted with 2 to 3 site engineers before release.

F.7 Pre-specified analysis plan

The plan is to be dated and saved in the repository before any response arrives. All items below are Derived Calculations from Expert Judgment ratings and would be reported with n.

1. Per item: number answering, median and interquartile range; frequency index (FI) and severity index (SI) with the same RII formula as the seed studies on the 1 to 5 scale; importance index FI x SI (the structure used by A11, an index the seed excludes); respondent-bootstrap 95% intervals (5,000 resamples) and bootstrap rank intervals.
2. Kendall's W across respondents (complete cases, tie-corrected) for frequency and for severity, and per group when a group has at least 8 respondents, with a permutation p-value. W measures consensus on the order, not the reliability of the instrument.
3. Reliability: Cronbach's alpha (Cronbach, 1951) with a confidence interval for convention only (the 15 causes are a checklist, not one construct, so a high alpha mostly reflects respondents who rate everything high); split-half reliability of the item order (1,000 random splits of respondents, Spearman-Brown corrected) is the reliability that matters.
4. Group differences per item (contractor side vs client / consultant side) by Mann-Whitney with Holm correction over the 15 items, with a rank-biserial effect size, only if each group has at least 10 respondents.
5. Comparison with the literature seed by Spearman correlation on the items with a pre-registered one-to-one mapping (about 10 of 15), exact p-value, described as exploratory. At n = 10 only a correlation of about 0.65 or more (two-sided 5%) could be detected, which is stated in advance.

- Optional short extra section, kept separate from the 15 items so that the main survey stays at about 10 minutes: respondents are shown the 22 site-fact rule conditions (Appendix C) and say for each whether it is a sensible warning sign (yes / no / unsure). This would endorse the conditions only; it would produce no thresholds and no numbers.

F.8 Sample-size honesty

The sample would be purposive and snowball (faculty, alumni, contractor contacts), so no population margin of error is claimed. With a planning standard deviation of one Likert point (Assumption AS-18, close to the 0.96 that M01's printed counts give), the half-width of one item's index is about 0.10 at $n = 15$ and 0.07 at $n = 30$, whereas adjacent seed risks differ by 0.001 to 0.02. Ten to fifteen responses are therefore a pilot; the target is at least 30, with the response rate (invited versus completed) and the respondent profile reported. For comparison, the seed surveys had $N = 44$ (M01), 201 (M06) and 163 (A14).

Appendix G. Second-coder protocol for the factor-to-risk mapping

PROPOSED, NOT YET CARRIED OUT. Every mapping of a survey factor to a library risk in Appendix A was made by one coder (Source: Expert Judgment). No second coder has been used, no agreement statistic exists, and the audit of 2026-10-08 found mapping decisions that are not consistent (finding F12; examples: M10 "poor site management", M06 "training", A02 "age", M14 "changing jobs"). No **mapping** or **risk** value has been changed since. This appendix is the protocol that the methods review proposed; it is to be executed before any mapping is altered and before the statistics are re-run.

G.1 What is coded

The 178 factor rows of `data/literature_seed.json` (135 direct, 43 related; the 4 unmapped rows are group-level or have no matching library risk). The current rows use 38 of the 46 library risks; the 8 risks added on 2026-10-08 have no rows yet. Any rows entered later (for example the 34 omitted A14 factors, or the omitted A10 and A21 items) are coded by the same procedure before entry.

G.2 Procedure

- Freeze the coding manual first.** For each of the 46 risks: name, scope, a one-sentence definition, two inclusion and two exclusion examples; the decision rule for "direct" (the factor is the same thing as the risk), "related" (the factor is a cause, consequence or part of the risk) and "none"; a rule for factors that are outcomes (for example "low productivity") and for factors that fit two risks.
- Choose coder 2.** A person who did not build the mapping: a team member who did not code, or a faculty member with masonry experience.
- Blind the sheet.** Export only the factor wording and the study context (country, building type) in random order. Hide coder 1's codes, the RII and the rank columns, so the coding cannot be driven by values.
- Calibrate.** Both coders code 15 to 20 rows from a study that is not in the agreement sample, discuss differences, amend the manual once, and freeze it again.
- Code all rows independently.**
- Compute agreement.** (a) Risk assignment: Cohen's kappa over 46 risks plus "none" (a total of 47 categories), with percent agreement and a bootstrap 95% interval over rows (Cohen, 1960). (b) Strength (direct / related / none): linearly weighted kappa. (c) The subset that actually feeds the seed (seed studies, direct rows): kappa and percent agreement on their own. Report the prevalence of each category, because kappa depends on it. Landis and Koch (1977) descriptors may be used as labels only.
- Apply the pre-committed rule.** If kappa is below 0.60 the manual is revised and a fresh sample is recoded; otherwise disagreements are resolved by discussion and every change is logged (row, old code, new code, reason, date).

8. **Re-run once.** The seed table and the statistics of Chapter 4 are recomputed once on the reconciled mapping, and the original and reconciled results are reported side by side. The choice of mapping is not to be made by which result looks better.

G.3 Output files (proposed layout)

File	Content
coding_manual_v1.md	The frozen manual (step 1), with a version and date.
coder2_sheet.csv	Blinded sheet: row id, factor wording, study context.
coder_codes.csv	Row id, coder 1 risk, coder 1 strength, coder 2 risk, coder 2 strength, reconciled risk, reconciled strength.
change_log.csv	Row id, old, new, reason, date, who.
kappa_report.md	Statistics of step 6 with n, prevalence and intervals.

G.4 What the result can and cannot show

A high kappa would show that a second person applying the same manual reaches the same codes; it would not show that the mapping is correct, because both coders share the manual. A low kappa would show that the manual, not only the coder, is ambiguous. Neither result changes the fact that the surveys are general building-construction surveys and not masonry-specific (Chapter 7).

Appendix H. Assumption register

Every value or rule in the tool that is chosen by the project team without supporting data is an Assumption (Source label “Assumption”) and is listed here, with where it is used, why it was made and which analysis, if any, shows how much the outputs depend on it. The register has 18 entries. It is deliberately longer than a typical assumptions list because the integrity rule of the project is that no placeholder may look like evidence. Sensitivity figures are Derived Calculations from `analysis/out/` (scripts `analysis/s1` to `s5`, one fixed random seed, all variants reported); they describe how the scoring rule and the seed bands respond to their inputs, not how any real site behaves. Where the figure comes from the ILLUSTRATIVE example it is itself ILLUSTRATIVE.

The factor-to-risk mapping is not listed as an Assumption because it is Expert Judgment (single coder, kappa pending; Appendix G). The Source labels used in this report are Literature, Historical Data, User Input, Expert Judgment, Derived Calculation and Assumption.

Table H.1. Assumption register. Source: tool code and data files named in the “Where used” column; sensitivity: `analysis/out/s1_summary.json`, `s2_summary.json`, `s3_summary.json` (Derived Calculation).

ID	Assumption	Where used	Why it was made	Sensitivity or check
AS-01	The seed’s equal-count class (1 to 5) is used as BOTH the probability class and the impact class of a risk with no entered numbers.	<code>app/literature_seed.py</code> ; every literature-tier chip	A survey importance index measures neither probability nor days; a placement needs some rule, and this is the simplest labelled one. It squares the rank (diagonal scores 1, 4, 9, 16, 25).	Banding alternatives change 30.0% (4 bands) to 80.0% (fixed RII cut-offs) of the 30 seeded classes (analysis S3).
AS-02	Five equal-count bands of six risks each over the 30 seeded risks (class 5 = most important).	<code>app/literature_seed.py</code>	Avoids absolute RII cut-offs that put almost all risks in one band; equal counts are a display convention.	3 equal-count bands change 40.0%, 5 equal-width bands 73.3%, fixed cut-offs 0.2/0.4/0.6/0.8 80.0% of classes (analysis S3).
AS-03	Items of one study that map directly to a risk are averaged first; the per-study means are then averaged with equal study weight.	<code>app/literature_seed.py</code>	Counts each study once; ignores sample size and between-study heterogeneity.	Dropping one seed study changes 6.7% to 48.1% of the classes of the risks that remain (analysis S3).
AS-04	Only ‘direct’ mappings of seed studies feed the seed; ‘related’ mappings and held-out studies are	<code>app/literature_seed.py</code>	Related rows are weaker matches; held-out studies are kept for the agreement check.	Adding related mappings changes 26.7% of classes (analysis S3).

ID	Assumption	Where used	Why it was made	Sensitivity or check
	excluded (default basis 'seed').			
AS-05	Ties in mean RII are broken alphabetically by risk ID (R-MTH and R-STO tie at 0.7000).	app/literature_seed.py	A rank must be a total order for equal-count bands.	Affects ranks of two risks only; not separately tested.
AS-06	Raw RII values from 1-4 and 1-5 scales (and one pooled index) are averaged without rescaling.	app/literature_seed.py	Scale details are not recorded for every study; rescaling needs the scale of each.	Rescaling (RII - 1/k)/(1 - 1/k) changes 6.7% of classes (analysis S3); within-study percentile rank changes 36.7%.
AS-07	Probability-class edges 0.2 / 0.4 / 0.6 / 0.8 (equal-width bins) for entered probabilities.	app/engine/matrix.py DEFAULT_P_EDGES	A conventional equal split of 0 to 1; not derived from data.	On the ILLUSTRATIVE example, alternative edge sets change 2 to 7 of 7 levels (analysis S2).
AS-08	Impact is the expected delay if the risk occurs divided by the planned duration, binned by four user-entered edges (no default).	app/engine/matrix.py	Makes impact comparable across activities of different length. The edges are User Input with their own Source; the example's edges (0.02 / 0.05 / 0.10 / 0.20) are ILLUSTRATIVE.	On the example, scaling all four impact edges by 0.5 to 2 changes the impact class of 1 to 7 of 7 risks but changes no level; the alternative edge set 0.2 / 0.4 / 0.6 / 0.8 changes 6 of 7 levels (analysis S2).
AS-09	Score = probability class x impact class; level thresholds 5 / 10 / 15 give Low (up to 5), Moderate (6 to 10), High (11 to 15), Extreme (16 and above).	app/engine/matrix.py matrix_level	A common multiplicative convention; its known weaknesses are discussed in Chapter 7 (Cox, 2008).	Over 2024 alternative threshold triples the median number of the example's 7 levels that change is 5 (IQR 1 to 6) (analysis S2).
AS-10	The score is symmetric: probability class and impact class are interchangeable.	app/engine/matrix.py	Consequence of the product; not a statement about risk.	Structural property of the rule (analysis S0/S1).
AS-11	A rule flag at level 'elevated' raises the probability class of a literature-tier risk by one, capped at 5.	app/service.py; labelled Assumption in all exports	Lets site facts move a seeded risk up the list; there is no source for the size of the step.	Not tested; option of dropping the step is a supervisor decision (Audit Corrections, F02 / F18).
AS-12	The rule conditions themselves (for example stock days at or below lead time) are logical conditions proposed by the team.	data/rules_masonry.json (all 22 rules carry Source = Assumption)	Observed shortfalls are the conditions a site engineer can check; practitioner endorsement has not been collected.	Not tested; planned in the expert survey (Chapter 8).
AS-13	A value exactly on an edge goes to the higher class; a relative tolerance of 1e-9 absorbs floating-point rounding (both axes).	app/engine/matrix.py EDGE_REL_TOL	Makes classification deterministic on decimal inputs such as 0.2.	Verified by tests (Appendix E). The example risk R-WX (p = 0.20) sits exactly on the first edge, which drives many sensitivity counts.
AS-14	Probability class 1 always gives level Low whatever the impact (1 x 5 = 5).	app/engine/matrix.py	Consequence of AS-09; the report lists impact-class-5 risks separately as a filter, not a score.	Structural property of the rule.
AS-15	Analysis design choices: sensitivity multipliers 0.5 to 1.5, 2,000 bootstrap resamples over 5 seed studies, permutation tests with 20,000 permutations, Holm correction, one fixed random seed (20261008).	analysis/*.py	Pre-specified so that no variant is chosen after seeing results; all variants are reported.	Not applicable; these are analysis settings, not tool behaviour.
AS-16	Example case inputs (probabilities, delay ranges, planned duration 16 days, site facts) are placeholders.	examples, app/service.py example_case()	A demonstration needs numbers; they are labelled ILLUSTRATIVE and are not evidence.	Sensitivity of the example is reported in Chapter 6 (analysis S1/S2).
AS-17	Optional decision layer: 10,000 simulation runs and random seed 12345 by default; no default cost, effect size,	app/analysis.py	Run count and seed are tool defaults; every economic number must be entered with a Source.	Decision stability under three seeds and cost-per-day sensitivity are produced by the tool itself.

ID	Assumption	Where used	Why it was made	Sensitivity or check
	probability or delay.			
AS-18	For planning the proposed expert survey, the standard deviation of one Likert item is taken as 1 point.	Appendix F	M01's printed response counts give about 0.96 (methods review); used only to state the precision to expect.	Planning figure only.

H.1 How to read the sensitivity column

The seed-band sensitivity shows that the class of a seeded risk depends on the banding rule and on which studies are included, which is why the tool shows seeded risks only as a labelled, uncounted literature tier. The example sensitivity shows that, in the example, the level responds to the probability class and to the thresholds far more than to the entered delay range, because the probability class caps the score. Neither result says that a real risk behaves in this way.

H.2 Items that are decisions, not assumptions

Some open points are decisions for the supervisor rather than assumptions: whether to drop the “+1 class for a rule flag” step and to show seeded risks without any level colour (finding F02 / F18), whether to enter the omitted A14, A10 and A21 factors and re-run the statistics (F03), whether to treat the M01 truncated list differently (F13), which corpus count to report (F08), and whether the registered title or the scope is to be changed (F07). They are recorded in `docs/Audit_Corrections_2026-10-08.md`.

Appendix I. User guide

This guide is for a site engineer or student who wants to install the tool and work through a case. It is summarised from `README.md` and `gui/README_GUI.md` and uses the ILLUSTRATIVE example case shipped with the tool. Every number in the example is a placeholder labelled Assumption; do not read it as site data or as evidence. The step results shown here were produced by running `app.service.run_case` on the example on 2026-10-08 and agree with `python -m app.cli run`; the screen layout is described from the GUI documentation and was not re-photographed for this guide.

I.1 What the tool does and does not do

The tool helps one engineer identify, register and rank delay risks for one brick or block masonry activity. It offers a library of 46 risks, 22 rules that read site facts you declare, a risk register in which each number carries a Source, and a 5 x 5 probability-impact matrix. It runs offline on your computer. It does not predict delays or productivity, it does not know anything about your site until you enter it, and it never turns a survey rank into a probability or a number of days. Risks you have not given numbers appear at a labelled “literature tier”, which is an ordinal starting position, not an assessment. An optional cost and decision screen works only from numbers you enter.

I.2 Install and start

Windows. Double-click `setup_windows.bat`. It creates a `.venv` folder, installs the requirements and starts the interface. Next time use `gui\run_gui.bat`. A built desktop program can be produced with `build_desktop_exe.bat` (a Windows executable that needs no Python on the target computer); building it was not part of this report's verification.

Any platform.

```

pip install -r requirements.txt
python -m gui                    # opens http://127.0.0.1:8765/
python -m gui --port 9000        # another preferred port
python -m gui --no-browser       # do not open a browser
python -m gui.qt_app             # PySide6 window instead of the
browser
```

Stop the browser version with Ctrl+C in the terminal. The server listens on 127.0.0.1 only; do not expose it on a network. Command-line use without the interface:

```
python -m app.cli example > case.json
python -m app.cli run case.json --out out      # writes register.md,
register.csv, result.json
python -m app.cli example-analysis > a.json    # optional decision
layer, ILLUSTRATIVE
python -m app.cli analyze a.json --out out      # writes analysis.md
```

To run the tests: `pip install -r requirements-dev.txt`, then `python -m pytest` (Appendix E).

AI suggestions work offline (evidence retrieval and matching library risks). If you add an Anthropic or Google key in Settings, the tool returns guarded candidate risks with evidence IDs; the key is stored outside the project folder. The AI may propose risk names and cite evidence. It never supplies a probability, a delay, a cost or a matrix placement, and numeric fields in its answers are rejected.

I.3 Step by step on the ILLUSTRATIVE example

Click **Load example** in the top bar (or open a case file with **Open**). A banner on every screen says ILLUSTRATIVE. The left rail has six screens.

1. **Case & Activity.** Project details, the activity name and the planned duration in working days with its Source. The example uses a planned duration of 16 working days (Assumption, ILLUSTRATIVE). Impact on the matrix is measured as a fraction of this duration, so enter your own value with its Source.
2. **Site facts & Rules.** Declare each fact as Yes, No, Unknown or a number. The 22 rules flag risks as relevant or elevated; a rule that lacks a fact is shown as not evaluable and is never guessed. Changing a fact adds any flagged risk that is missing from the register, with probability and delay left empty. Opening or loading a case adds nothing: a saved case is restored exactly. In the example, six risks are flagged elevated or relevant by the rules that fired (table below).
3. **Risk register.** One card per risk, with probability, delay range (PERT, triangular, uniform or fixed), a Source for each, status, evidence chips, and the text fields owner, early-warning sign, response type (avoid / reduce / transfer / accept), action and review date. The side panel lists the library and the AI suggestions. A risk with a missing number stays in the register, off the matrix, with a note, unless the literature seed can place it, in which case it appears as a grey dashed chip labelled “literature tier (Assumption)”.
4. **Risk matrix.** Enter the four impact bin edges (fractions of the planned duration) and their Source. The 5 x 5 grid places every complete risk and a table lists probability class, expected delay if it occurs, impact class, score and level. This is ordinal prioritisation only. The example has 7 risks.
5. **Evidence.** The curated evidence records; the search reaches the whole corpus including the bulk literature harvest.
6. **Export.** Download `register.md`, `register.csv`, the case JSON, the matrix (`matrix.svg`) and `report.html`.

Top bar: case name, unsaved-changes indicator, input-problem counter, **New**, **Open**, **Save**, **Load example**, light / dark theme. Shortcuts: Ctrl+S save, Ctrl+O open, Esc closes pop-ups. A value that is present but wrong (probability outside 0 to 1, a delay with minimum above maximum, a number with no Source) is shown as a problem and the register keeps its last valid state until it is fixed.

Result of the example (as produced by the tool)

Table I.1. Matrix rows for the ILLUSTRATIVE example: planned duration 16 days (Assumption), impact edges 0.02 / 0.05 / 0.10 / 0.20 of the planned duration (Assumption, ILLUSTRATIVE), probability edges 0.2 / 0.4 / 0.6 / 0.8 (Assumption, tool default), thresholds 5 / 10 / 15 (Assumption, tool default). Expected delay is

the mean of the entered PERT range, $(a + 4m + b) / 6$. Level counts: 1 High, 5 Moderate, 1 Low. Source: `app.service.run_case` on the example case (Derived Calculation from Assumption inputs, ILLUSTRATIVE).

Risk	p (as entered)	p class	Expected delay if it occurs (days)	Impact class	Score	Level
R-MAT	0.5	3	3.50	5	15	High
R-LAB	0.35	2	2.50	4	8	Moderate
R-RWK	0.25	2	2.33	4	8	Moderate
R-WX	0.2	2	2.67	4	8	Moderate
R-PLAN	0.1	1	1.33	3	3	Low
R-SAFE	0.3	2	2.50	4	8	Moderate
R-SKILL	0.22	2	2.33	4	8	Moderate

Table I.2. Rules that fired on the example’s declared facts. Source: `app.service.run_case` (rules from `data/rules_masonry.json`, Assumption).

Rule	Flags risk	Level
RL-LAB	R-LAB	elevated
RL-MAT	R-MAT	elevated
RL-MAT2	R-MAT	normal
RL-WX	R-WX	normal
RL-RWK	R-RWK	elevated
RL-PLAN	R-PLAN	elevated
RL-SKILL	R-SKILL	elevated
RL-WX2	R-WX	elevated
RL-PACE	R-PLAN	elevated

Reading the table: R-MAT has the highest level (High, p class 3 x impact class 5 = 15) because its entered probability and its expected delay of 3.5 days (a fifth of the planned duration or more) are the largest; the five Moderate risks all share the cell p class 2 x impact class 4. Several of those classes depend on a value sitting exactly on a class edge (R-WX has $p = 0.20$, which is on the first probability edge and so falls in class 2); a tiny change would move it. This is one reason the level is a prioritisation aid and not a measurement (Chapter 6).

I.4 Working with a real activity

1. Enter your own planned duration, impact edges and Sources before anything else; replace every ILLUSTRATIVE value, and delete the example’s banner text from the notes.
2. Declare the facts you know; leave “Unknown” where you do not. Rules never guess.
3. Review each flagged risk. Add risks from the library or from the AI suggestions that fit the activity; remove any that do not.
4. For every risk you care about, enter probability and delay with a Source. Use “Expert Judgment” for your own estimate, “Historical Data” for numbers from your own past activities, “User Input” for a measured or contractual value. The tool does not check that a number is sensible.
5. Fill in owner, early-warning sign, response type, action and review date. These are text and are part of the register.
6. Export the register and keep the case JSON. The exports print each number with its Source, and print the matrix settings as Assumptions.

I.5 Optional cost, options and decision analysis

The command `python -m app.cli analyze` (and `POST /api/analyze`) runs a Monte Carlo schedule simulation, delay-cost and EMV analysis and a comparison of the responses you enter, with a command of **AUTHORIZE MITIGATION, ACCEPT RISK, NO ADMISSIBLE OPTION** or **NO RESPONSE EVALUATED**. It needs the cost of one delay day, a cost for each response and the effect of each response, all entered by you with a Source; it has no default costs or effect sizes, never uses literature-tier placements as probabilities, and words its output as “preferred under the stated criterion”, never as optimal. It was added on 2026-10-08, is not in the navigation of the interface, and has been verified only as software, not against real projects. The earlier labour-productivity models are not used by the interface.

I.6 Known limits to keep in mind

- The tool is a single-user, local program on a development-grade server.
- The probability-class edges and level thresholds are fixed assumptions; the impact edges are yours and have no default.
- The literature seed comes from general building-construction surveys; it is a starting position only. Replace it with your own numbers or with expert-survey results.
- Retrieval of evidence is lexical (word overlap), not semantic; many harvested records have only an abstract.
- The Google (Gemini) AI path was checked against the live service; the Anthropic path was not (no key was available).
- Defects listed as “needs code change” in `docs/Audit_Corrections_2026-10-08.md` section 4 may still be present.

Appendix J. Glossary and abbreviations

Table J.1. Terms used in this report. Definitions follow the use in this project, not necessarily a standard.

Term	Meaning in this report
Activity	One masonry work package (brick or block walls of a floor or part of a building), the unit for which a case is built.
Agreement check	A comparison of the order of risks in one survey with the order of the literature seed, using rank correlation on the risks both have. It is exploratory and low-powered; it is not validation.
AAC	Autoclaved aerated concrete (block).
Assumption	A Source label for a value or rule chosen by the project team with no supporting data. Every assumption is listed in Appendix H.
Band, class	One of five ordinal groups (1 to 5). The seed puts the 30 seeded risks into five bands of six by mean RII. Probability and impact are also classified 1 to 5.
Case	The JSON file that holds one activity, its site facts, risks, entered numbers and sources (Appendix D).
CPWD	Central Public Works Department (India); its labour norms are cited in the legacy productivity code, which is out of scope.
Delay (in the tool)	Extra working days of delay if the risk occurs, entered by the user as a range (PERT, triangular, uniform or fixed).
Derived Calculation	A Source label for a number computed from other labelled numbers; it inherits ILLUSTRATIVE if its inputs do.
EMV	Expected monetary value: the probability-weighted cost of an outcome. Used only in the optional decision layer, on user-entered numbers.
Expert Judgment	A Source label for a value or mapping that comes from a person’s professional judgment. The factor-to-risk mapping is Expert Judgment by one coder.
Fact	A site condition entered as Yes / No / Unknown or as a number; the input of a rule.
Held-out study	One of the 6 surveys in the seed file that does not feed the seed and is used only for the agreement check (M10, M15, M14, A10, A21, HAS25).
Holm correction	A step-down adjustment of p-values for several tests, controlling the chance of any false positive (Holm, 1979).
Historical Data	A Source label for numbers from the organisation’s own past projects. None has been collected.
ILLUSTRATIVE	Marker on the example case: every number is a placeholder, not evidence and not site data.
Kappa (Cohen’s)	Chance-corrected agreement of two coders on categories (Cohen, 1960). Planned for the factor-to-risk mapping (Appendix G); not computed.
Kendall’s W	Coefficient of concordance of several rankings (0 = no agreement, 1 = complete).
Library	The 46 candidate masonry delay risks (Appendix B).
Literature	A Source label for a value taken from a published paper.
Literature tier	The labelled ordinal position (1 to 5) given to a risk from the

Term	Meaning in this report
	literature seed when no numbers are entered; an Assumption, shown as a grey dashed chip, not counted in level totals.
Mapping	The assignment of a survey factor to a library risk: direct, related or none.
Matrix (5 x 5)	A grid of probability class by impact class whose cells are given a level by the product of the classes.
Monte Carlo	Repeated random sampling of risk occurrence and delay to build a distribution of project duration. Offered only as an optional layer on entered numbers; it is not used on survey values.
P90	The duration not exceeded in 90% of simulated runs (decision layer only).
PERT	A three-point delay distribution (minimum a, most likely m, maximum b) with mean $(a + 4m + b) / 6$.
Planned duration	The planned working days of the activity; the denominator of the impact fraction.
Register	The list of risks of one activity with their numbers, sources, status and (as text) owner, early-warning sign, response type, action and review date.
RII	Relative importance index: sum of respondents' weights divided by (highest weight x number of respondents). It measures how important respondents rated a factor, not its probability or days of delay.
Rule	A logical condition on facts that flags a library risk as relevant or elevated (Appendix C).
Seed (literature seed)	The default ordering of library risks by mean RII from the 5 seed studies; an ordinal starting position and an Assumption, to be replaced by expert and site data.
Seed study	One of the 5 surveys (M01, A02, M06, A14, A15) that feed the seed.
Source label	The tag on every number: Literature, Historical Data, User Input, Expert Judgment, Derived Calculation or Assumption.
User Input	A Source label for a number typed in by the user for the case at hand.
WBS	Work breakdown structure; used in the earlier, larger scope (Chapter 1, scope revision), not in the current tool.